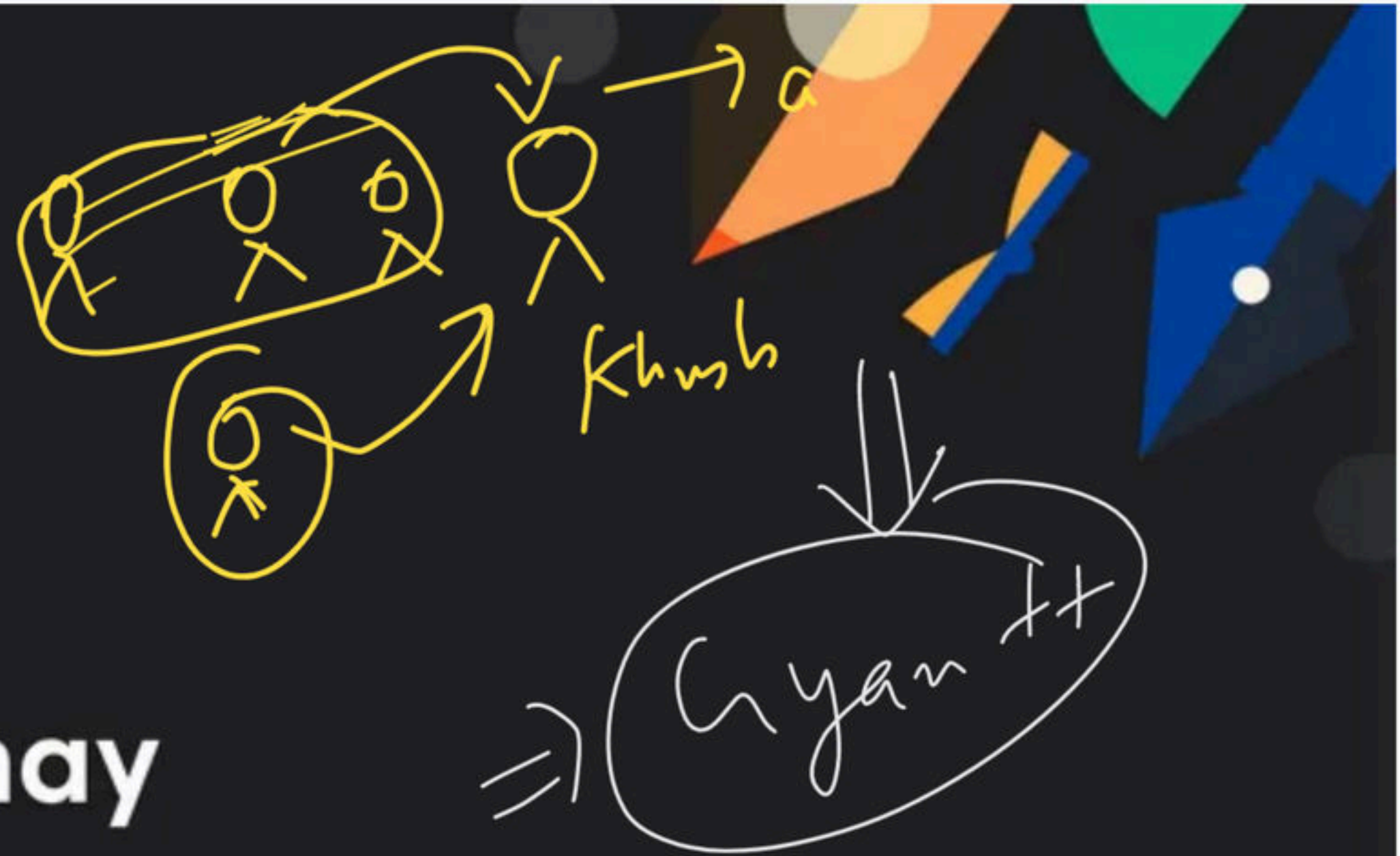
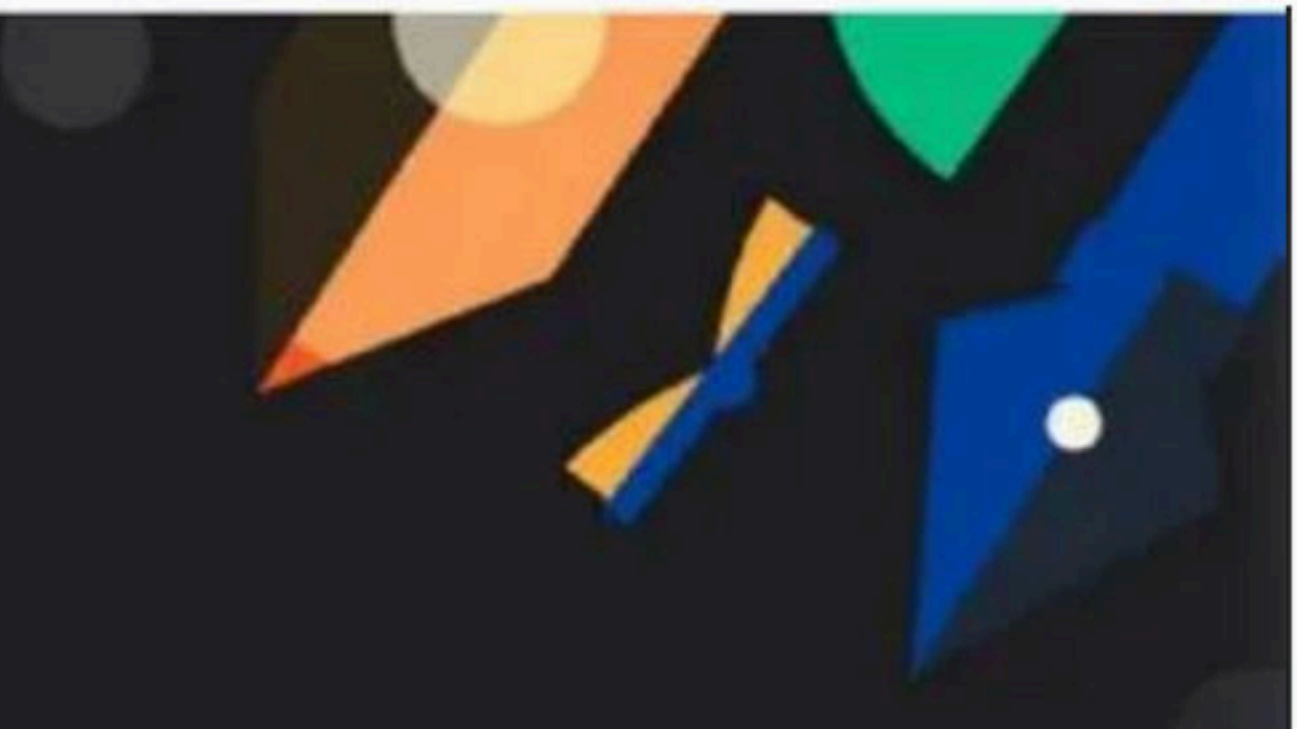




Doubt Class with Lakshay

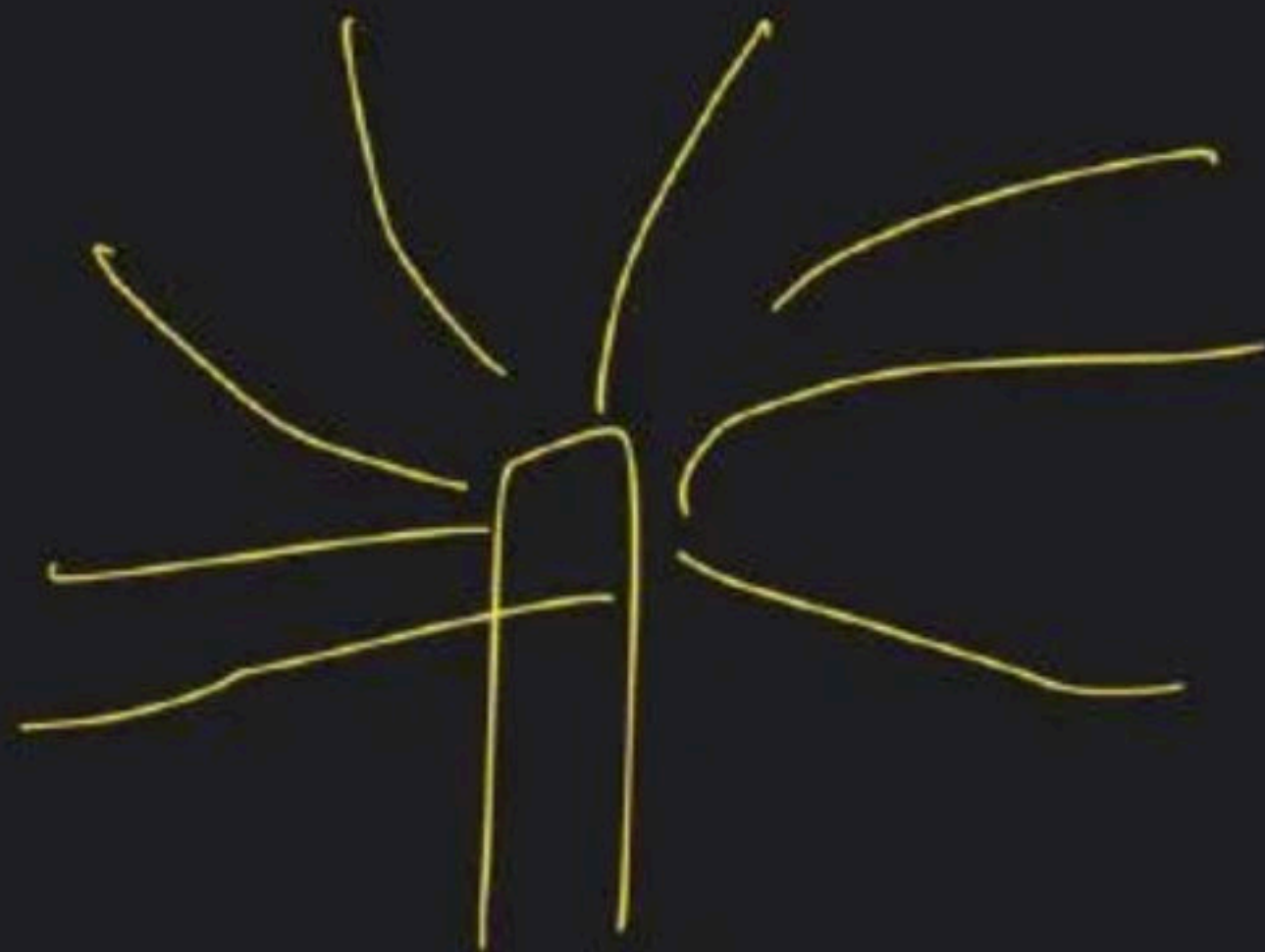
Special class



Trees Class - 1

Special class

→ Trees → D.S → non-linear





$n^{0,1,2}$
Binary tree

Level 0

Root

ancestor

node
↳ data
↳ next

Level 1
child of 10

sibling

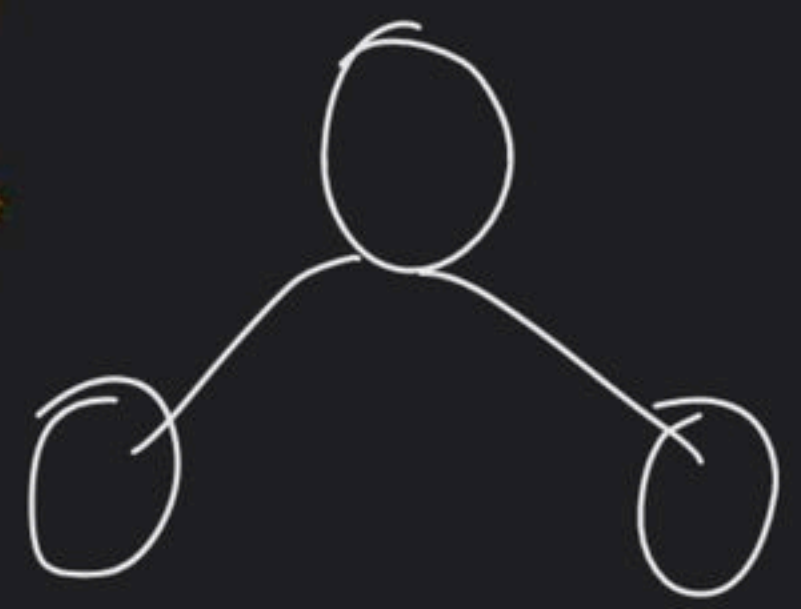
child of 10

Level 2

leaf node

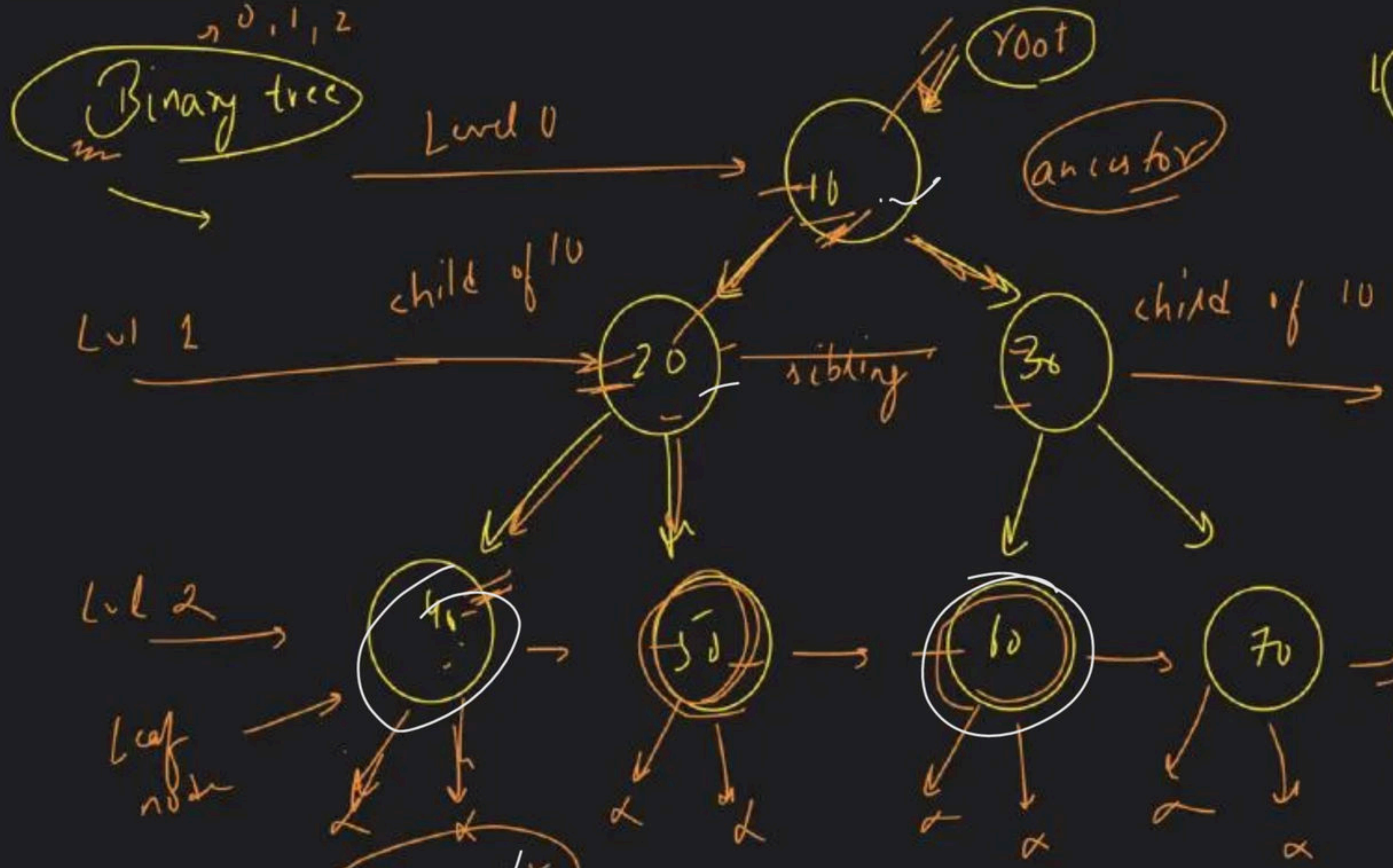
successor

descendants



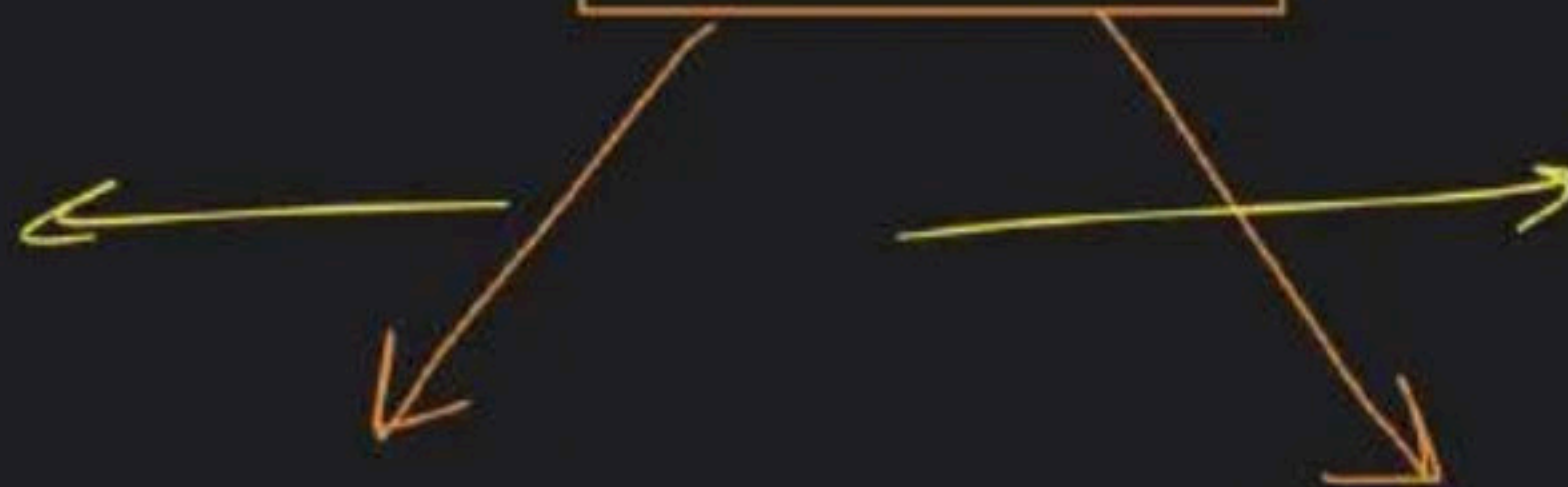
Binary Tree

at most 2 child

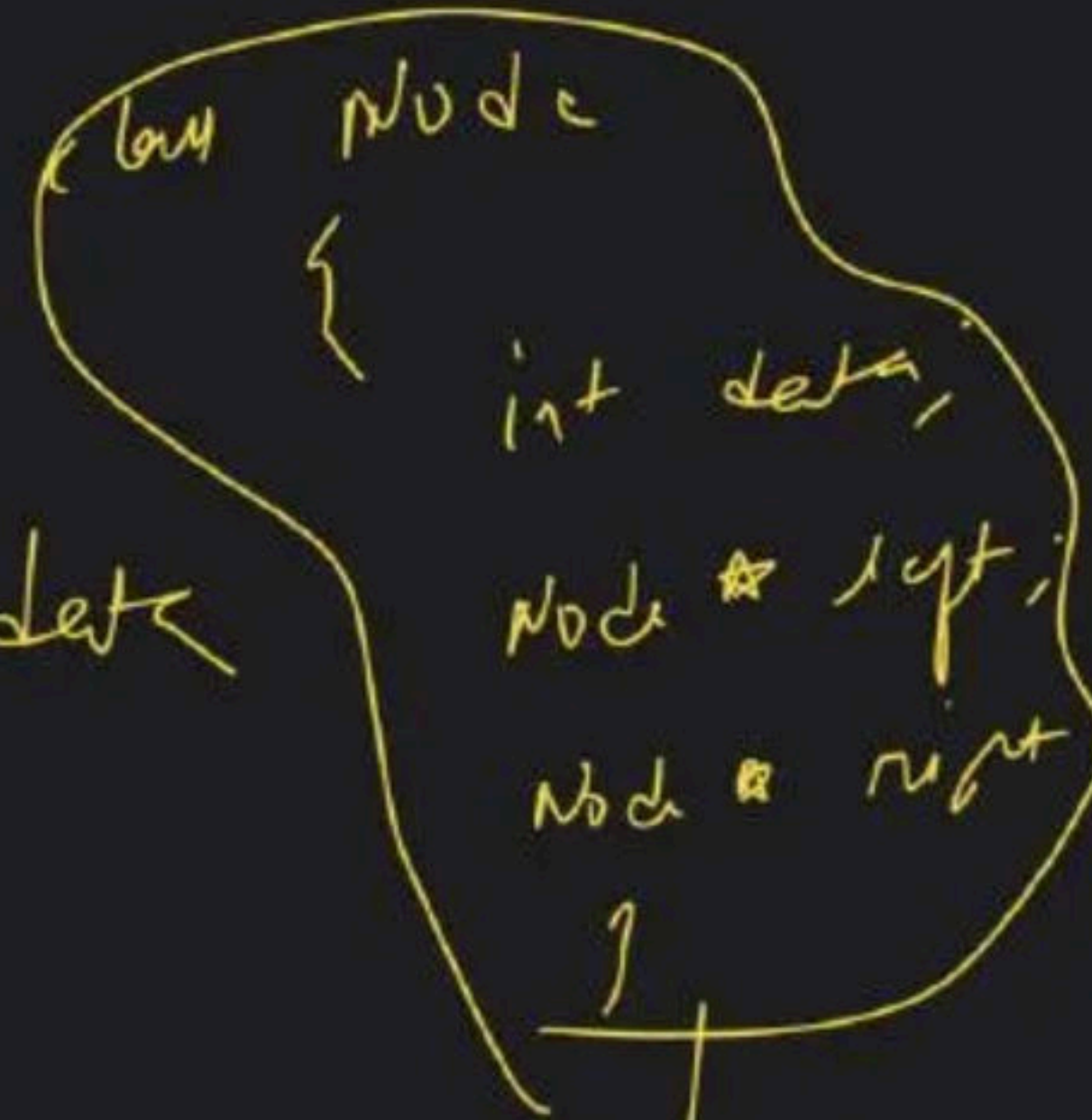


Node

Left child

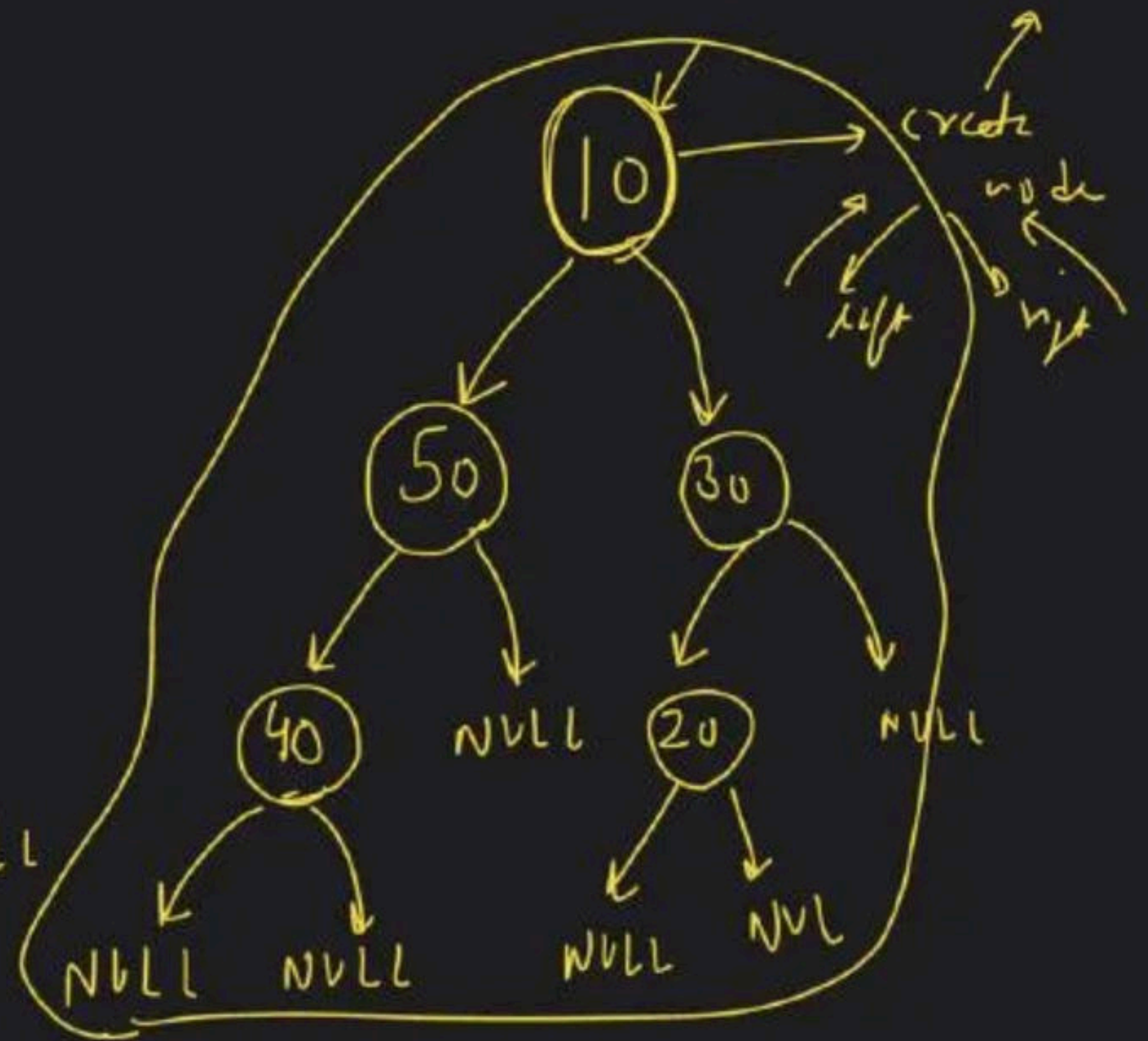
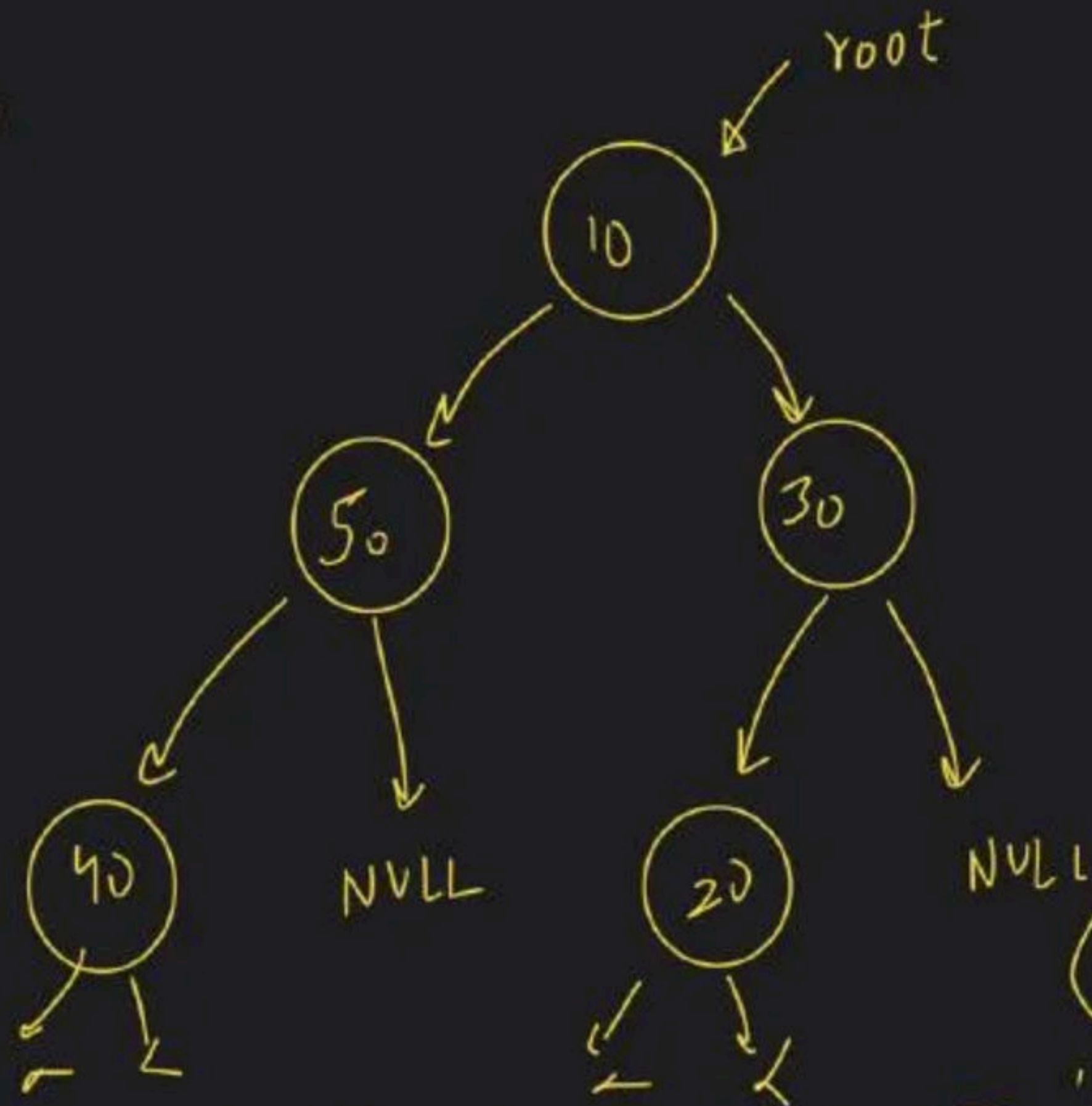


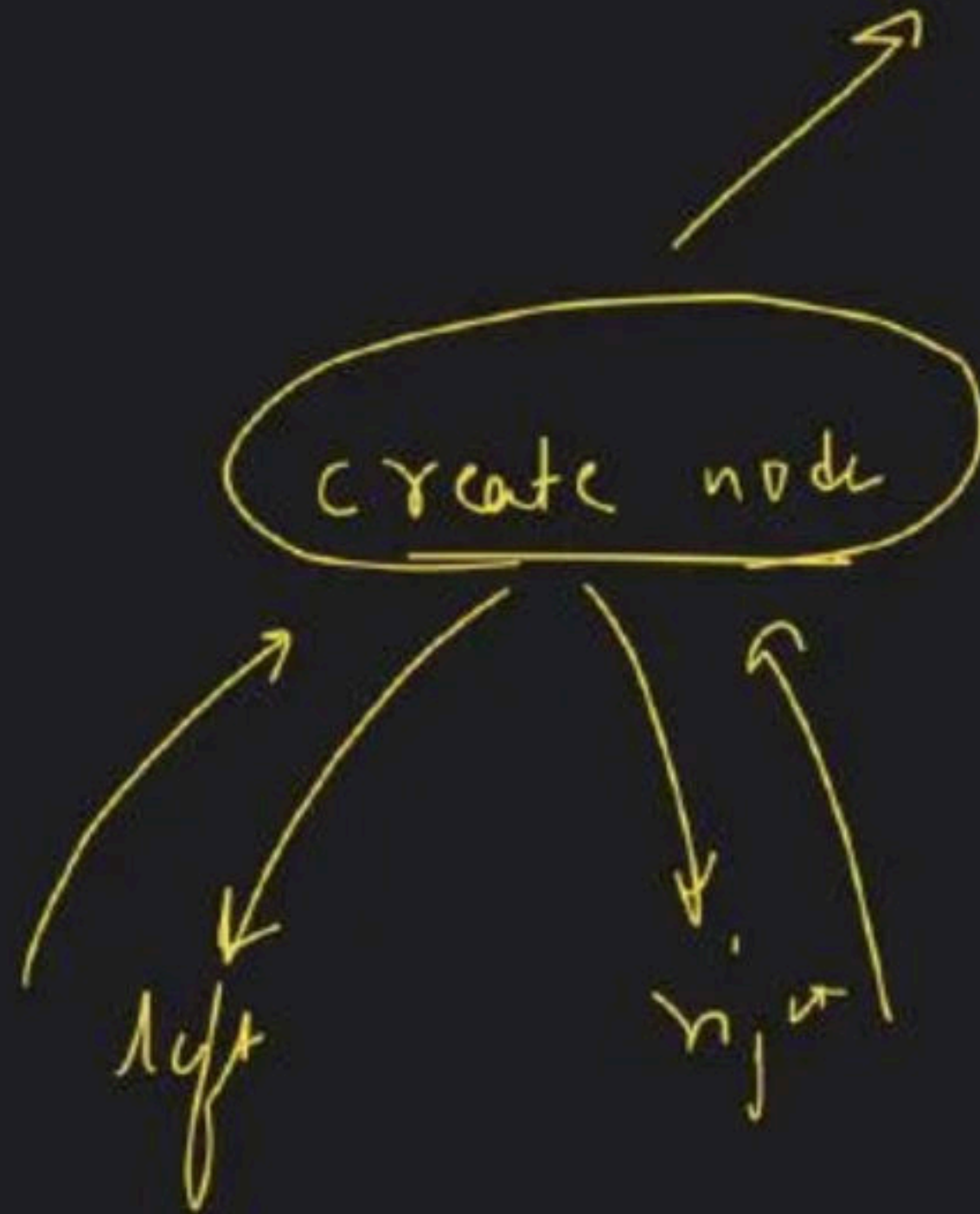
Right child



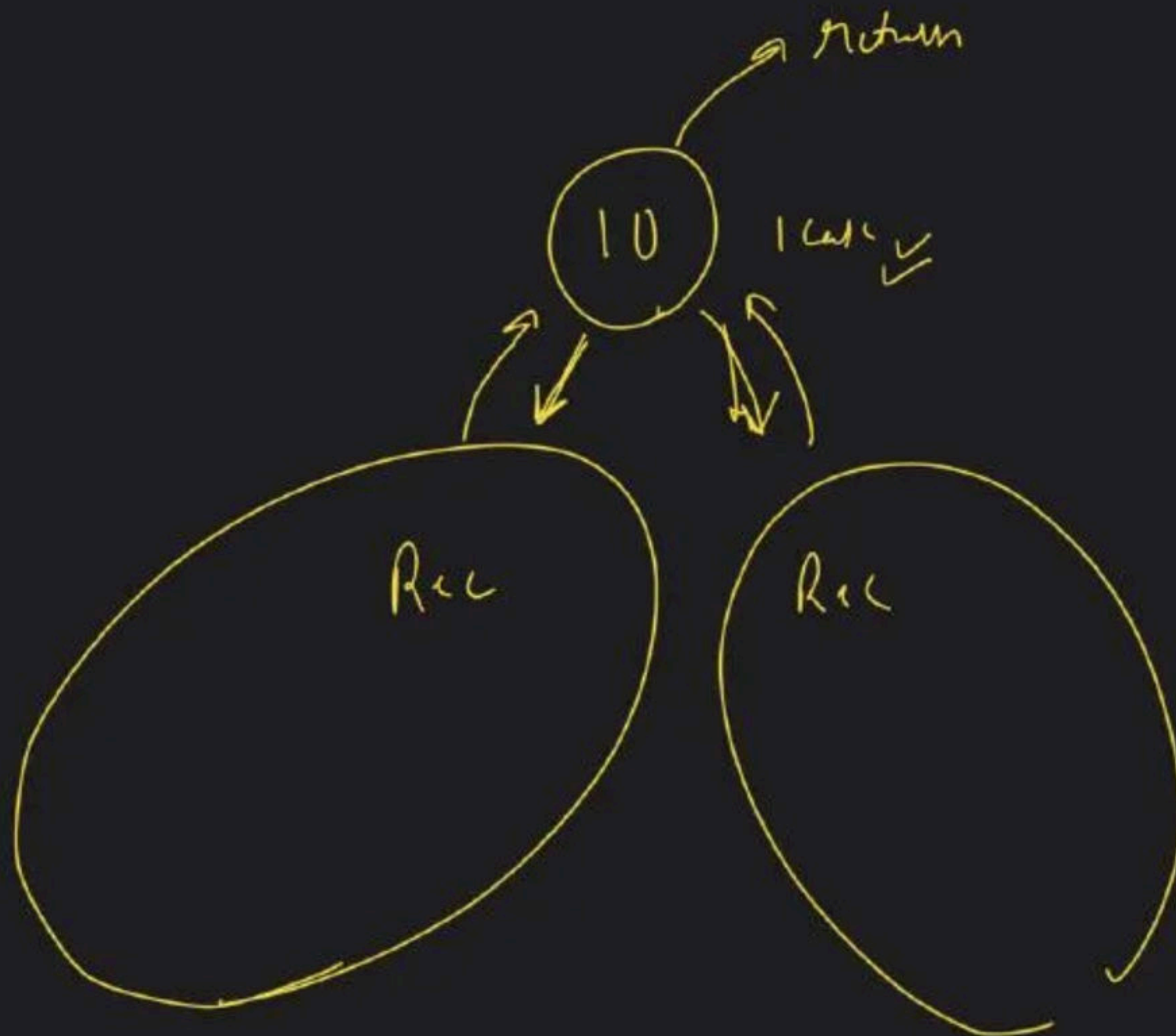
DLL

→ Tree





10



2 min
Break

Traversal

99%

node
count

left subtree

right subtree

N L R

preorder

inorder

postorder

L N R

L R N

level order traversal

Morris traversal

Assignment =

Boat

99%

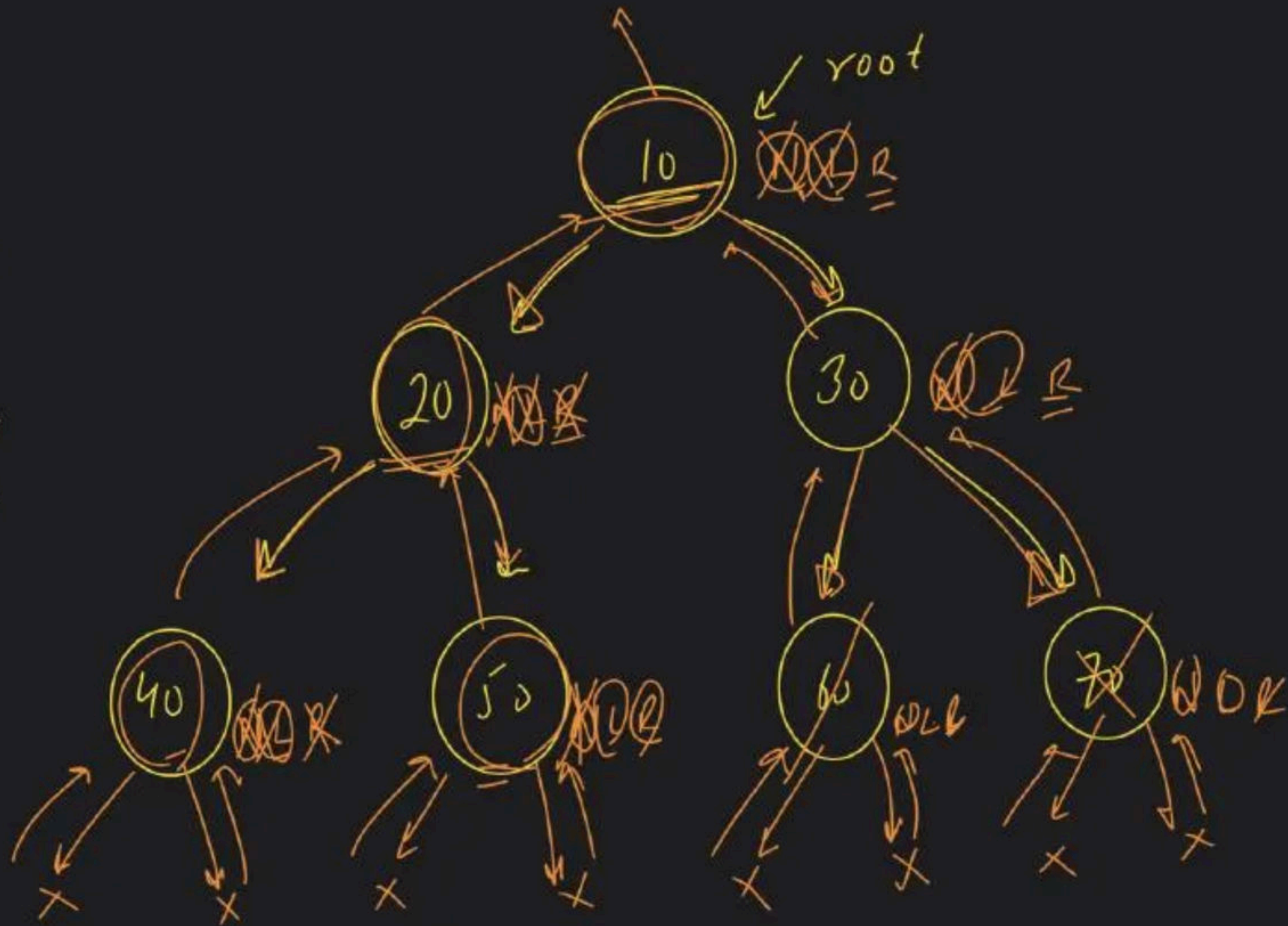
Queue → inhi
se ho

0/1
 ↓
 10-
 20/
 40-
 50-
 30-
 60-
 70/

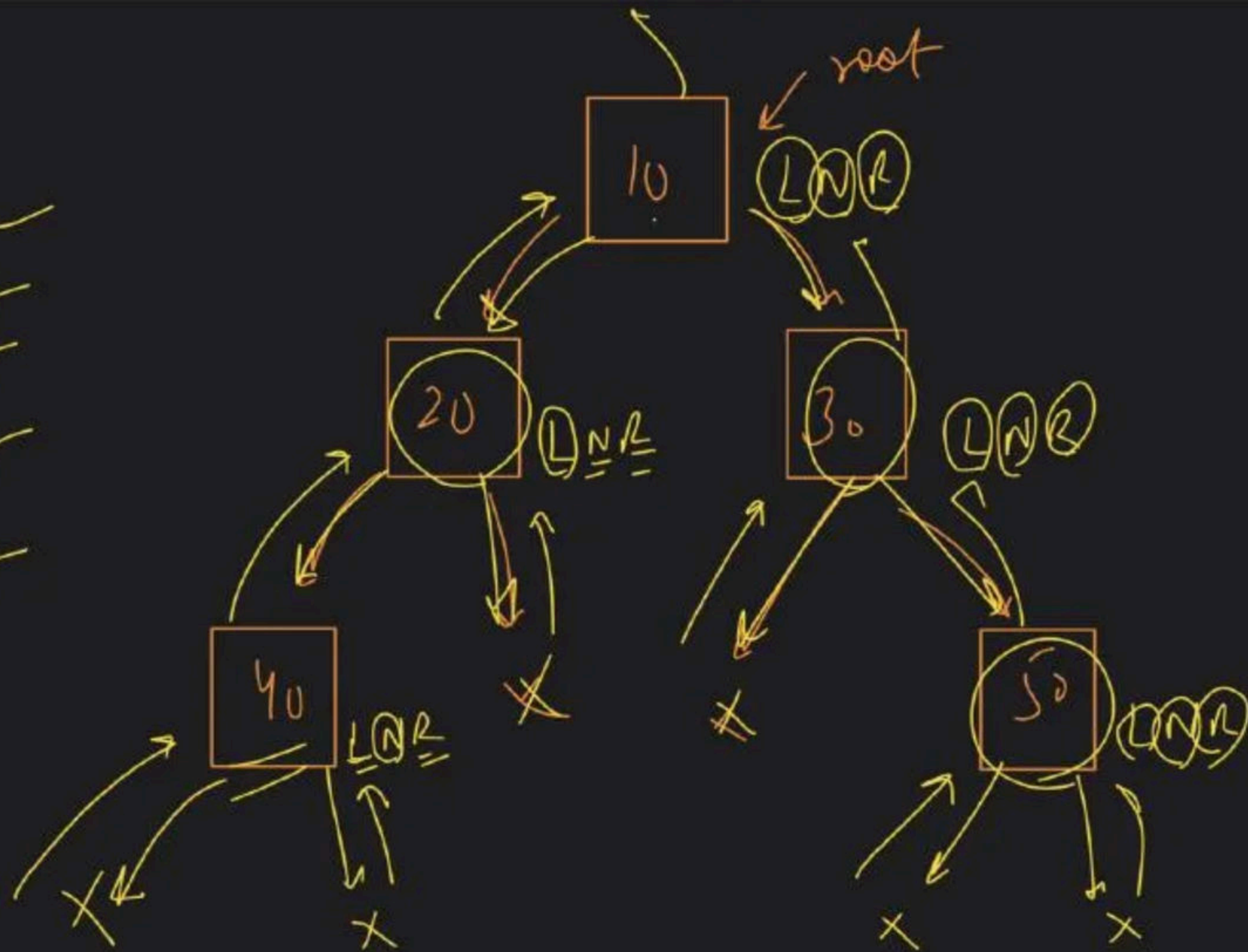
pycodeLL

NLR

NLR



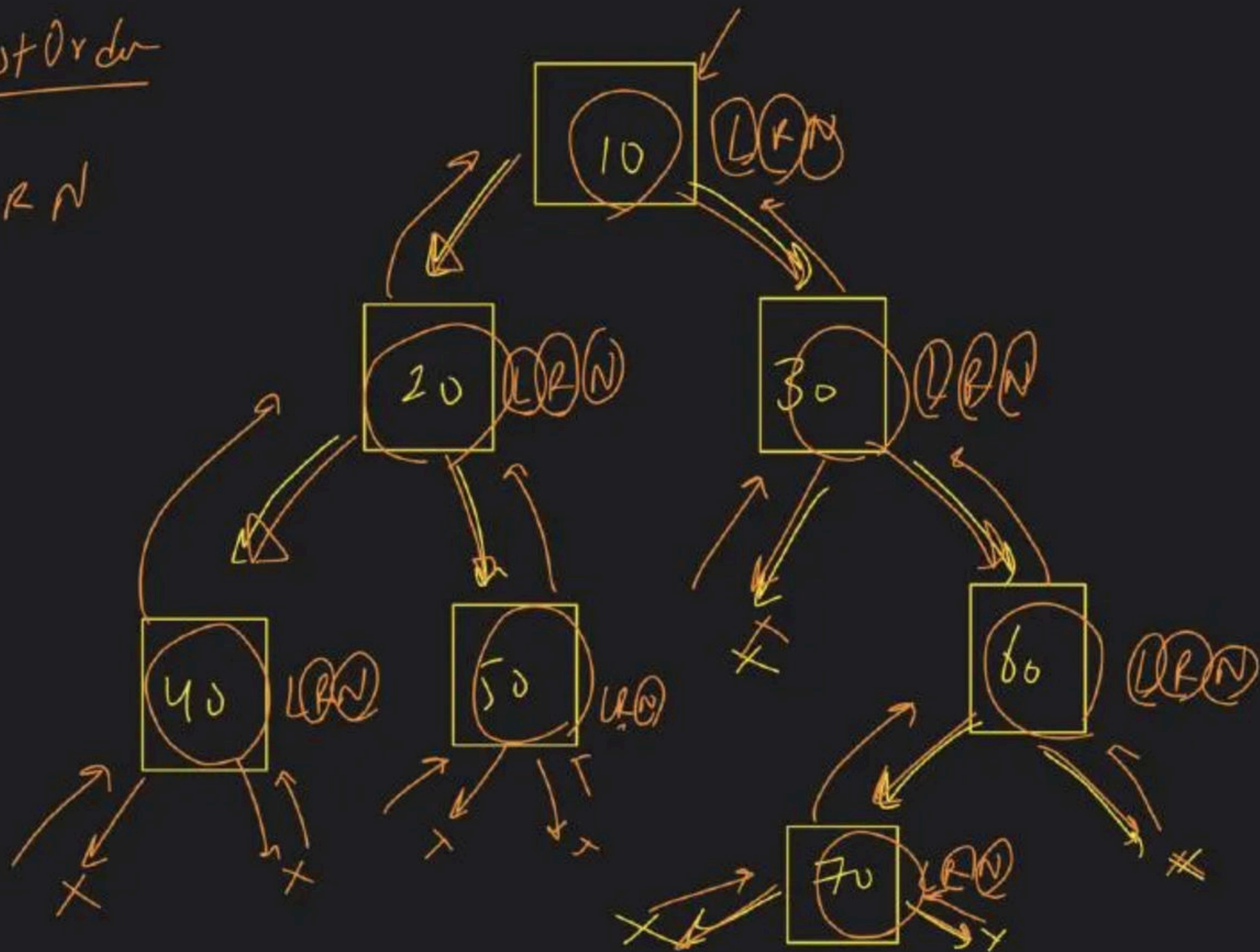
40 —
20 —
10 —
30 —
50 —



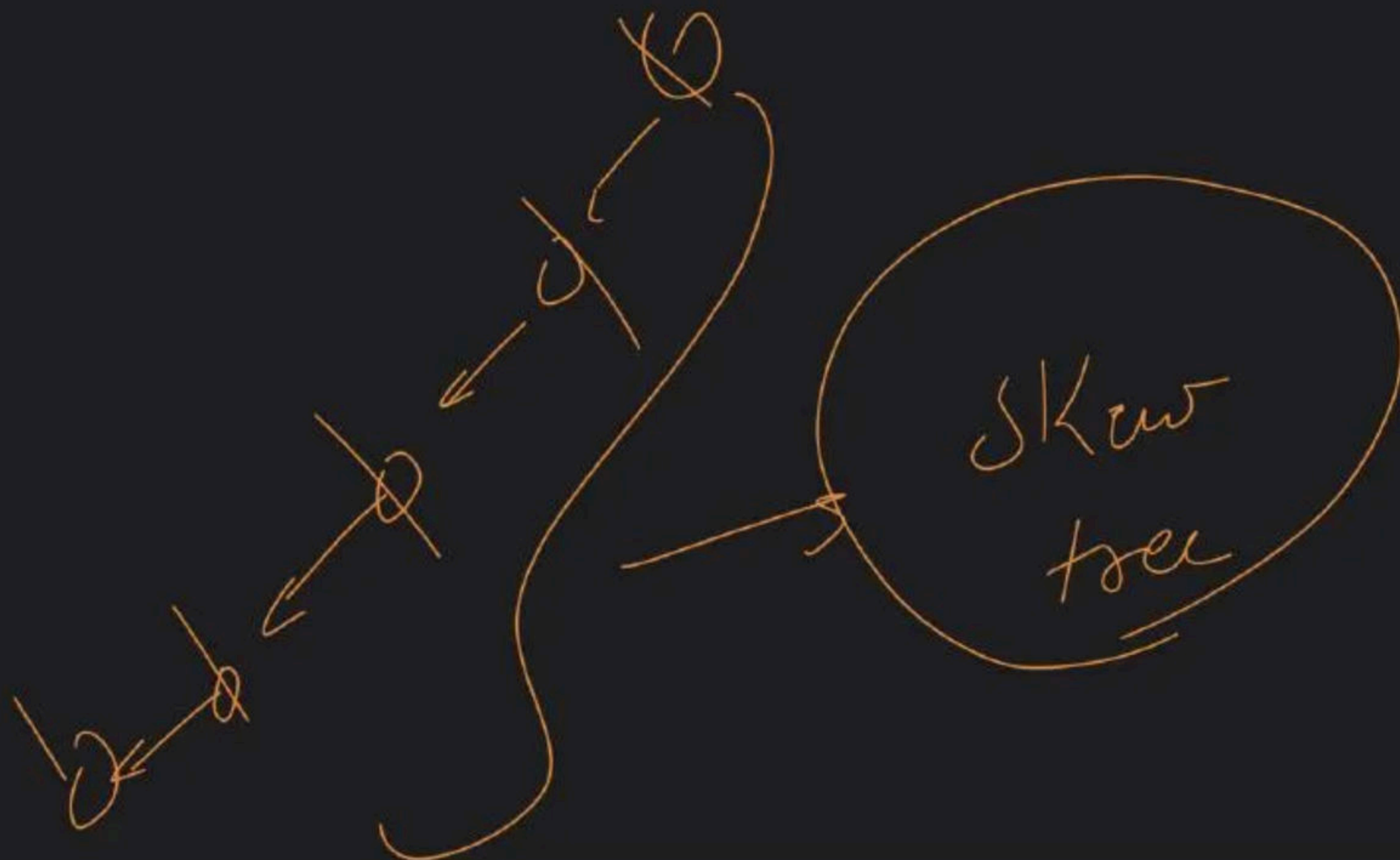
inorder
↓
LNR

PostOrder

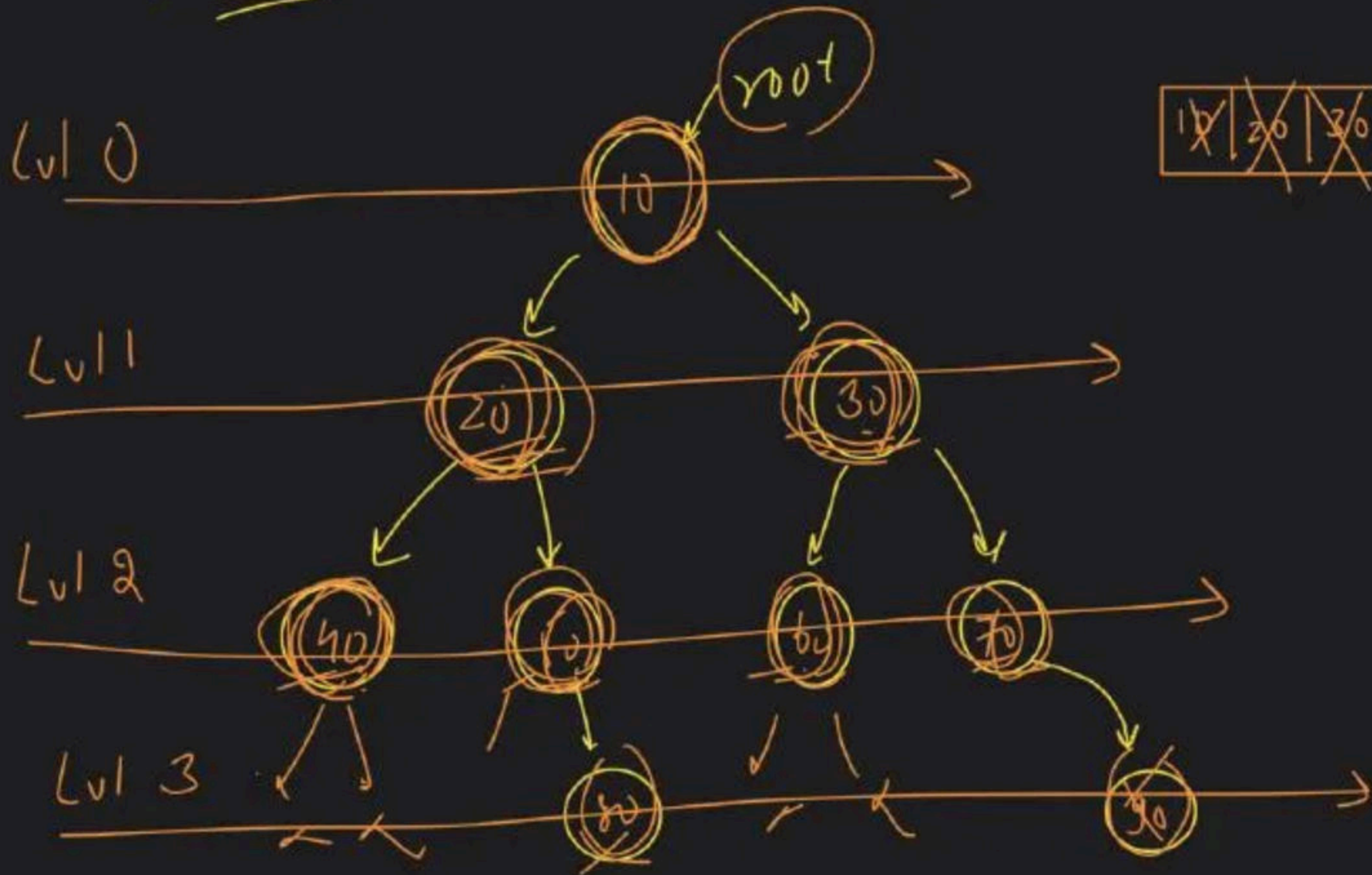
L R N



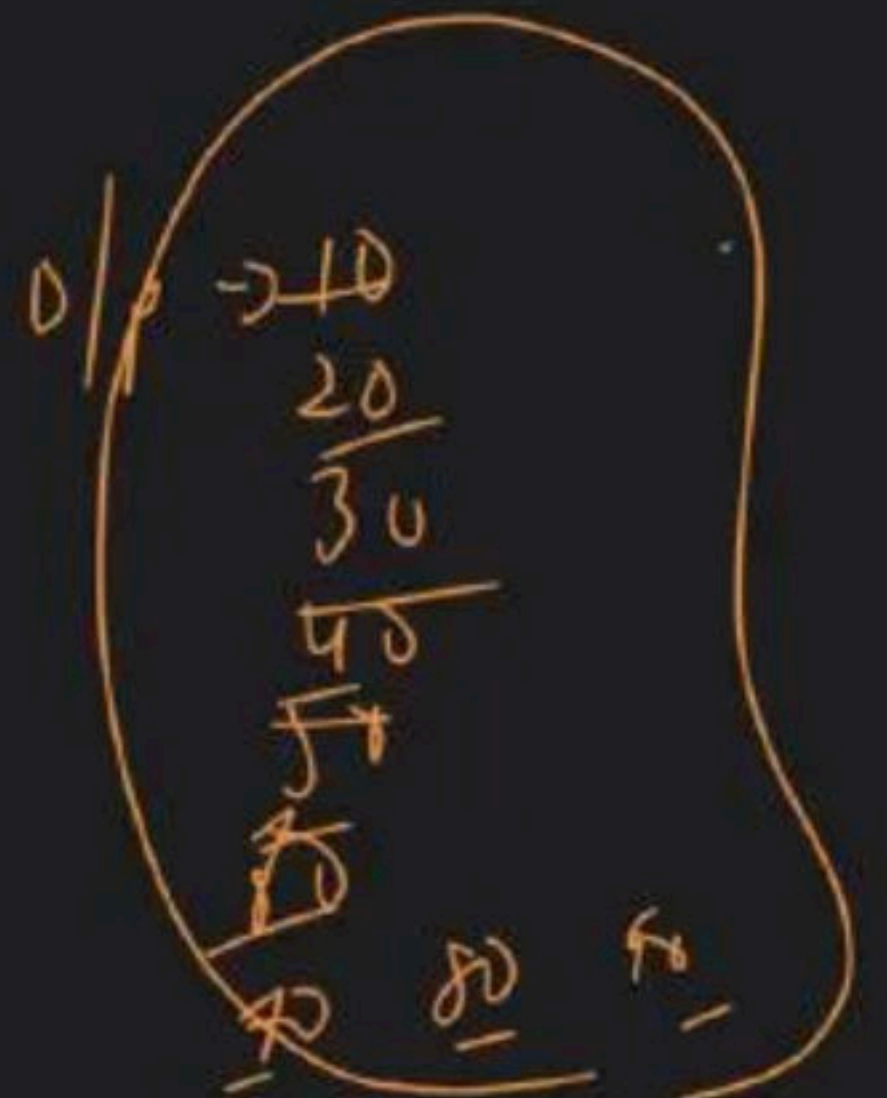
40
50
20
70
60
30
10



→ Level Order Traversal :-



10	20	30	40	50	60	70	80	90
---------------	---------------	---------------	---------------	---------------	---------------	---------------	---------------	---------------





① $\text{pop} \rightarrow F \Rightarrow$ ~~80~~ ~~90~~

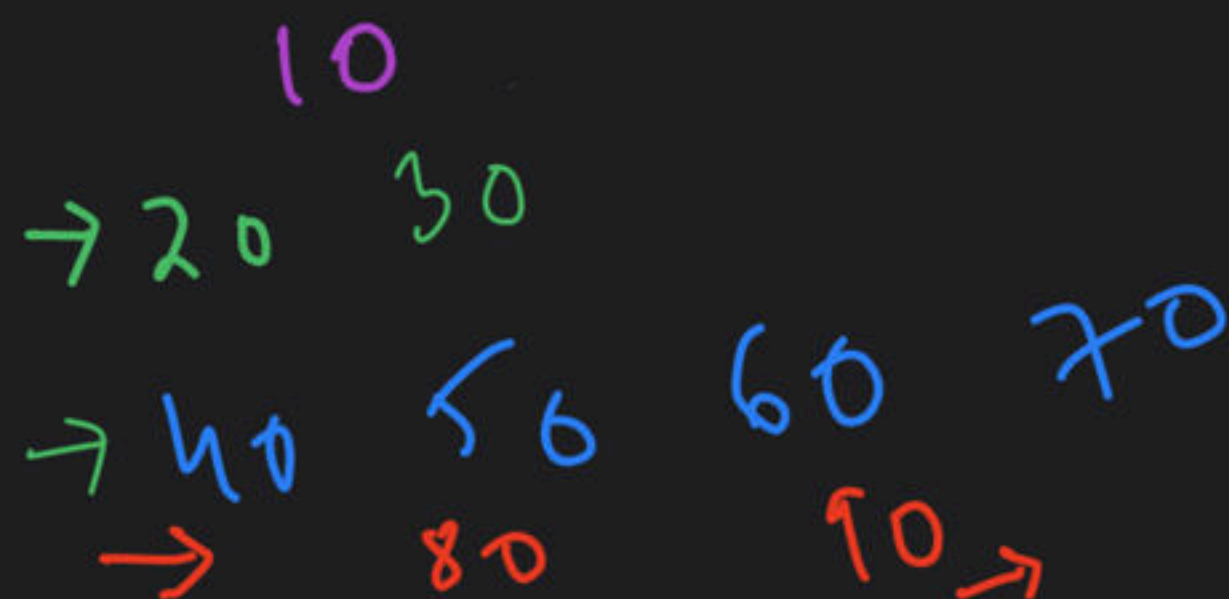
(2) Print 10

③

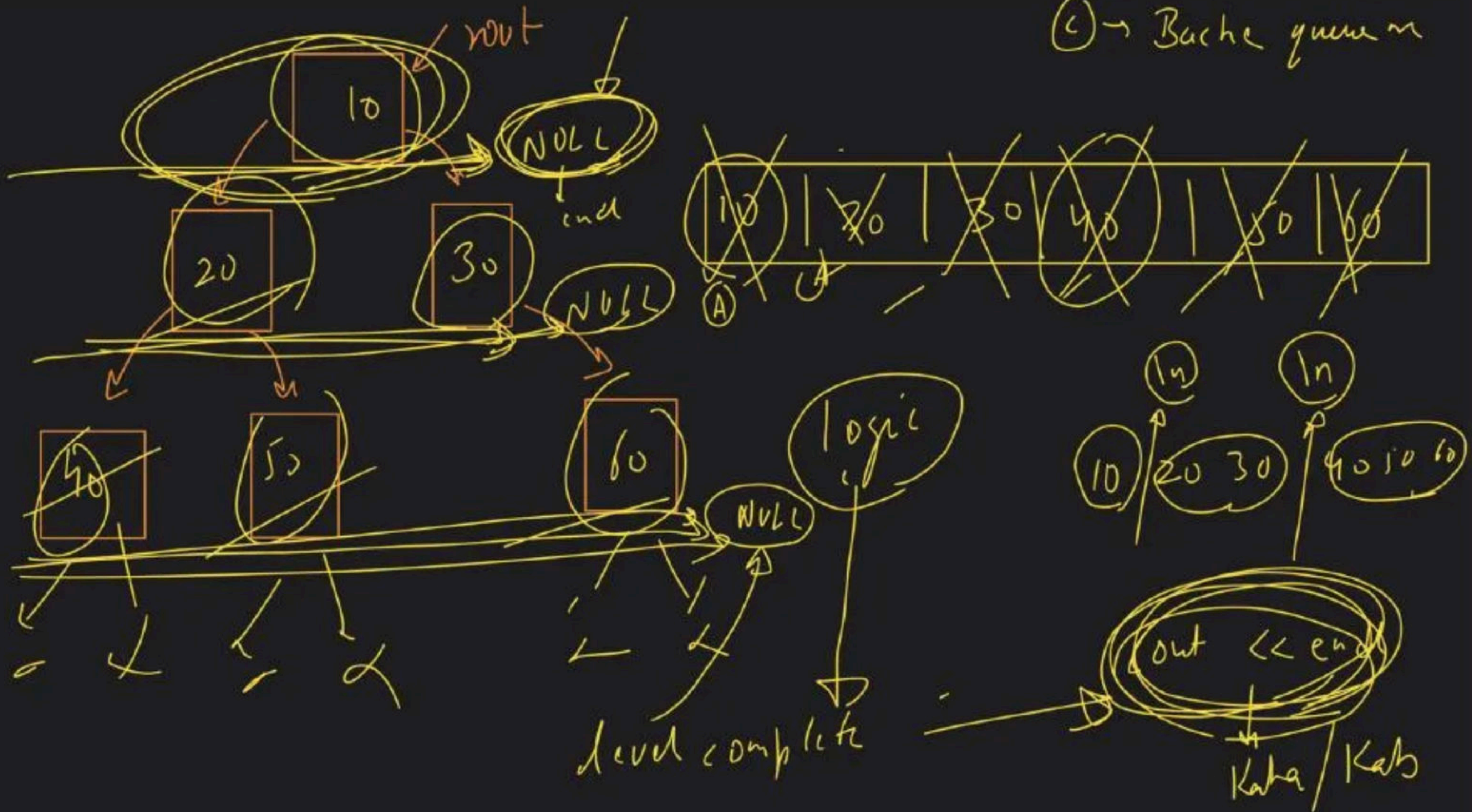
```
if (F → left)
```

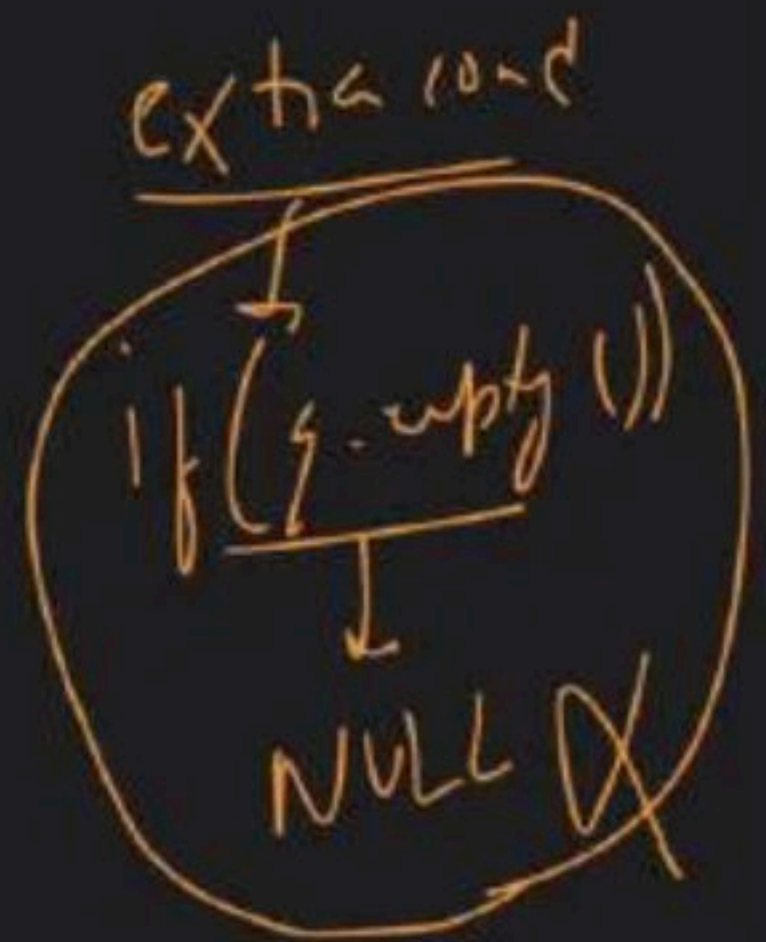
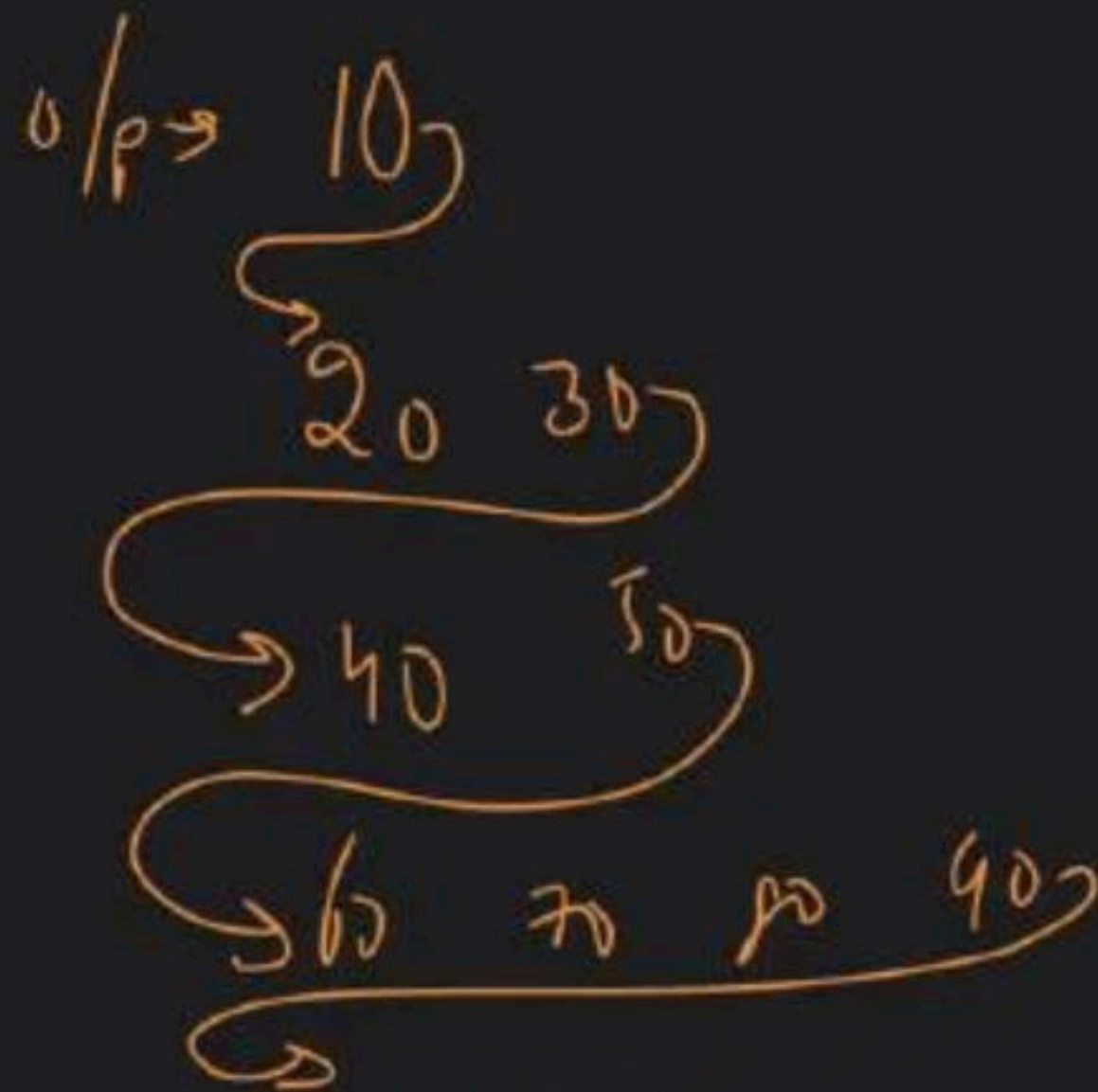
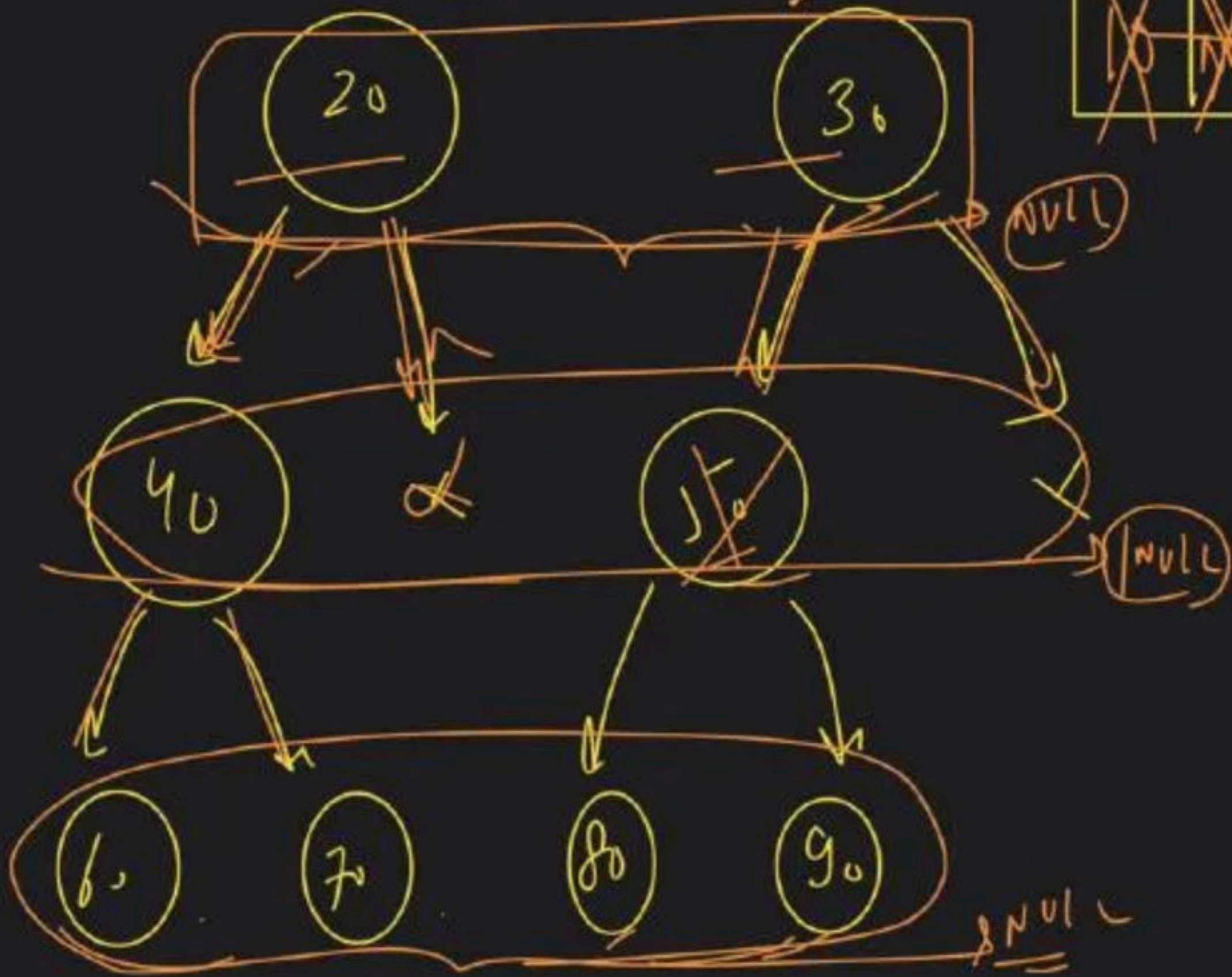
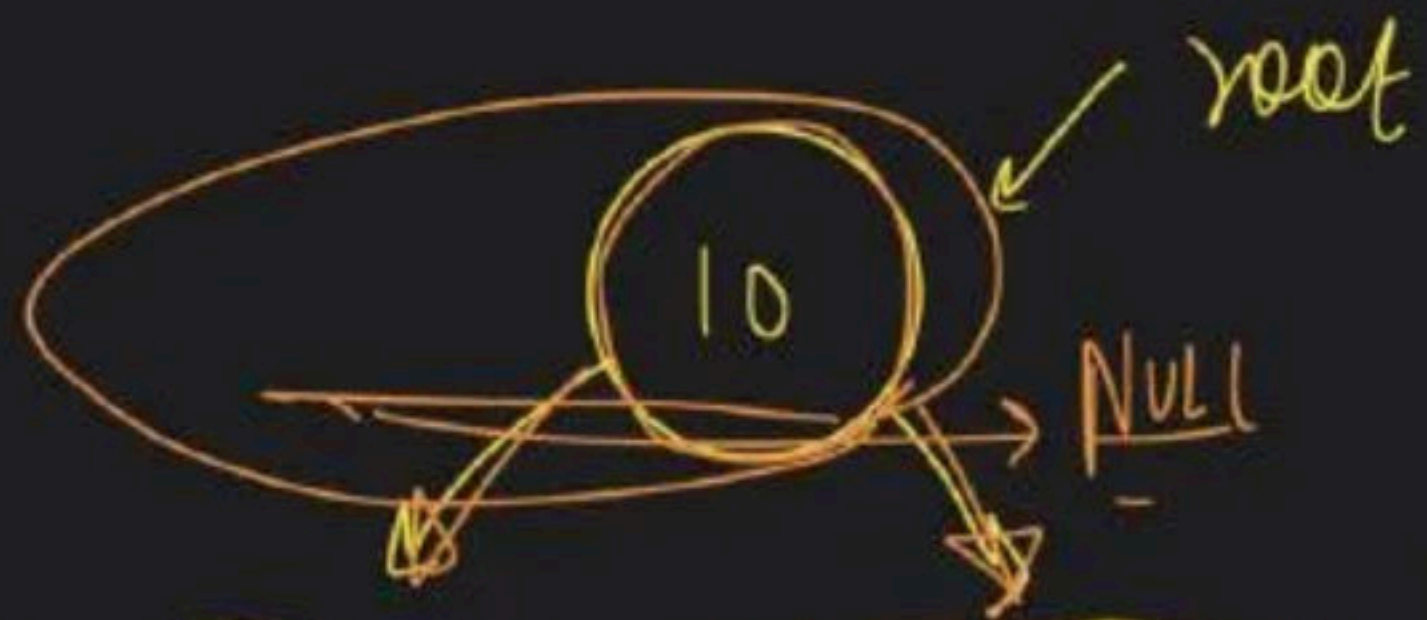
Pushy

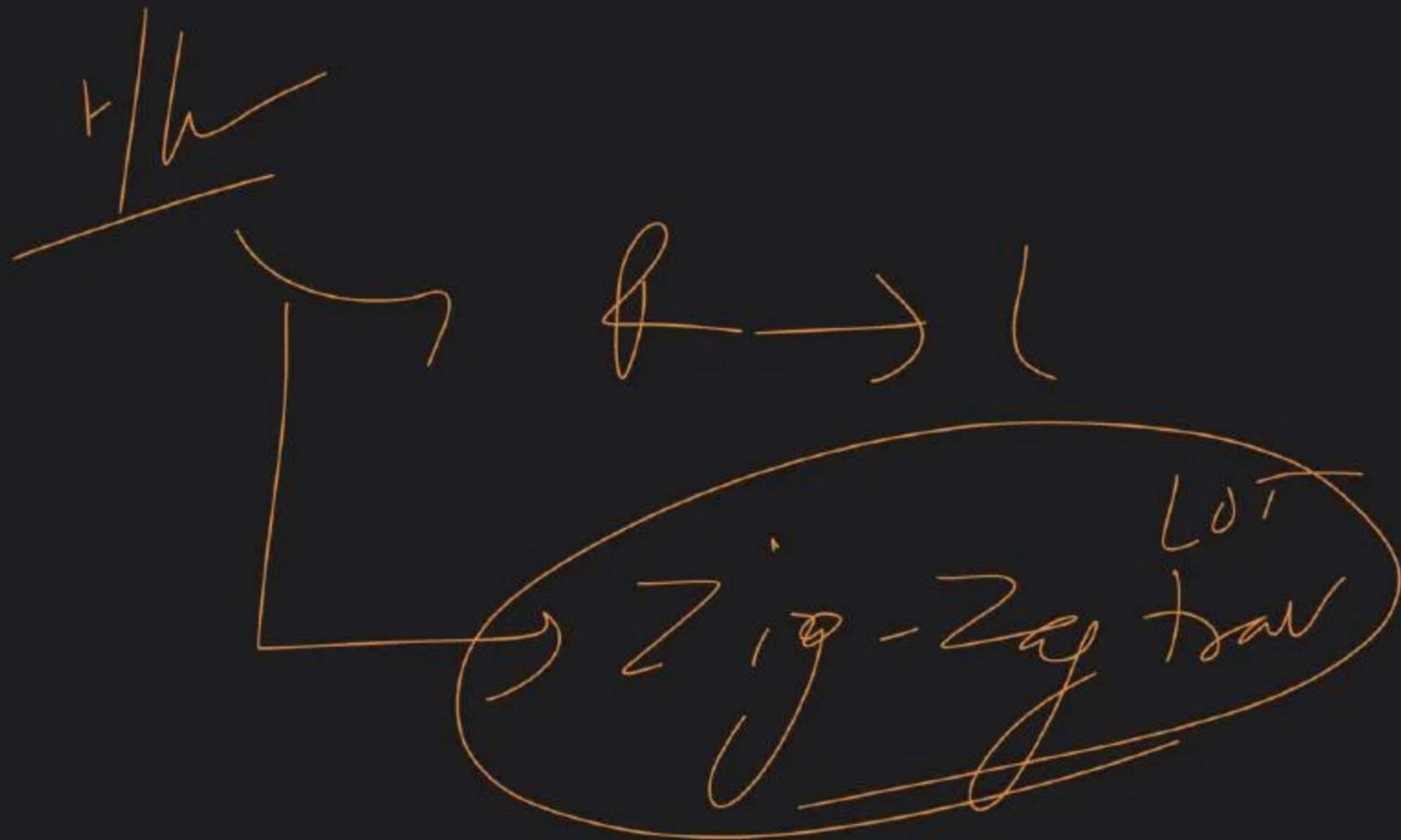
~~if (F $\rightarrow$ Right) $\rightarrow$ Push~~



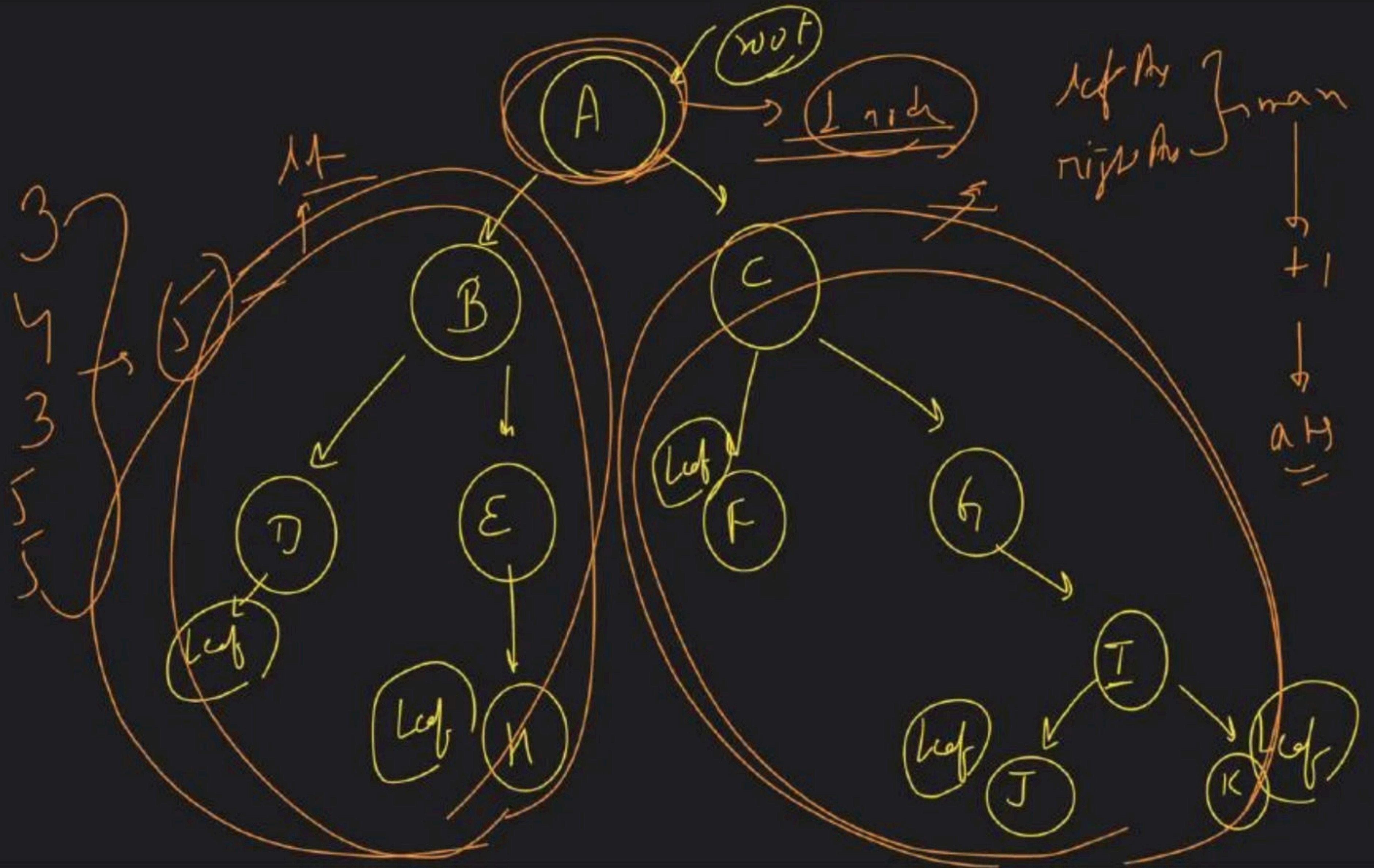
(c) → Bachhe ghumna







→ Height of Tree





→ Diameter of a tree



A horizontal line representing a tree trunk. A double-headed arrow is drawn across the line, indicating a measurement. A vertical line segment with an arrowhead at the bottom points downwards from the center of the trunk.

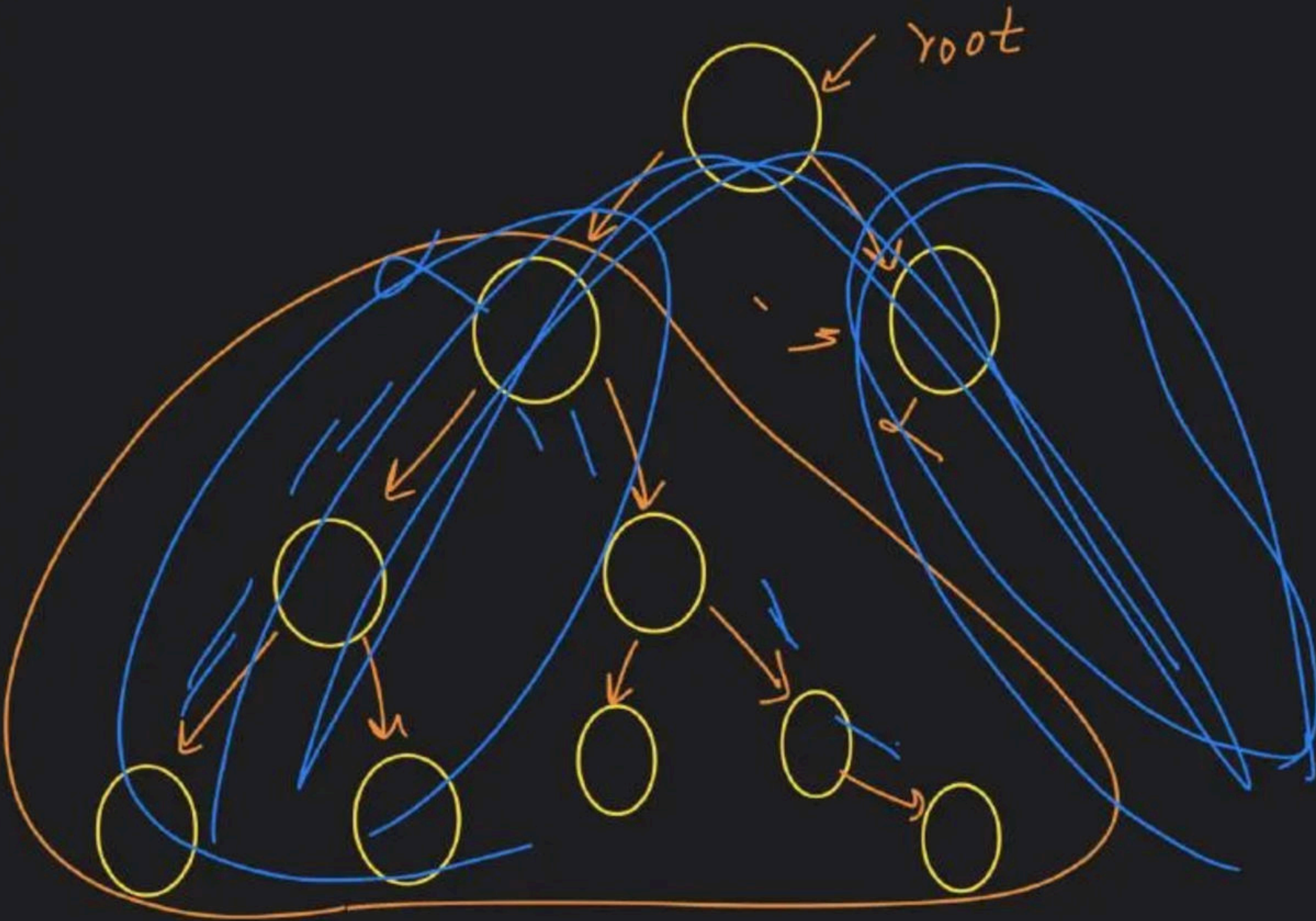
feet

Diameter



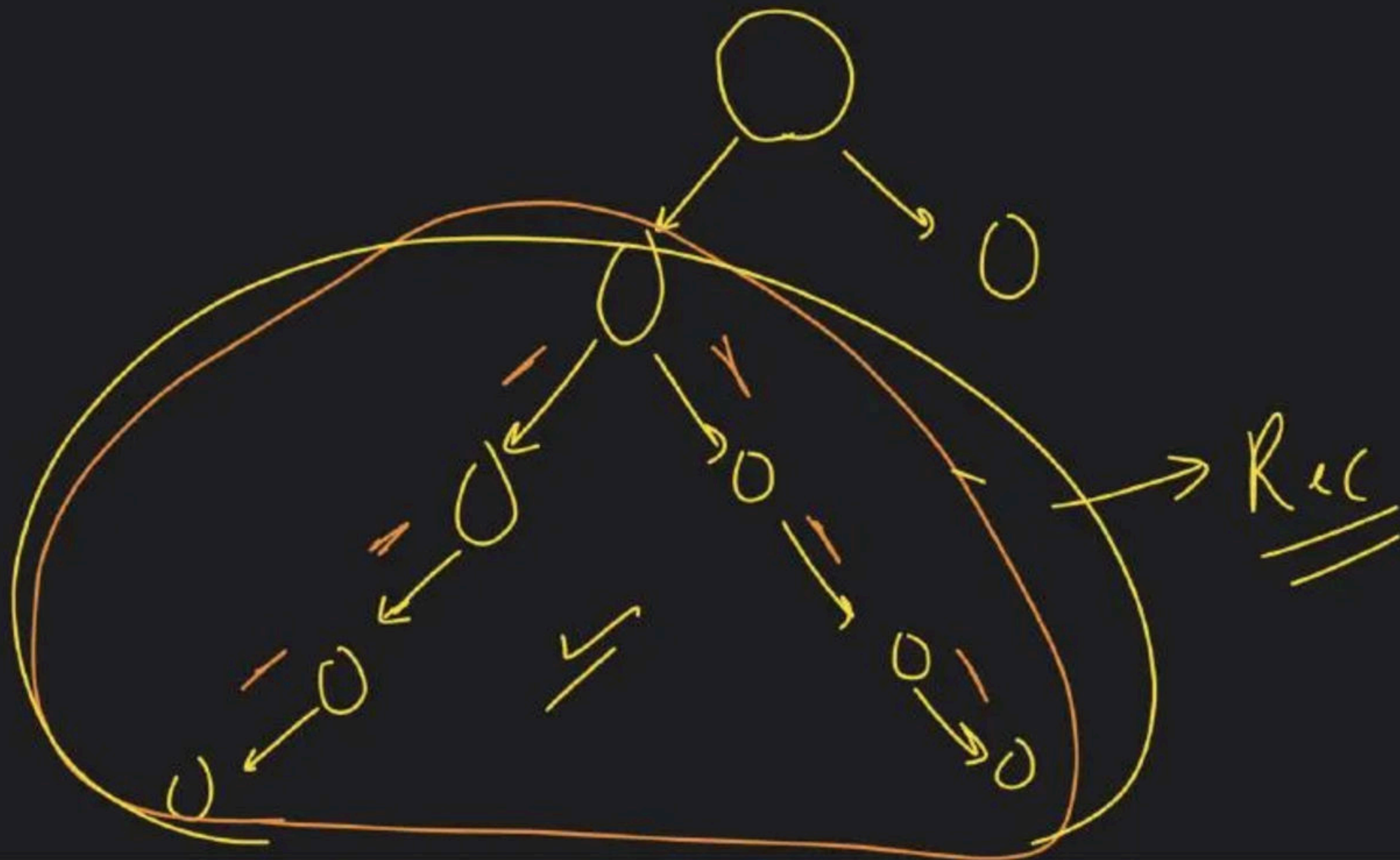
A circle with a vertical line passing through its center. The line is labeled 'H/w'.

normal



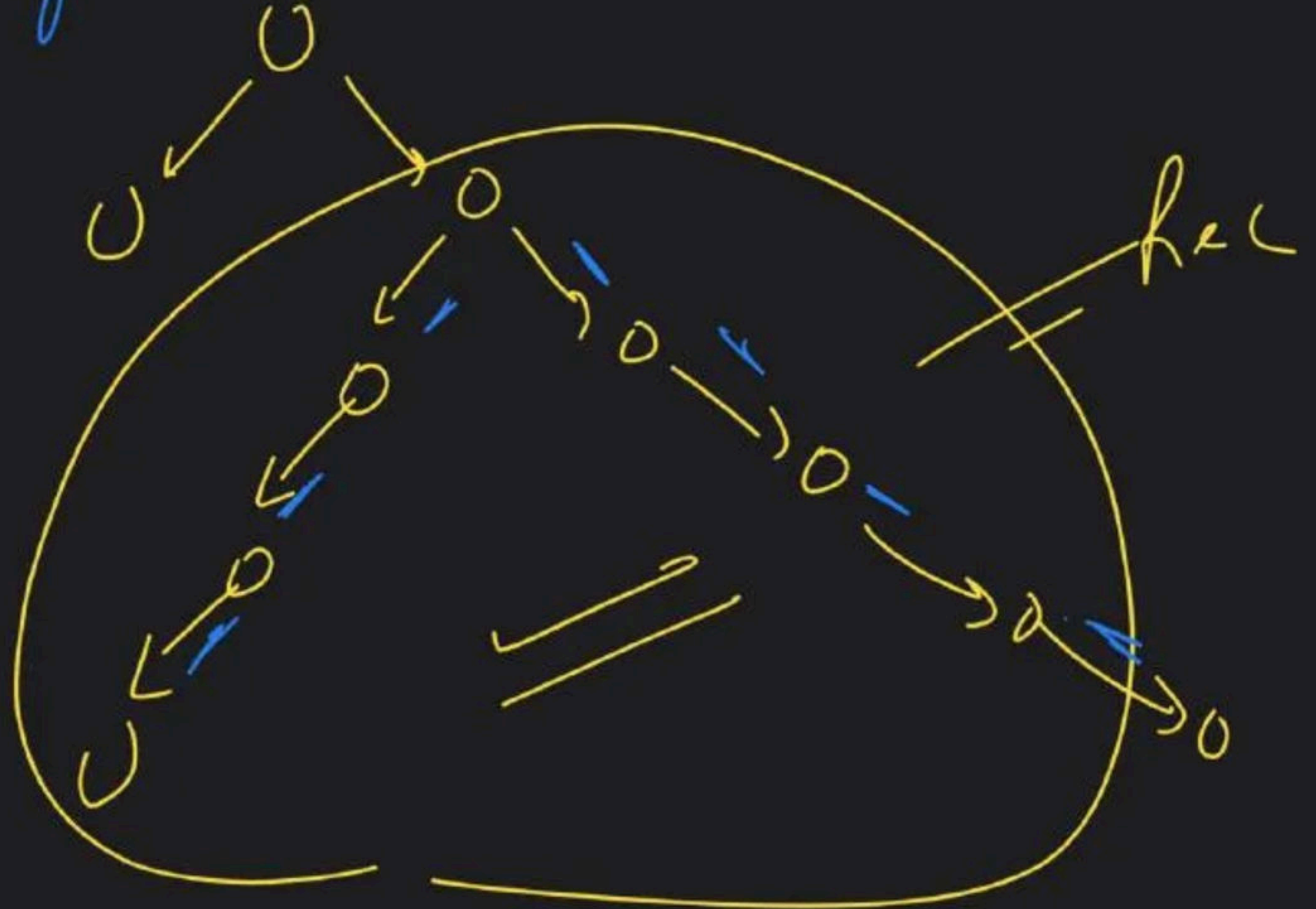
o/p $\rightarrow$ left subtree
o/p $\rightarrow$ right subtree
o/p $\rightarrow$ combine

op¹ → left subtree → ay

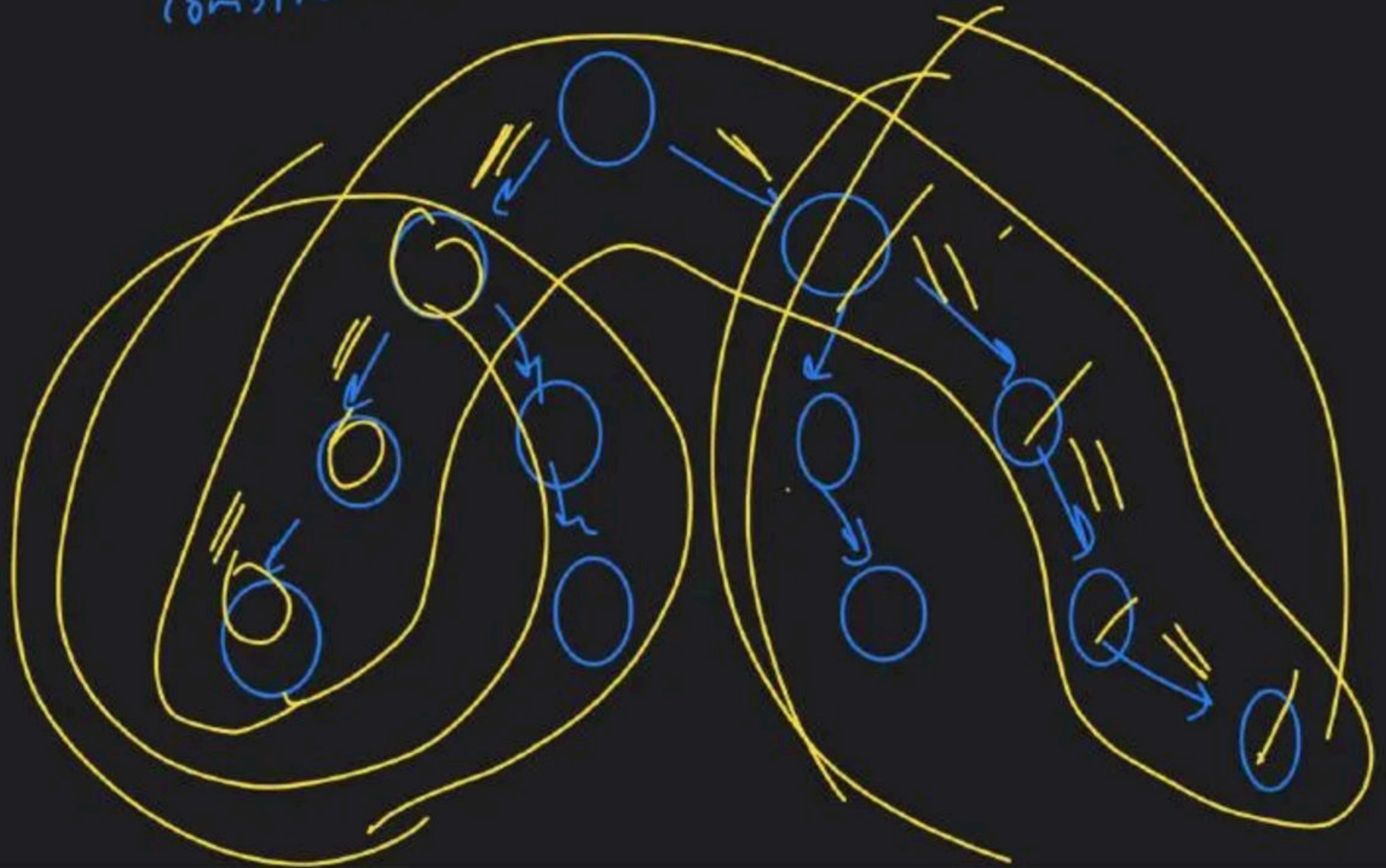


o/p 2 $\rightarrow$

right subtree $\rightarrow$ any
()



op3 $\rightarrow$ left Δ Right $\rightarrow$ ans $\rightarrow$ left + right
combination



⑥

Diameter pair

$op1 = 4$
 $op2 = 3$
 $op3 = 6$

root

$h \rightarrow$ height

$O(n^2)$

$op1 = 1$
 $op2 = 1$
 $op3 = 4$

$op1 = 0$
 $op2 = 1$
 $op3 = 2$

$op1 = 0$
 $op2 = 0$
 $op3 = 1$

$op1 = 0$
 $op2 = 0$
 $op3 = 1$

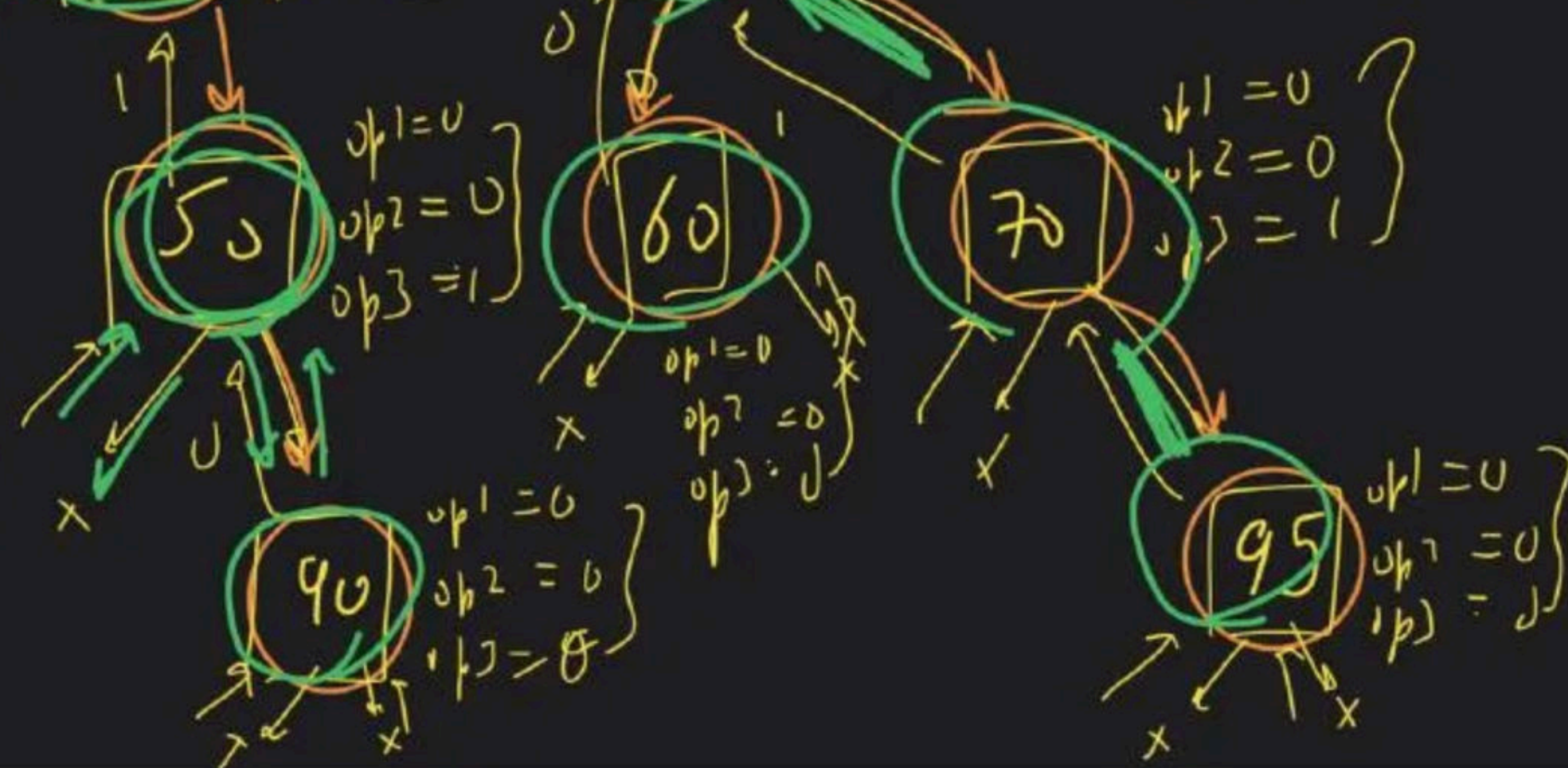
$op1 = 0$
 $op2 = 0$
 $op3 = 1$

$op1 = 0$
 $op2 = 0$
 $op3 = 0$

$op1 = 0$
 $op2 = 0$
 $op3 = 0$

$op1 = 0$
 $op2 = 0$
 $op3 = 0$

$op1 = 0$
 $op2 = 0$
 $op3 = 0$



int Left 4

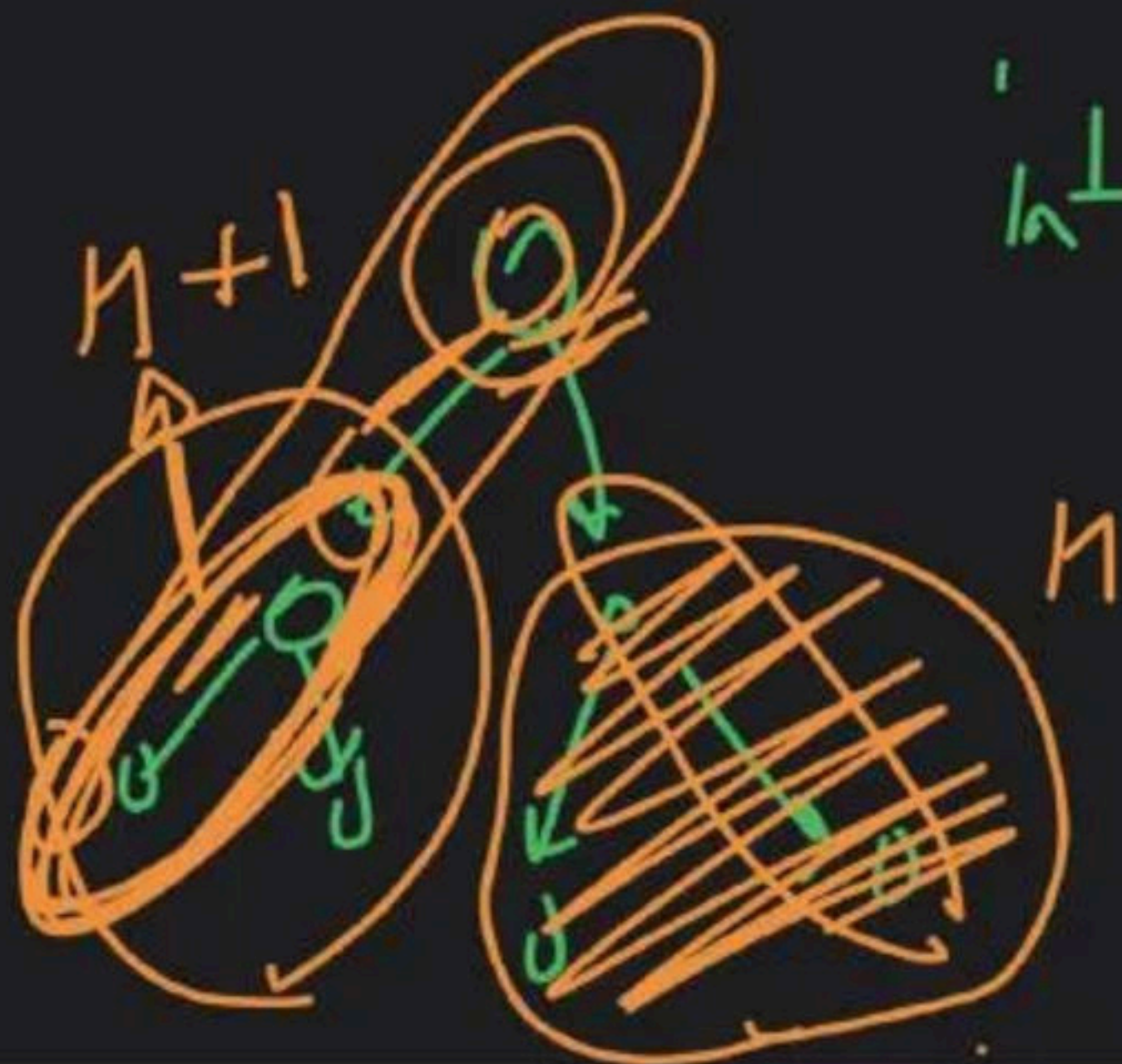
int Right 4

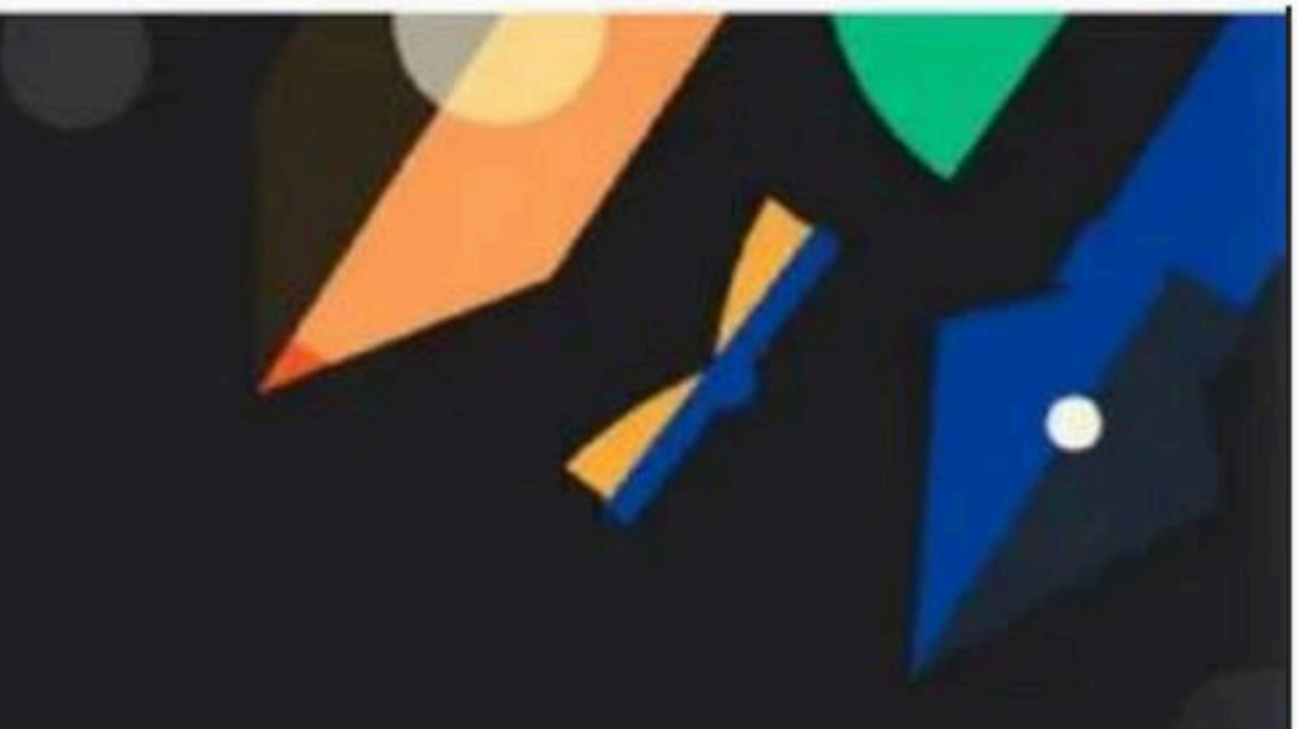
int $\rightarrow$ total $\rightarrow \max(L, R)$

?

$\uparrow$

$\begin{matrix} + & 1 \\ \hline \hline \end{matrix}$





Trees Class - 2

Special class

↳ create Tree

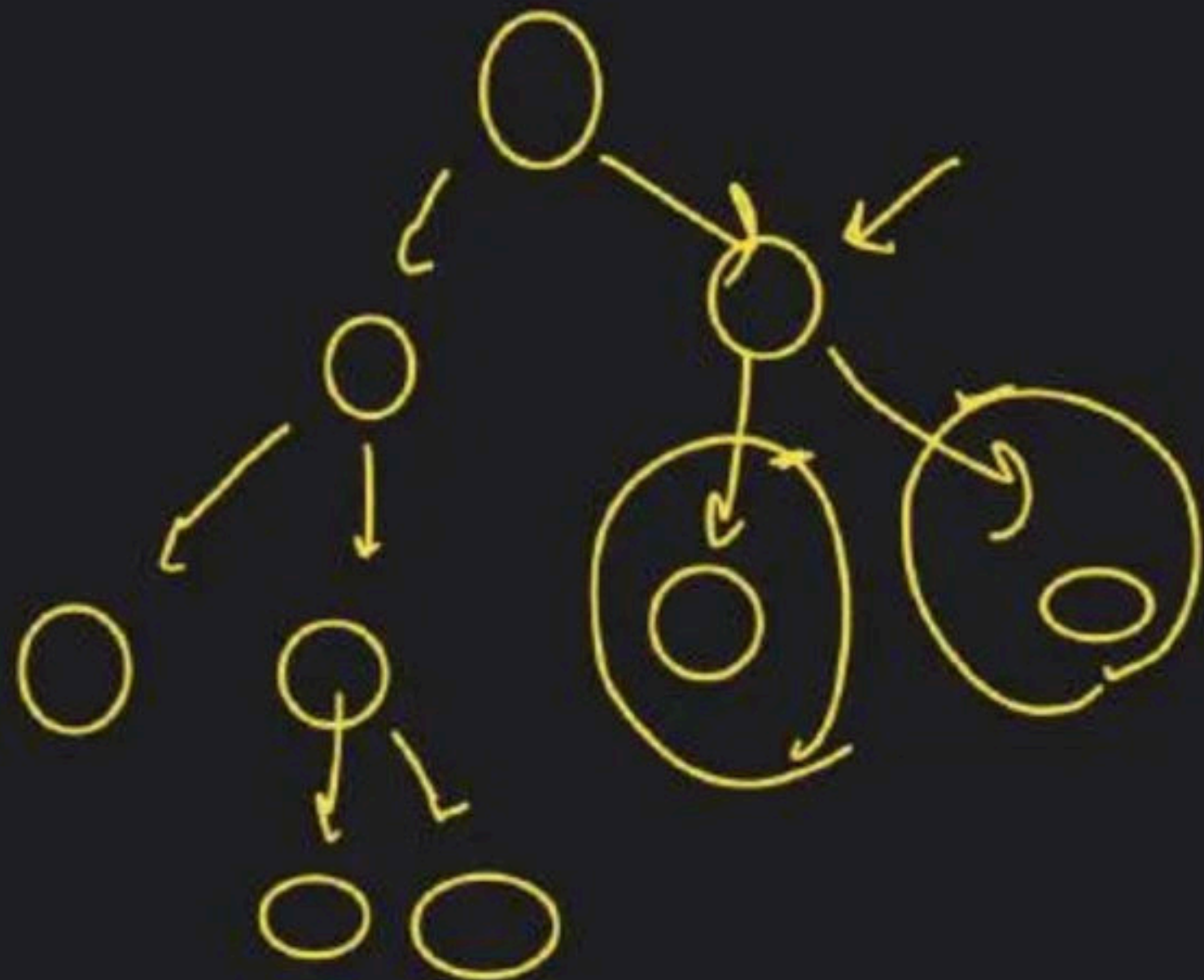
↳ traversals

↳ height / diameter ✓

→ Check whether tree is Balanced or not

Balanced → for node → abs (lh - rh) ≤ 1

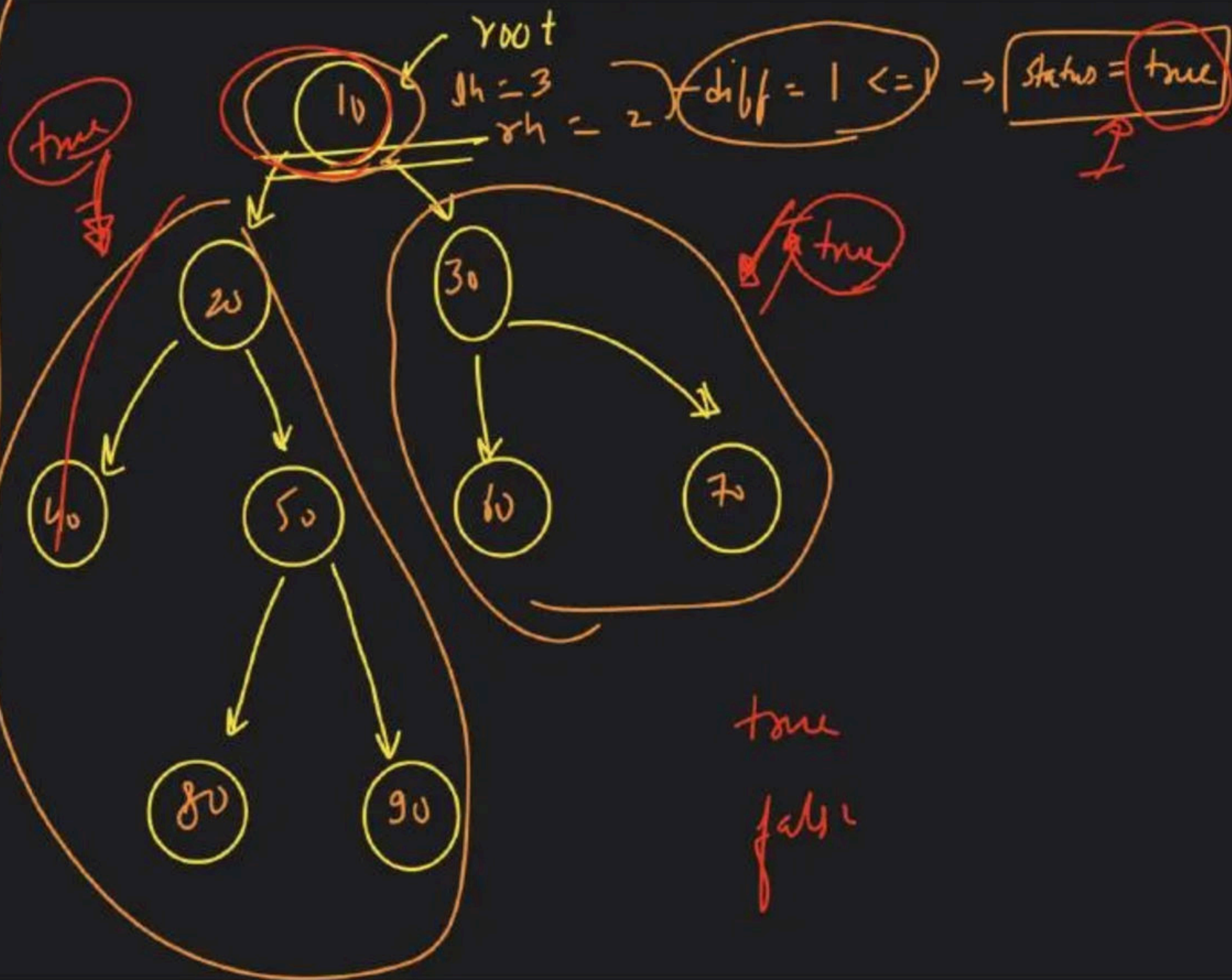
0, 1, 2

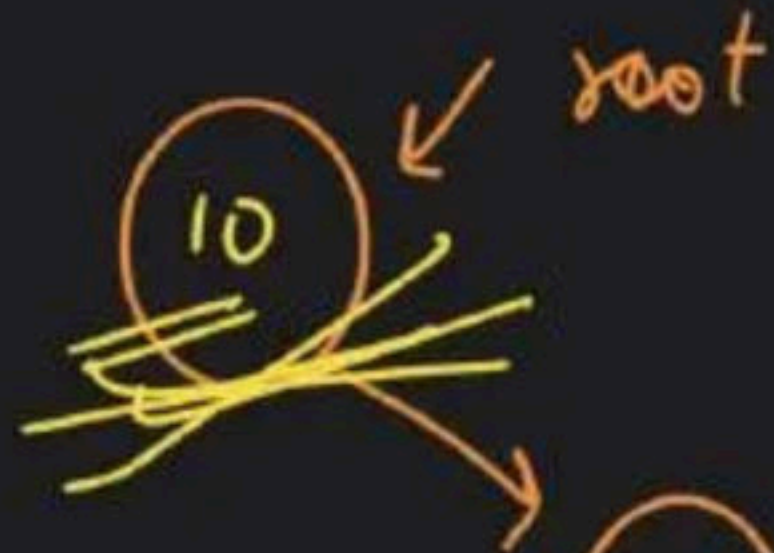


Balanced Tree

each node

also $(lh - rh) \leq 1$





$$\frac{n \times (n+1)}{2}$$

$$O(n^2)$$



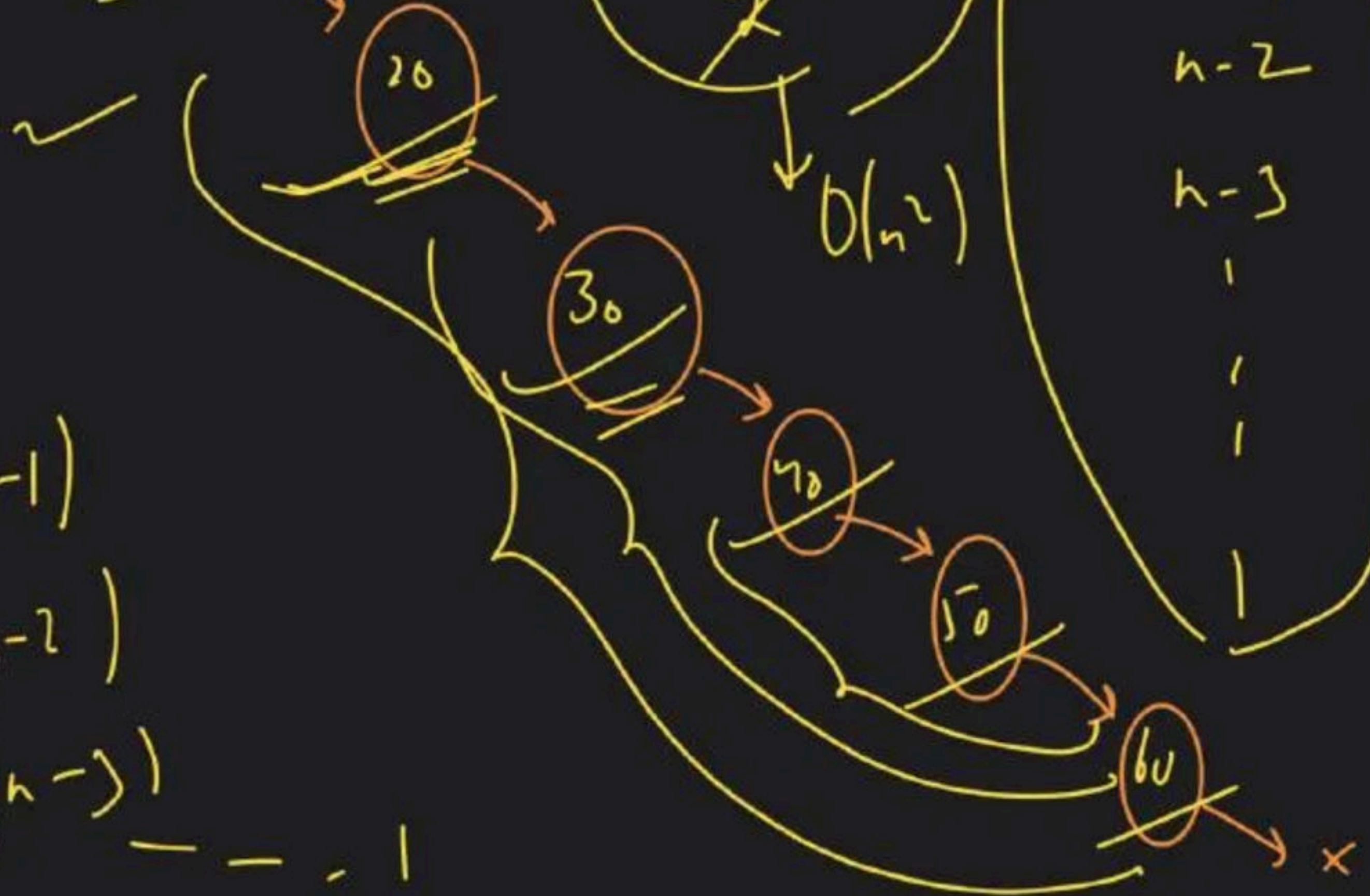
$$h \rightarrow O(n)$$

$$O(n-1)$$

$$O(n-2)$$

$$O(n-3)$$

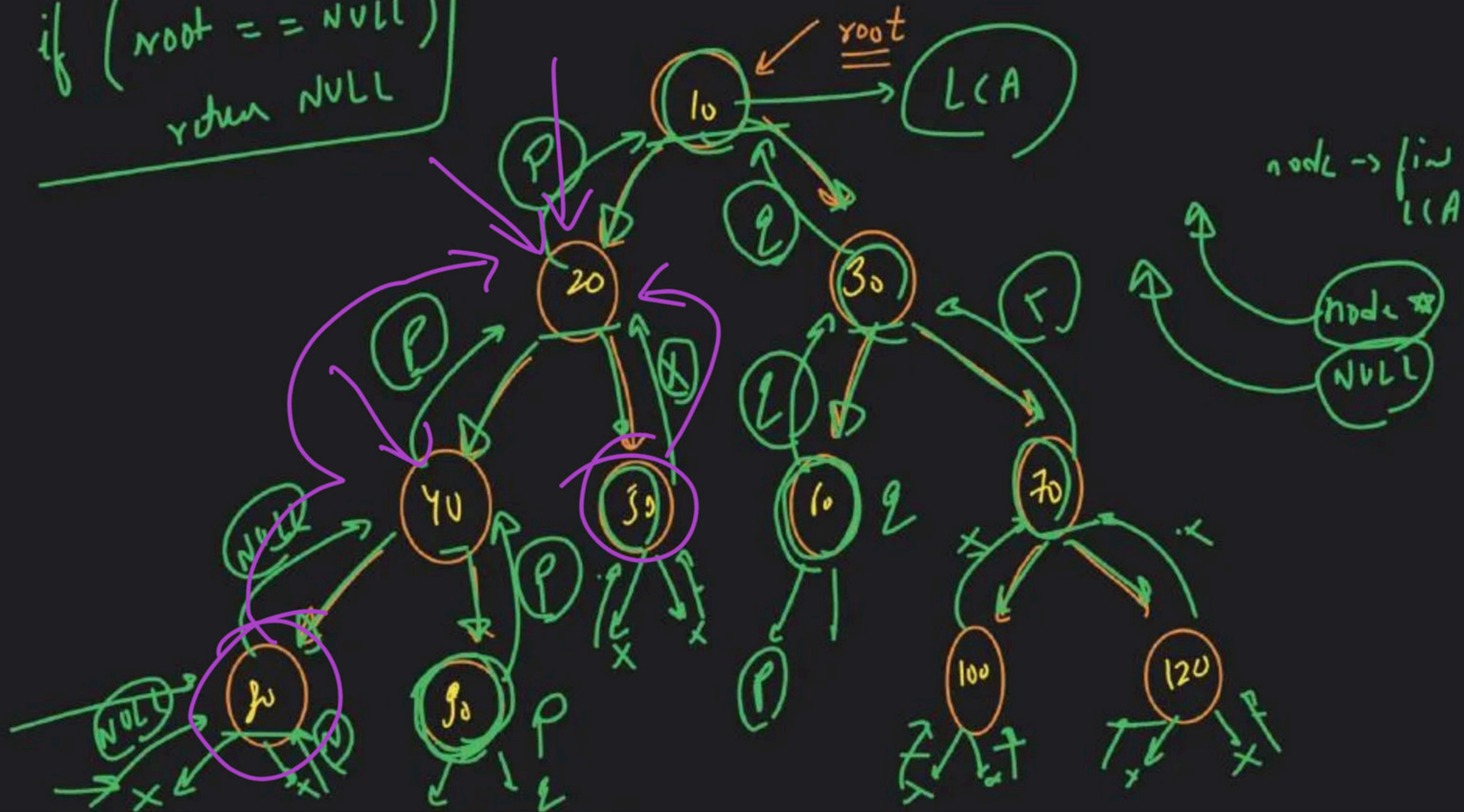
...

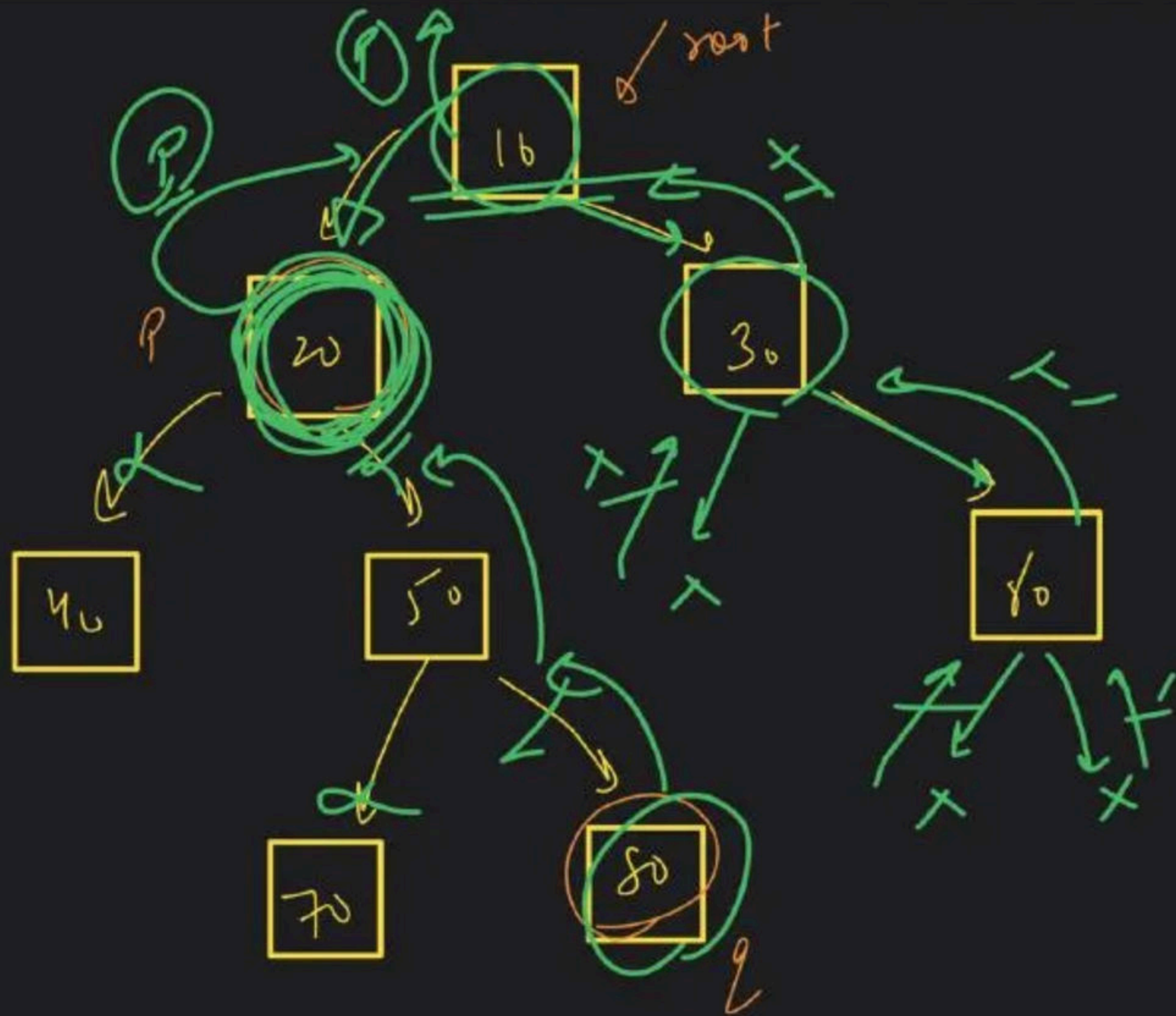


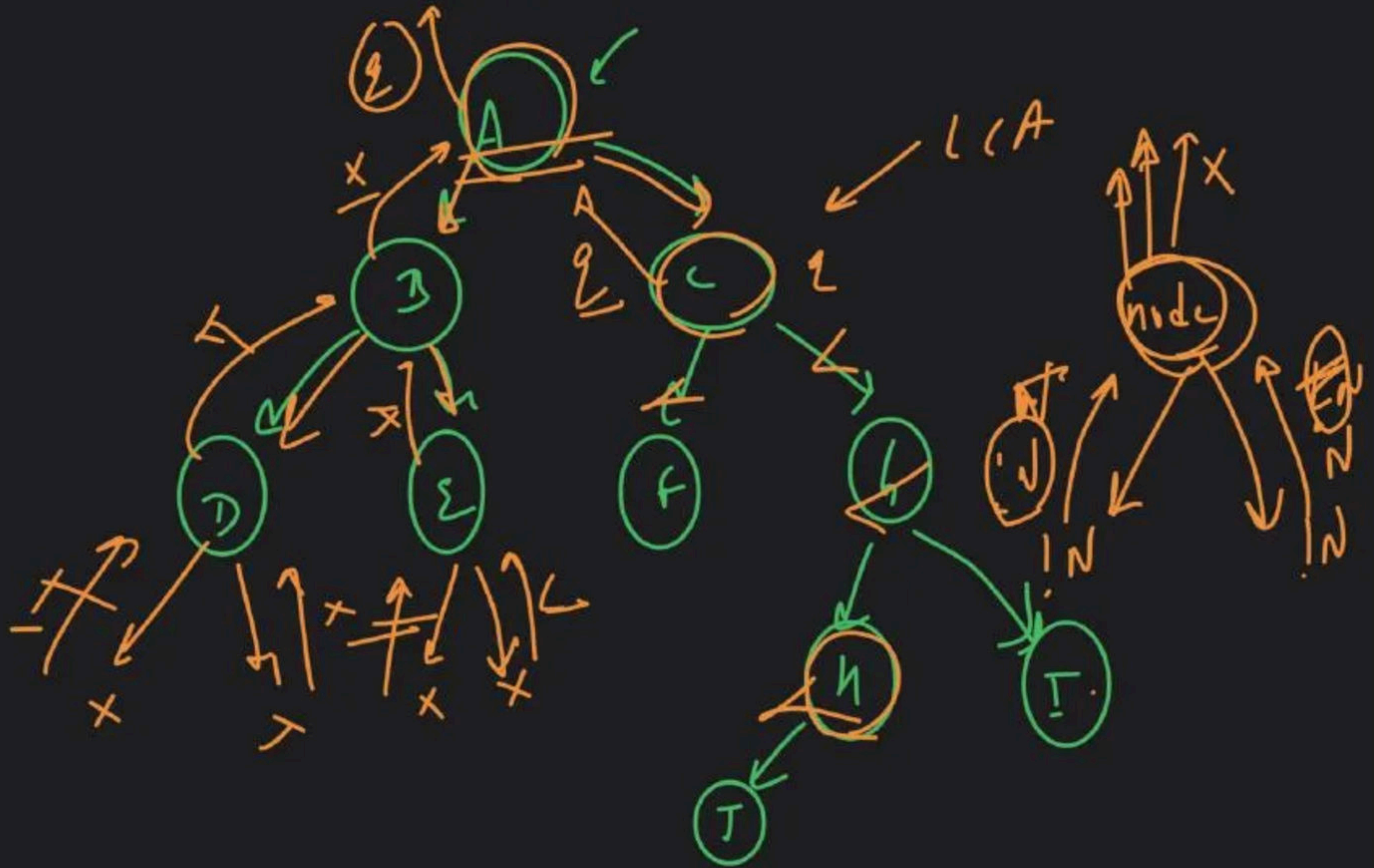
⇒ LCA of Binary Tree

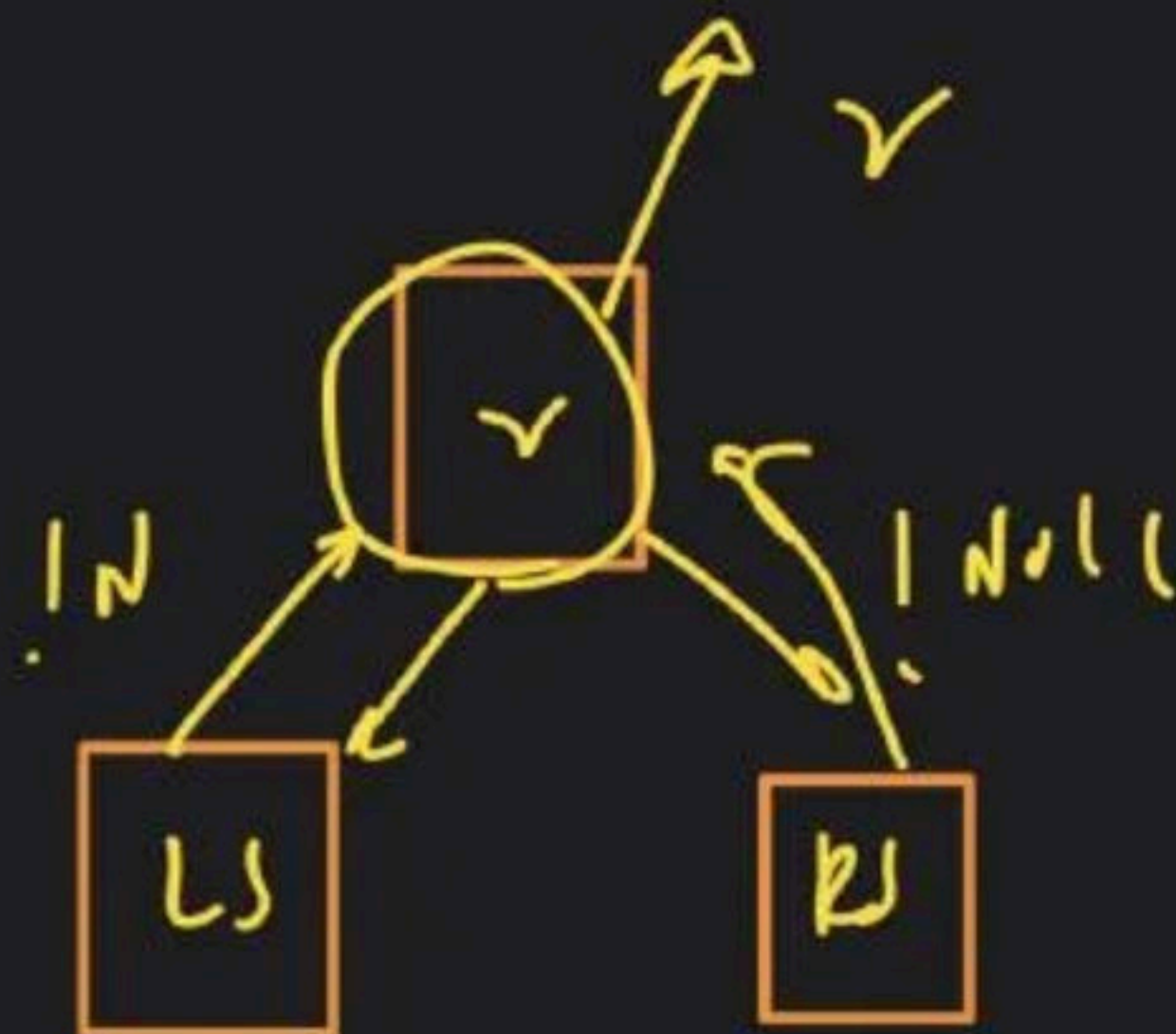
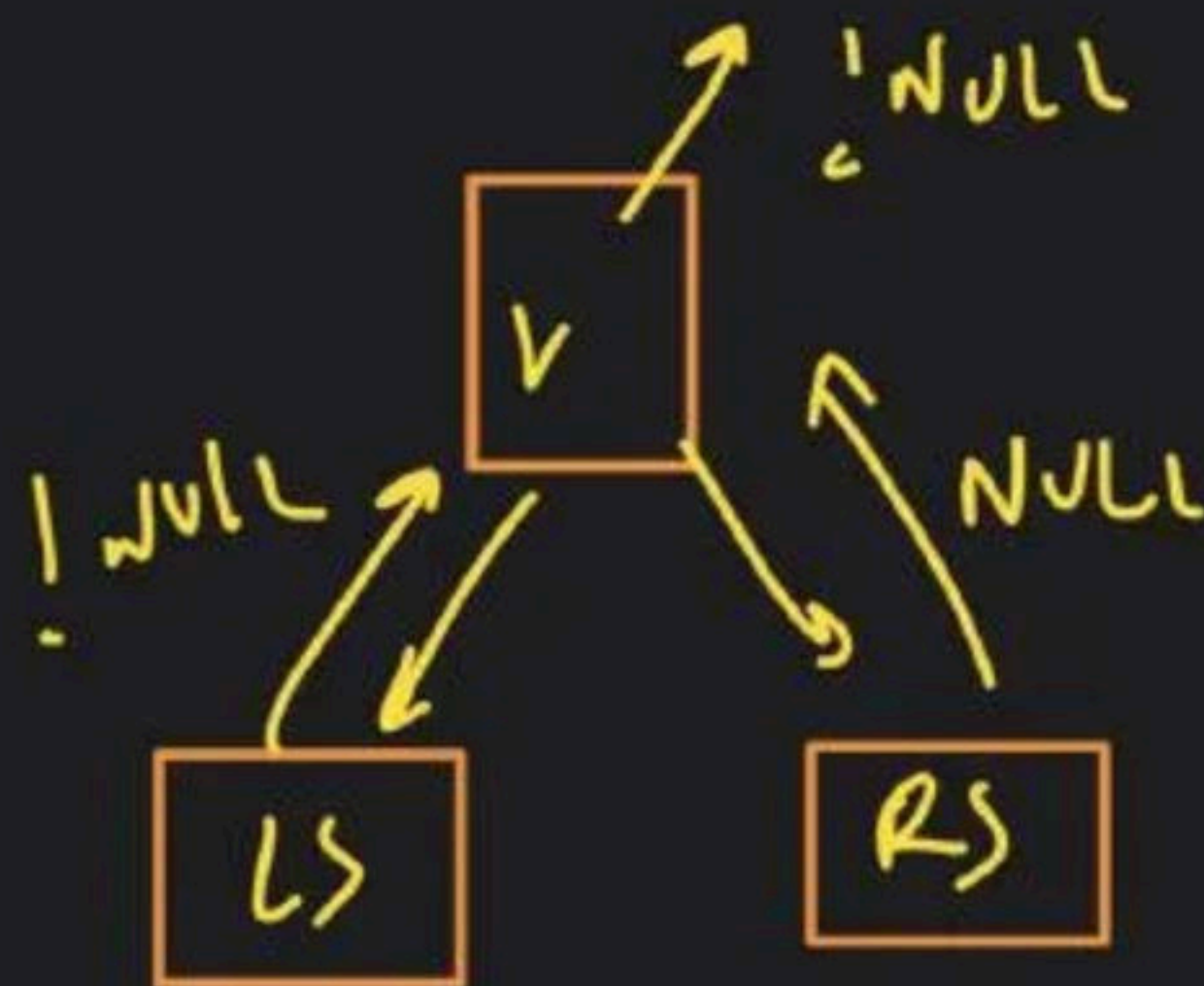
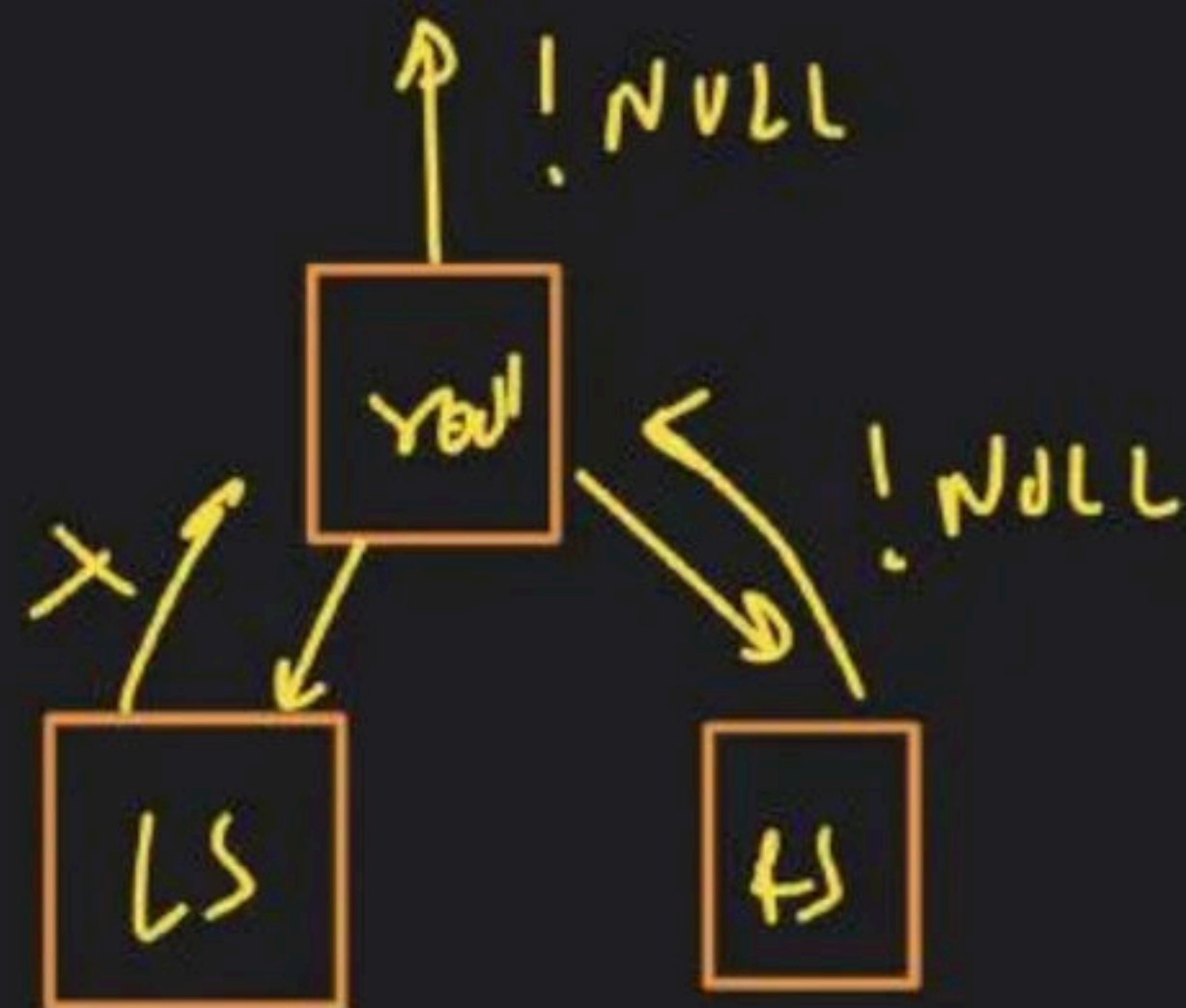
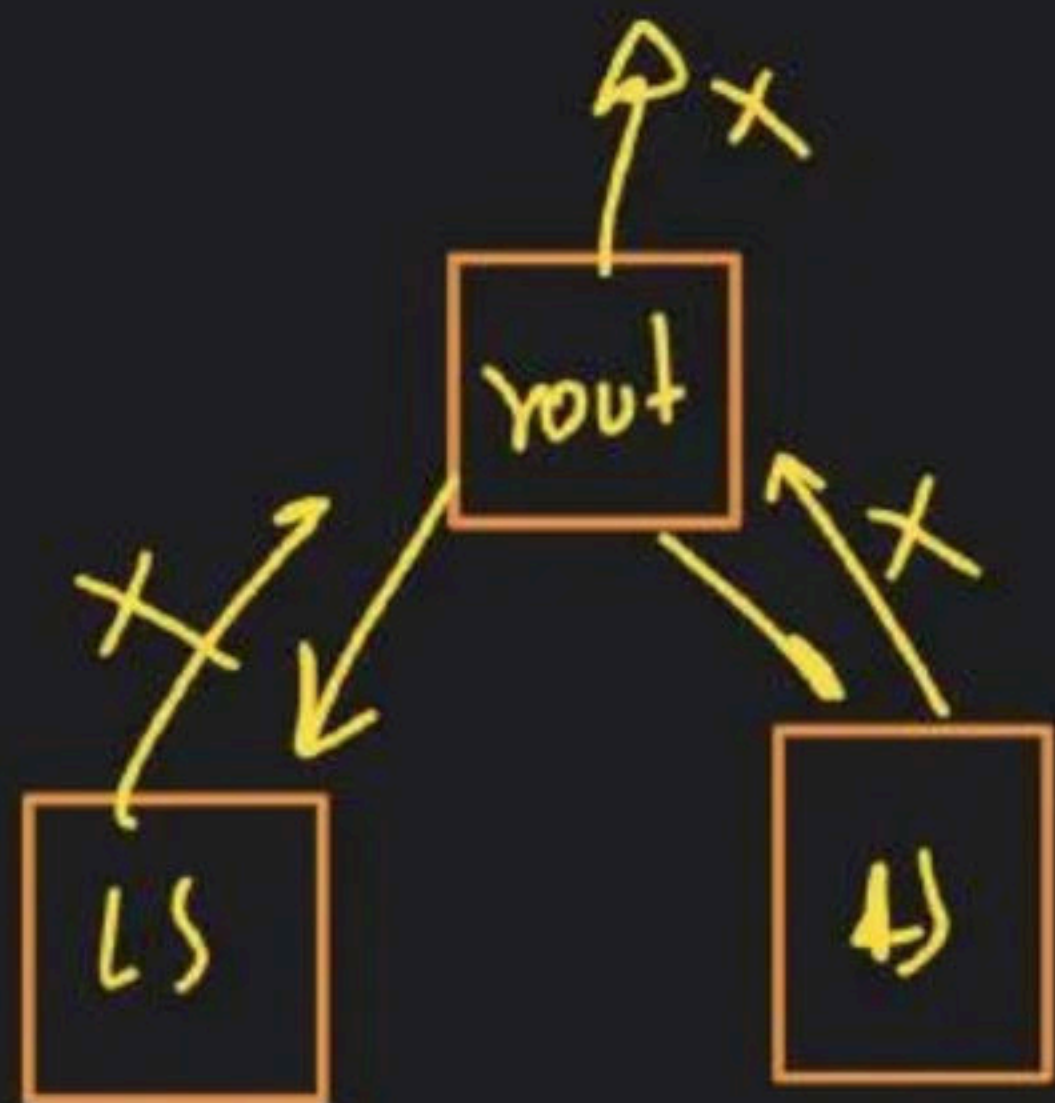
Lowest
Common
Ancestor

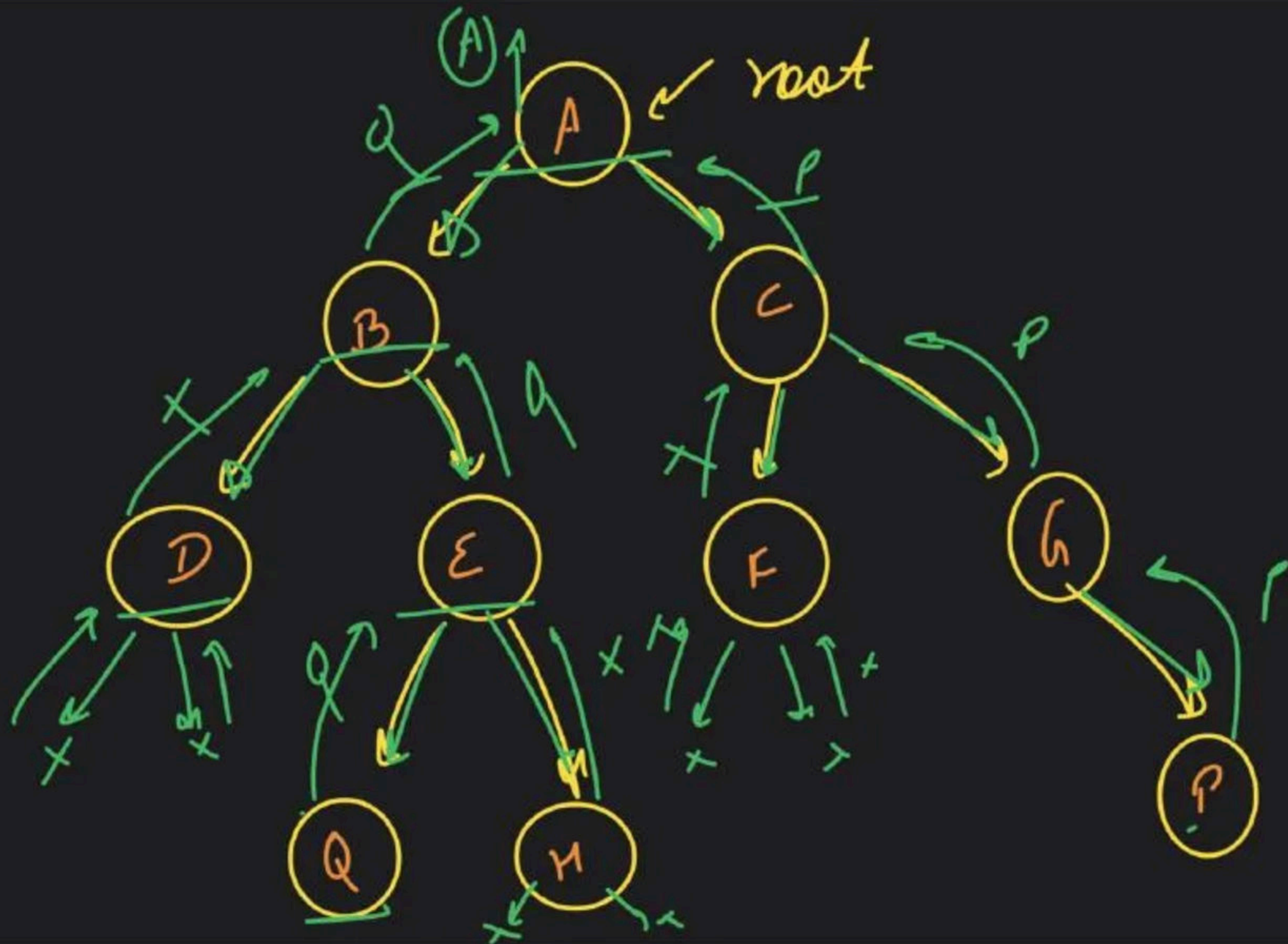
if (root == NULL)
return NULL

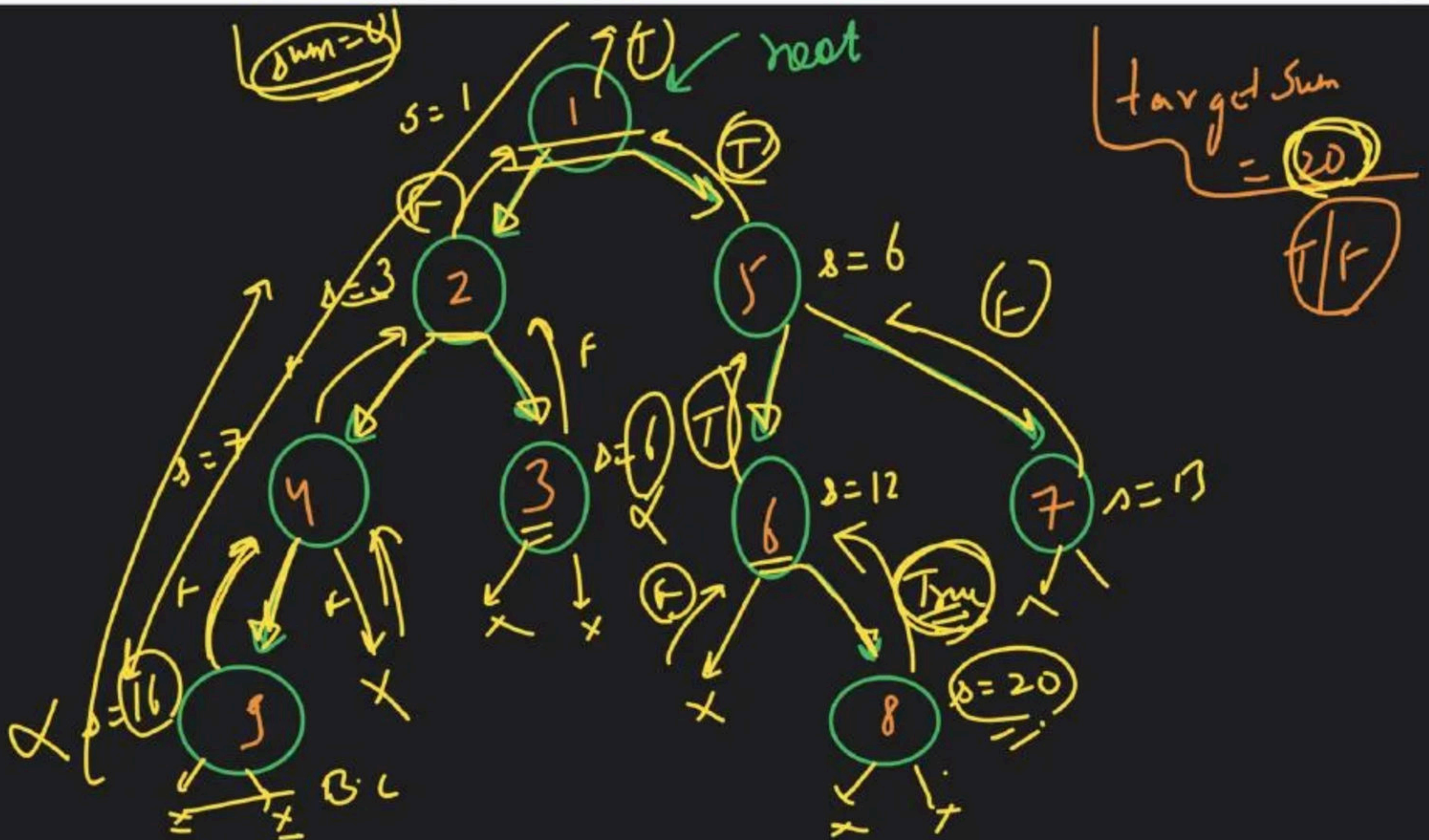






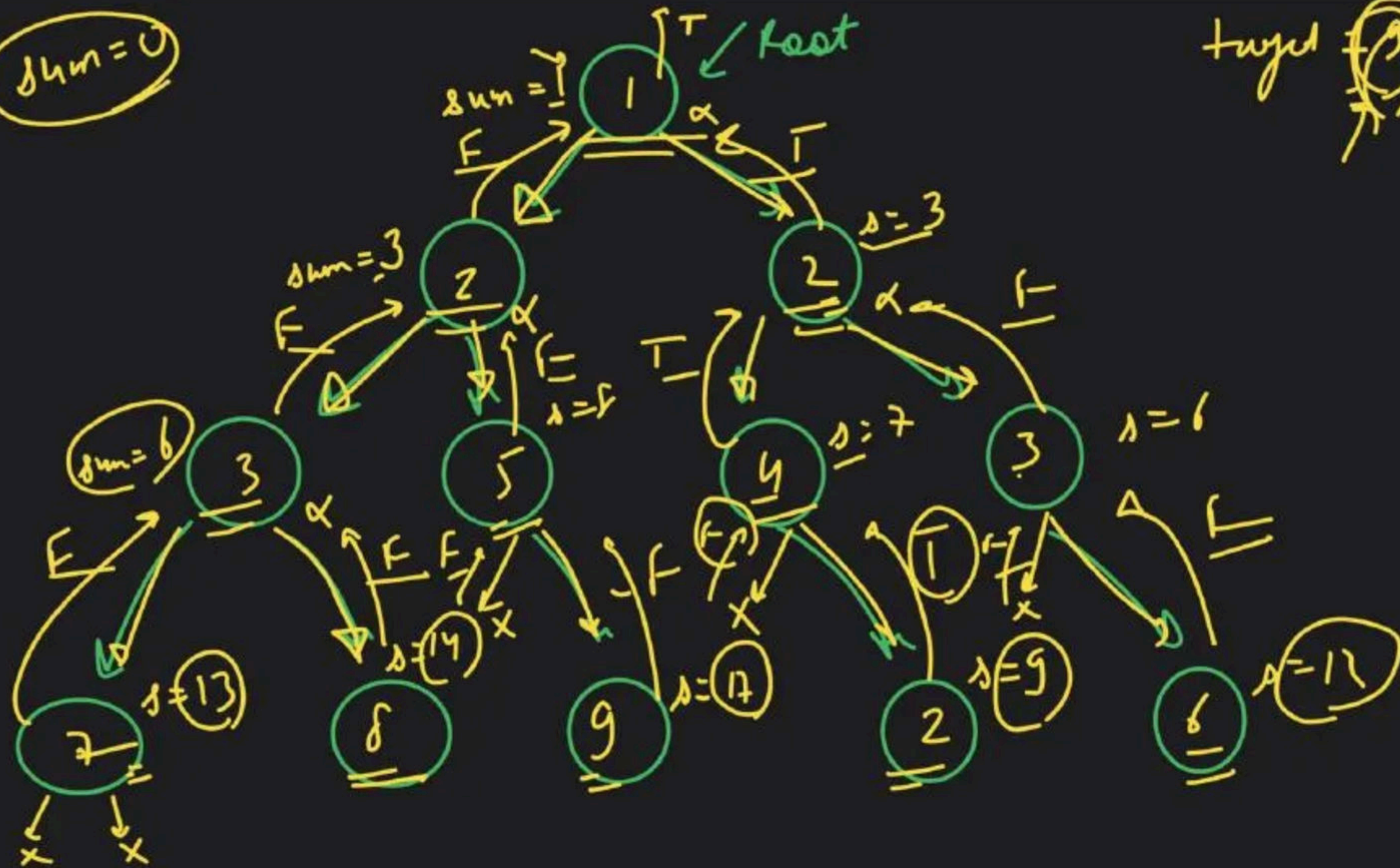






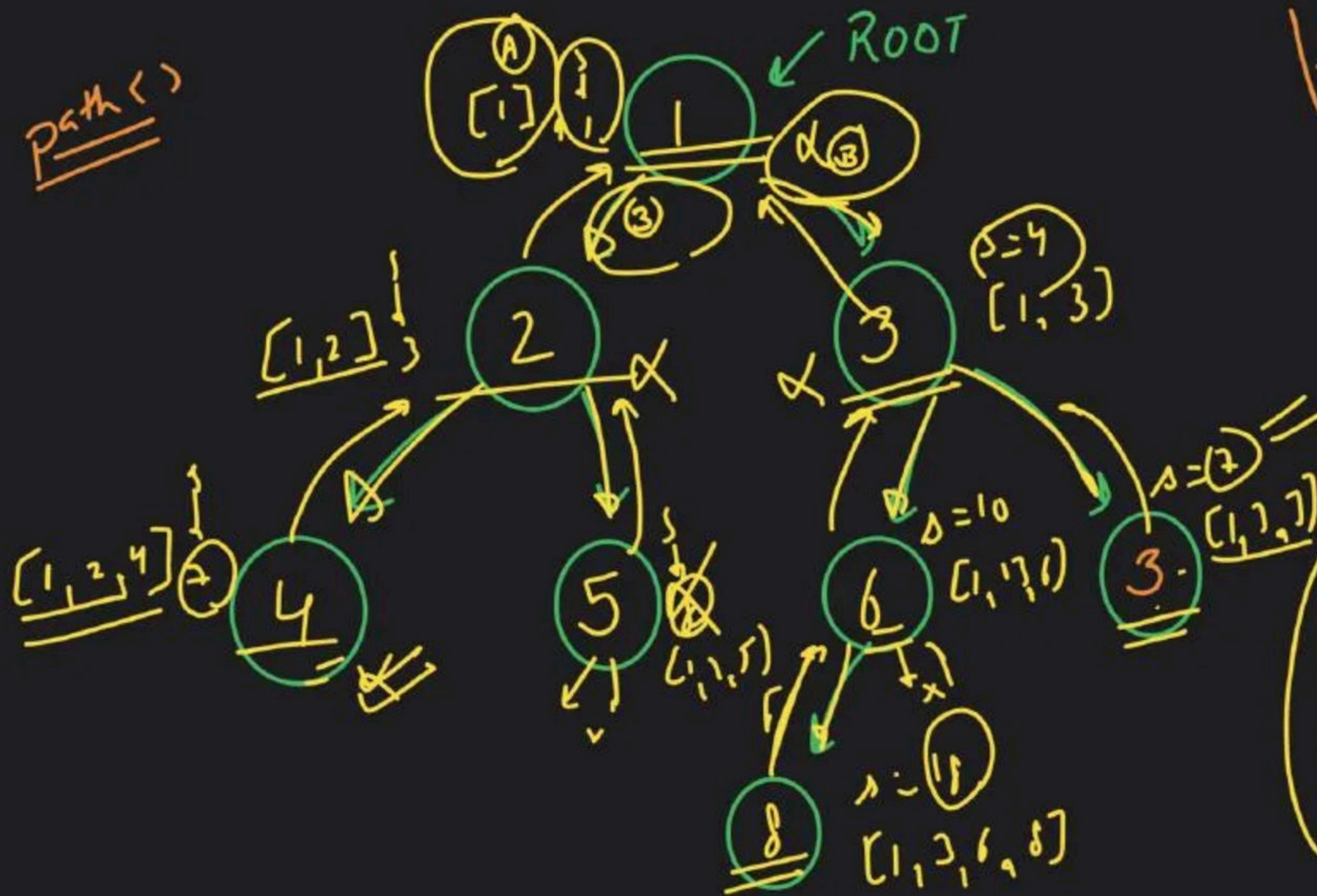
$\delta sum = 0$

target ± 5



path()

target = 7



ans
↓
 $[1, 2, 4]$
 $[1, 3, 3]$

Adobe



Kad - Extra

Class.



8:30 Pm



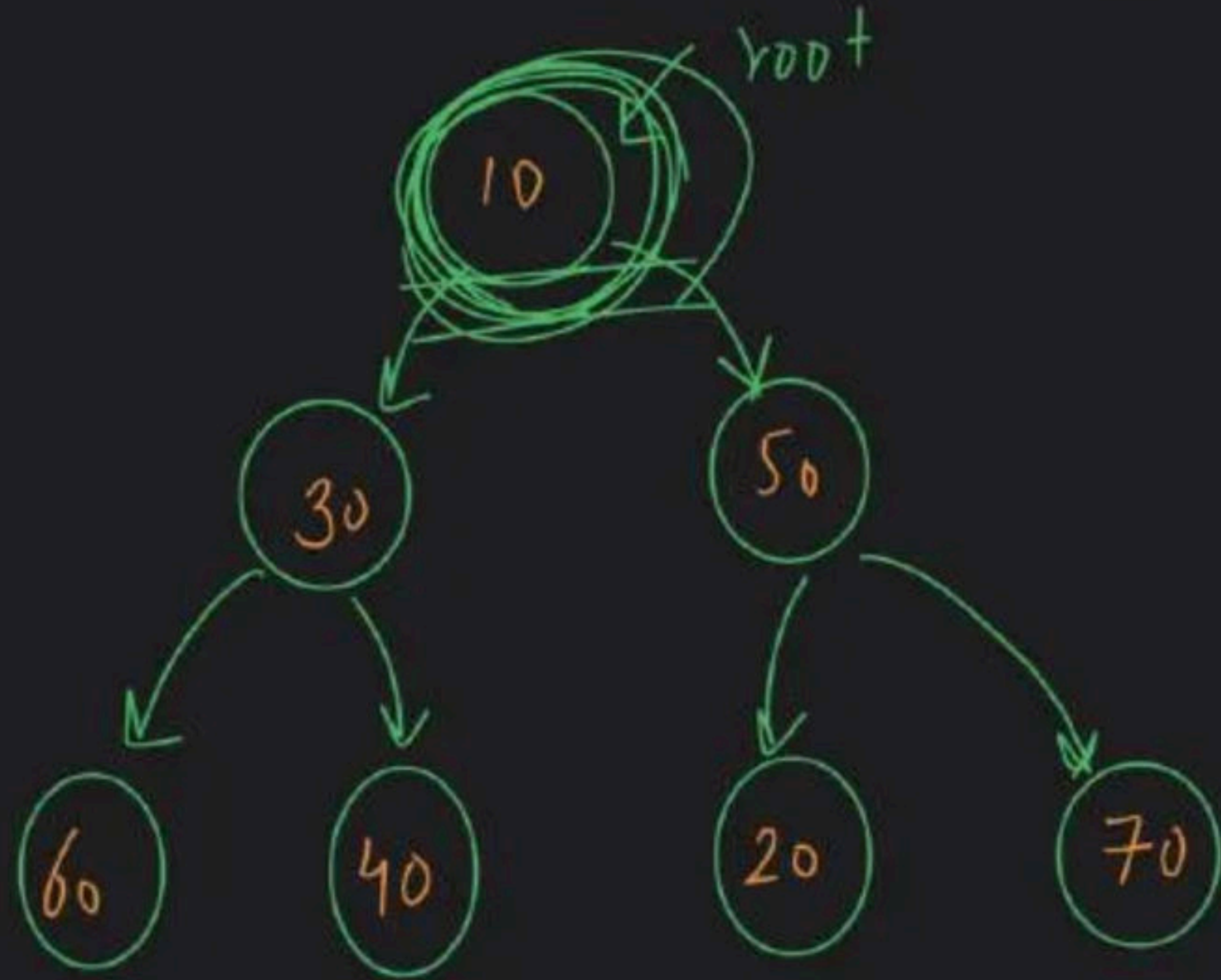
Yash

Path Sum - III

Trees Class - 3

Special class

→ Construct a tree from inorder & preorder traversal



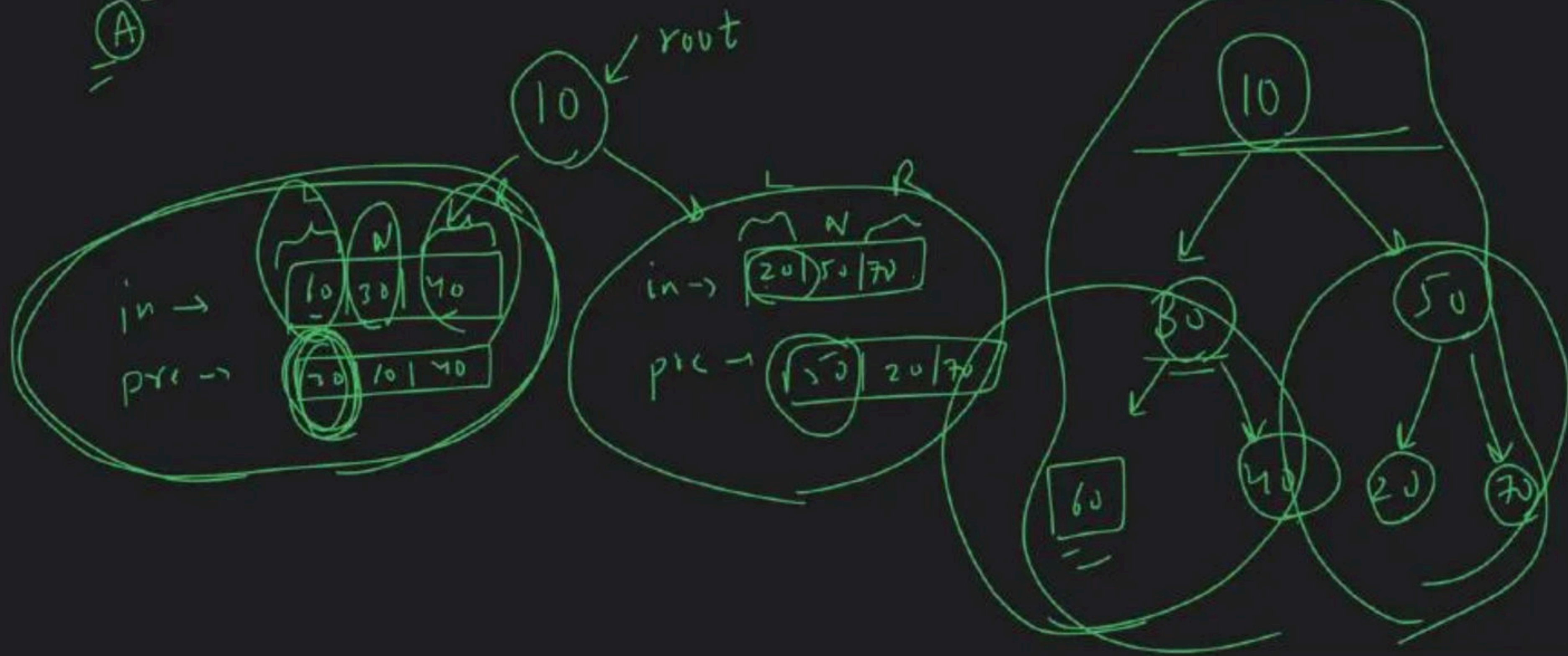
(LNR)
inorder → 60 30 40 10 20 50 70

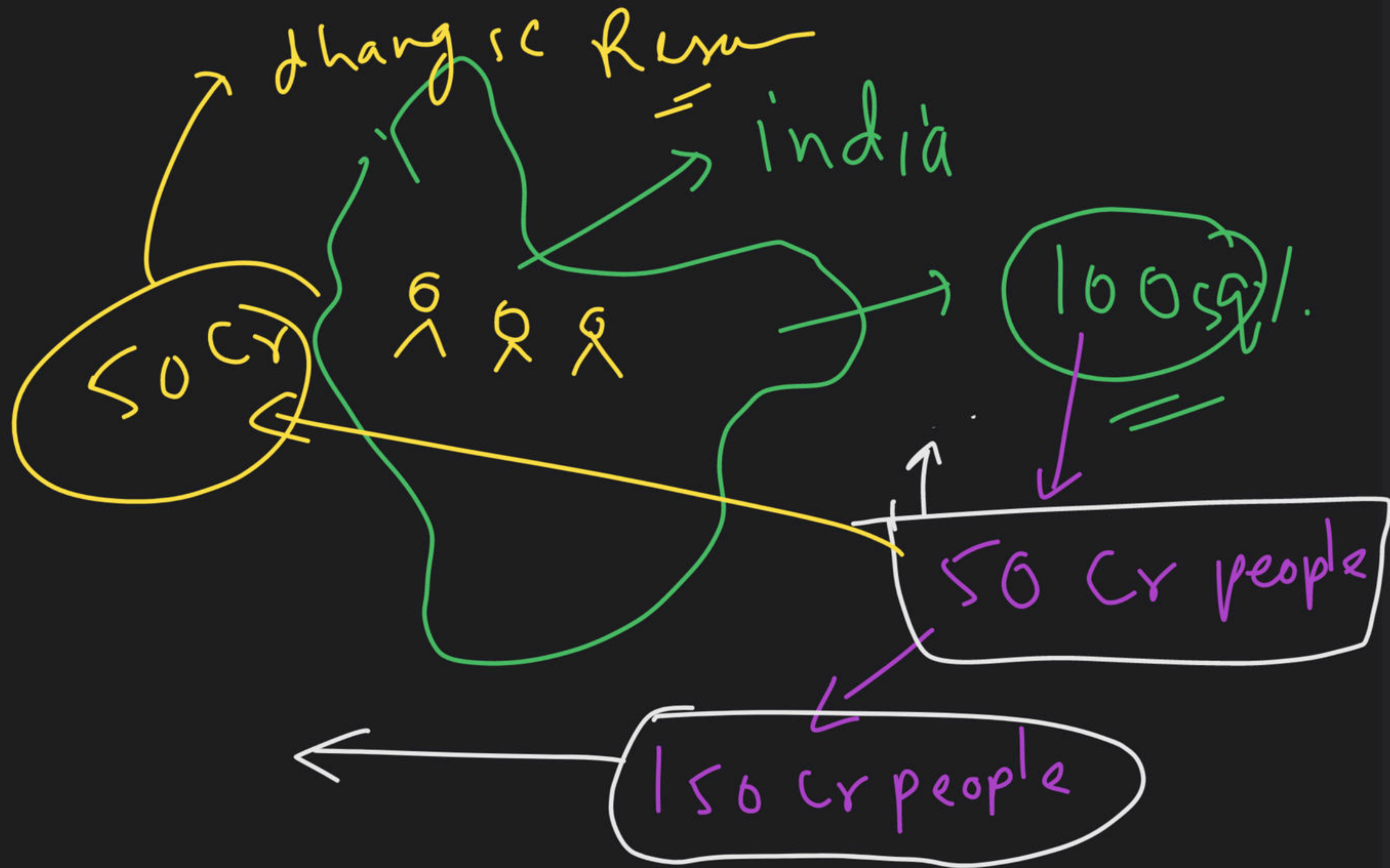
(NLNR)
preorder → 10 30 60 40 50 20 70
root

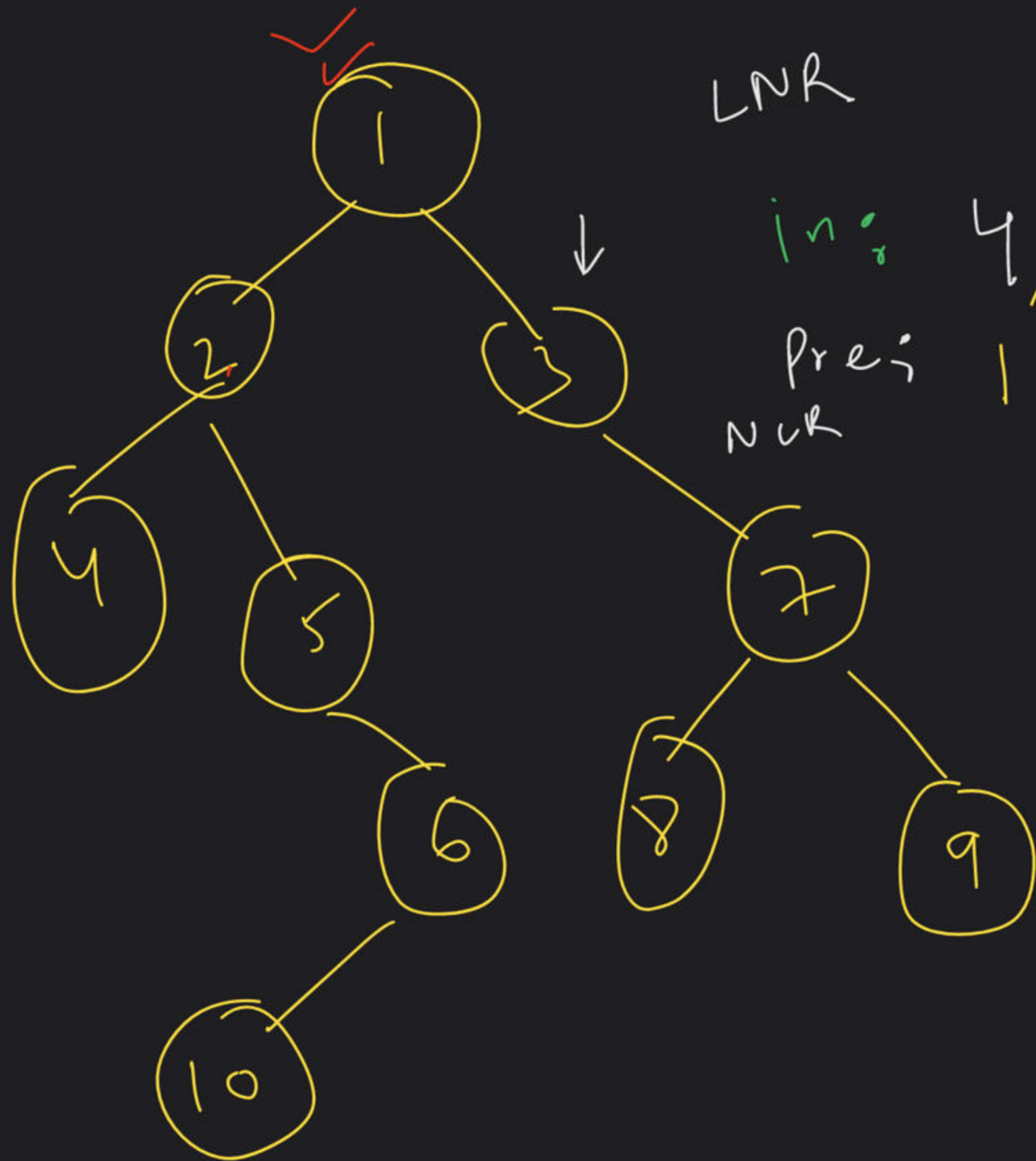
= inorder $\rightarrow$ [10, 30, 40, 10, 20, 50, 70] $\rightarrow$ [L N R]

= preorder $\rightarrow$ [10, 30, 60, 40, 50, 20, 70] $\rightarrow$ [N L R]

A







LNR

in →

in:

4, 2, 5, 10, 6, 1, 3, 8, 7, 9

Pre:

1, 2, 4, 5, 6, 10, 3, 7, 8, 9

NCR

in: 4 | 2 | 5 | 10 | 6 | 1 | 3 | 8 | 7 | 9

Pre: 1 | 2 | 4 | 5 | 6 | 10 | 3 | 7 | 8 | 9

→ LNR

→ NLR

✓ ① Pre[indent+1] = in den →
 ↓
 Node - 1 indent + 1 =

mid = 5

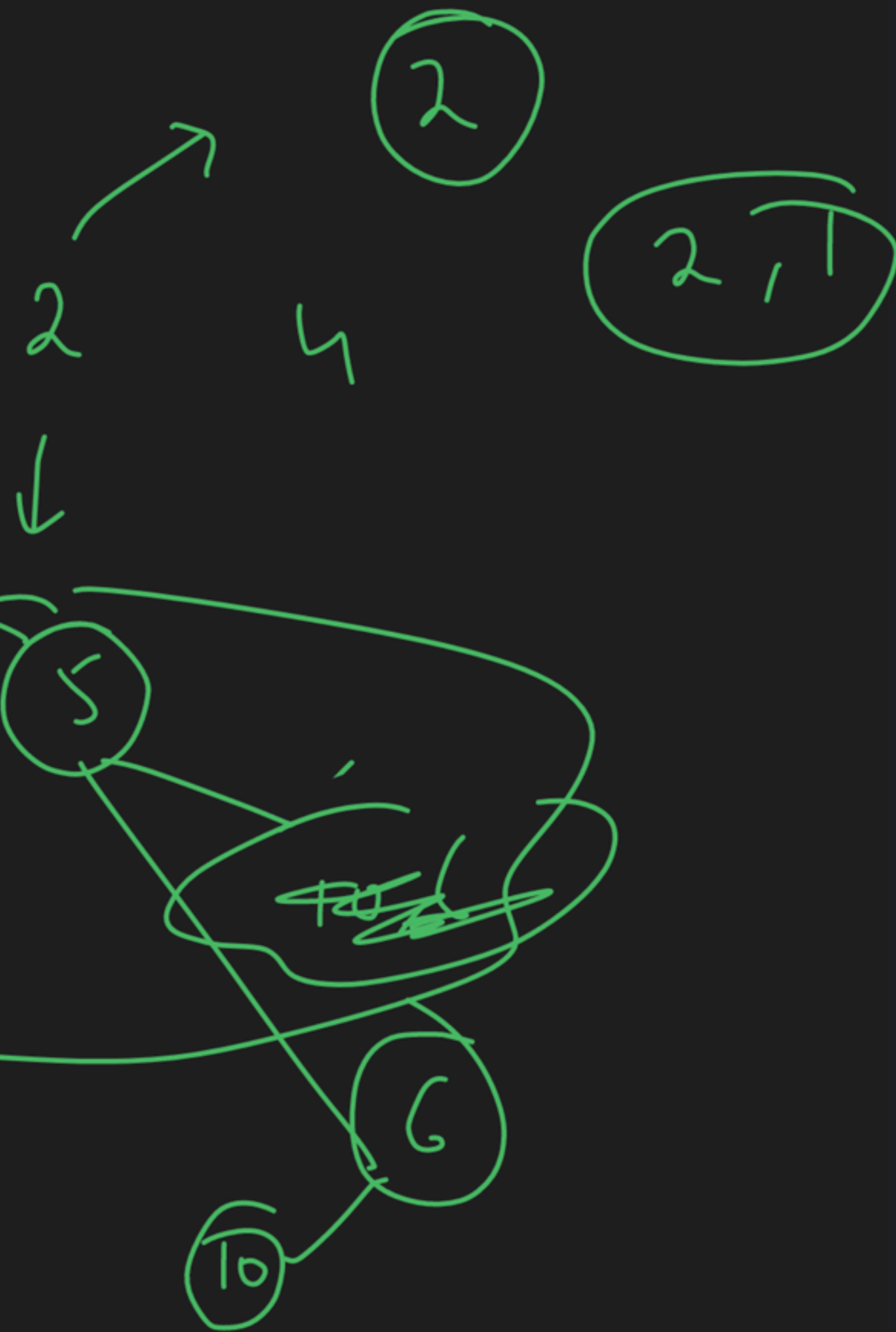
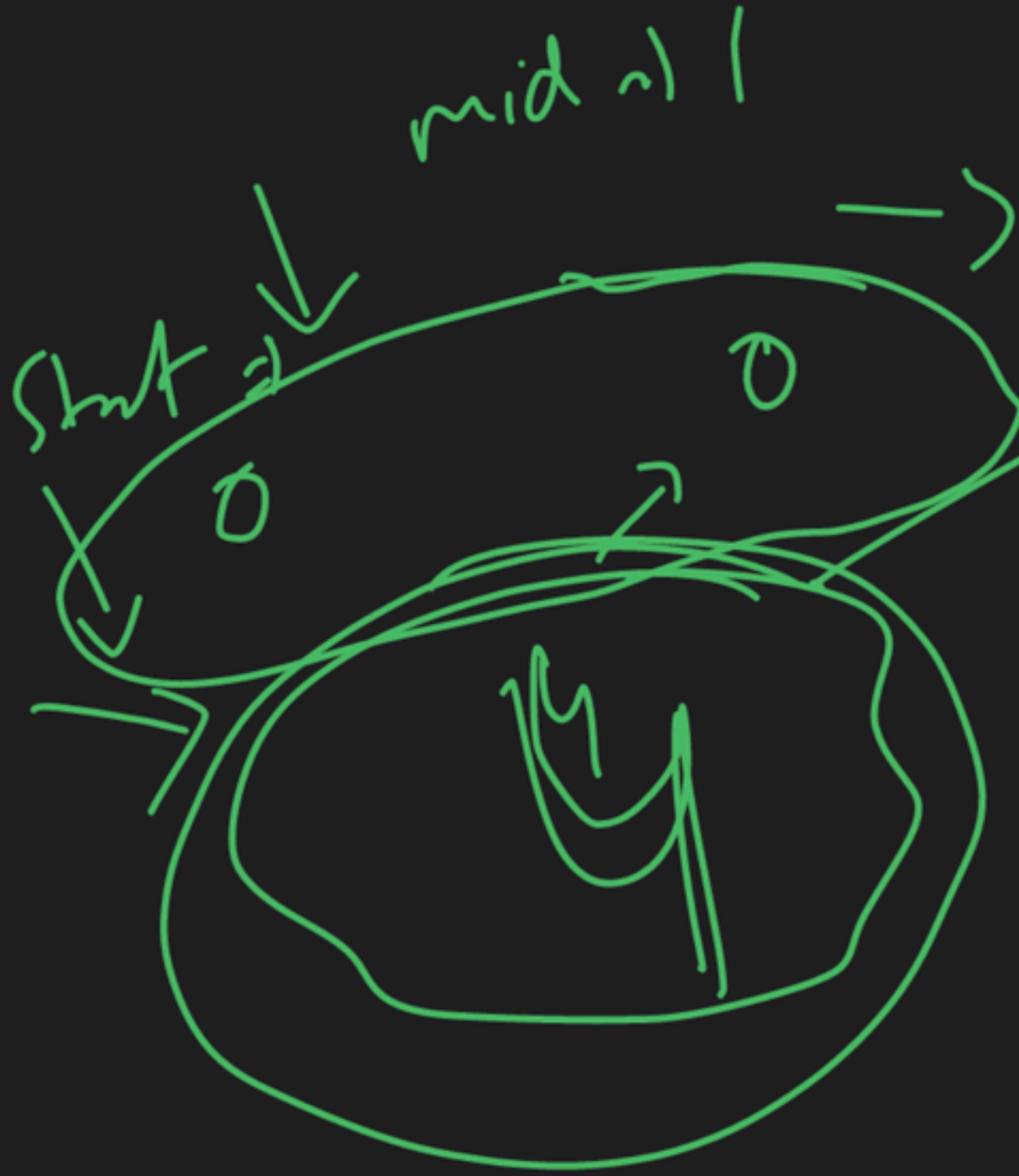
s → 0

start = 0
 end = 9



- 1

4 | 2 | 5 | 10 | 1
↑ ↑

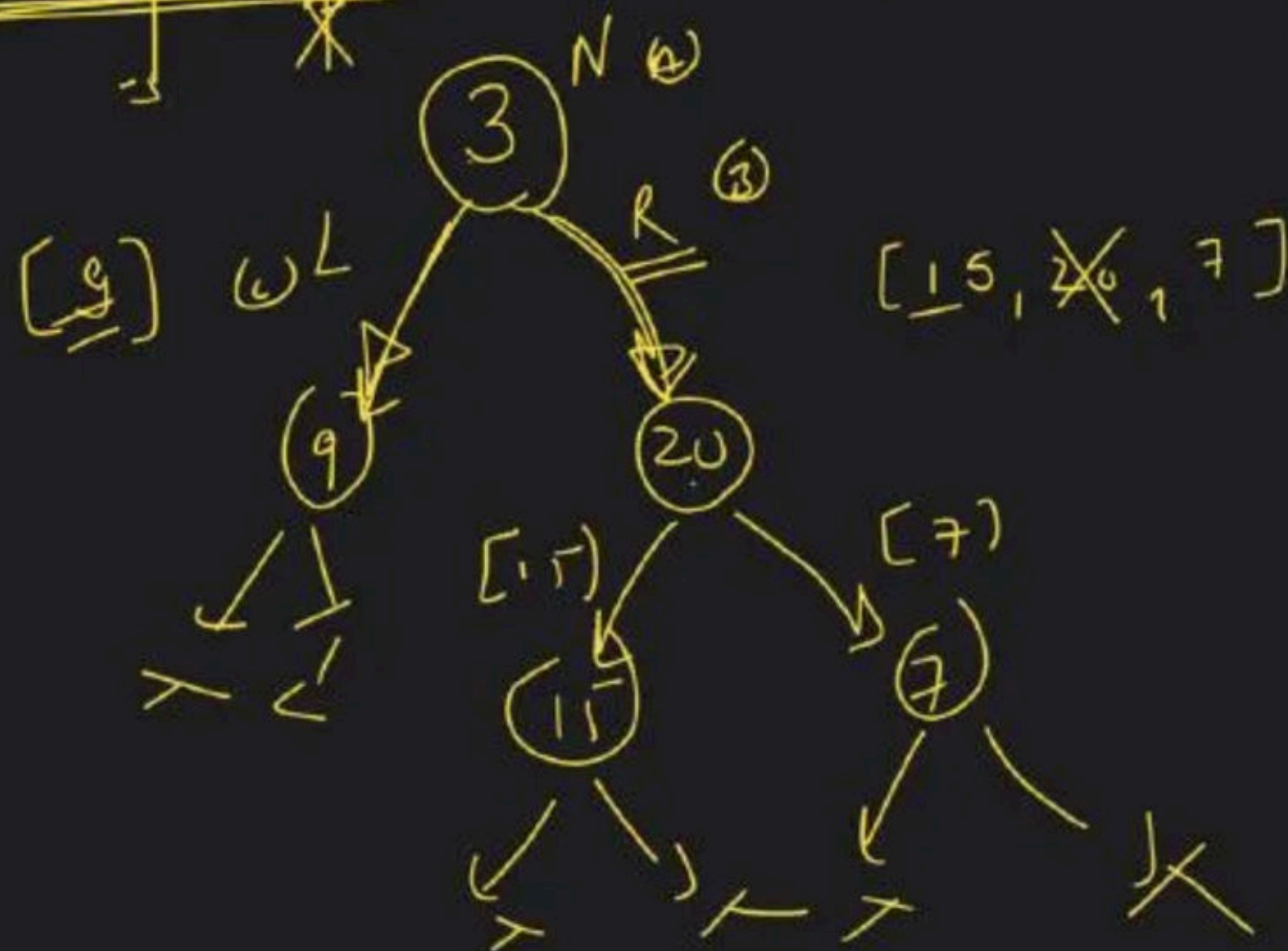


in $\rightarrow$ $\overset{L}{\boxed{9}} \cancel{2}, \overset{R}{\boxed{15, 20, 7}} \rightarrow \underline{L} \textcircled{N} R$

post = $\boxed{9}, \boxed{15}, \boxed{7}, \boxed{20}, \boxed{3} \rightarrow \underline{L} R \textcircled{N}$

in/post

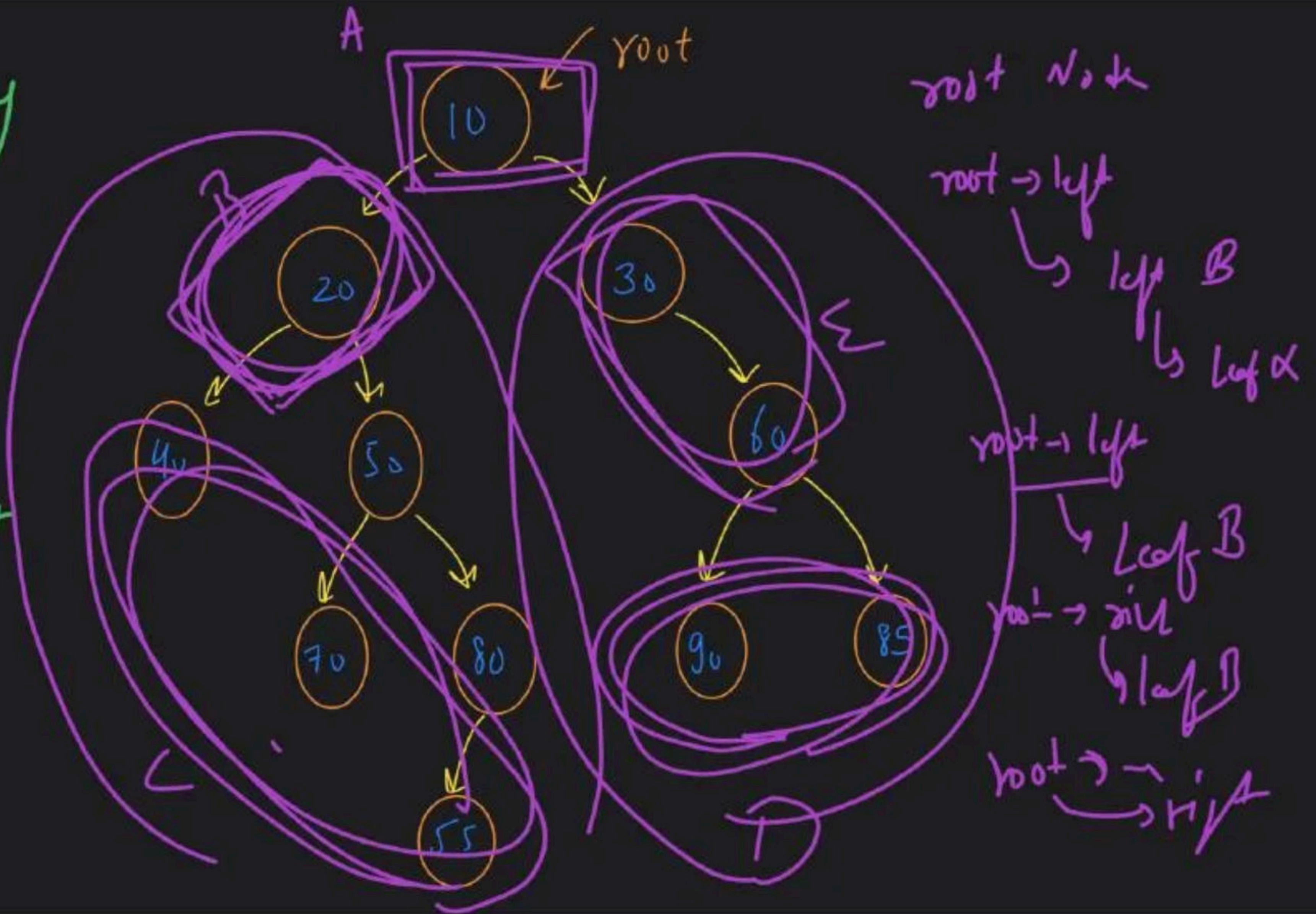
right subtree
left subtree



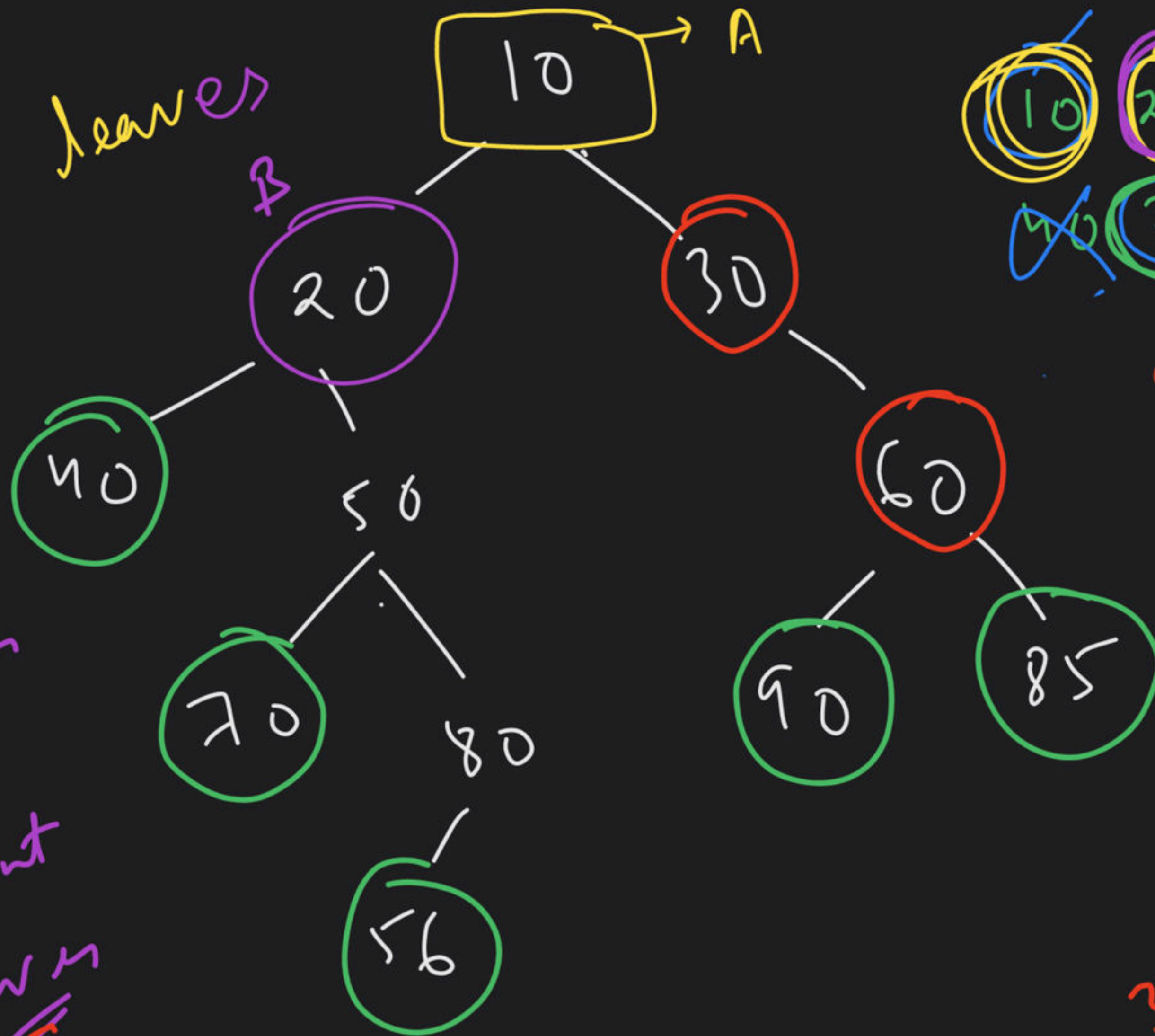
Boundary

$$\vdash \text{Left}^B$$
$$\hookrightarrow \text{Lef } \mathcal{B}$$

↳ Rev Rigel



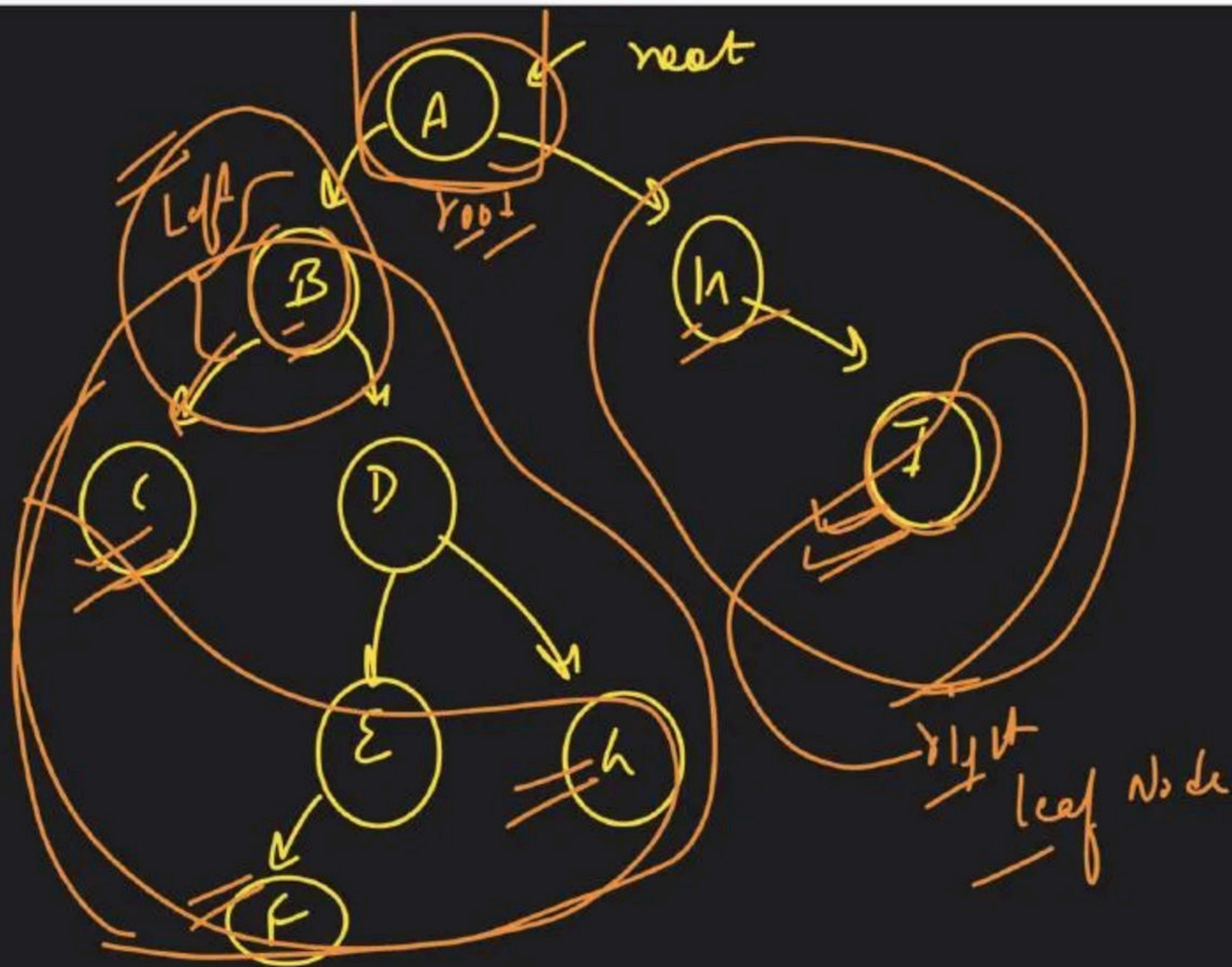
Left B
↓
do not print leaves

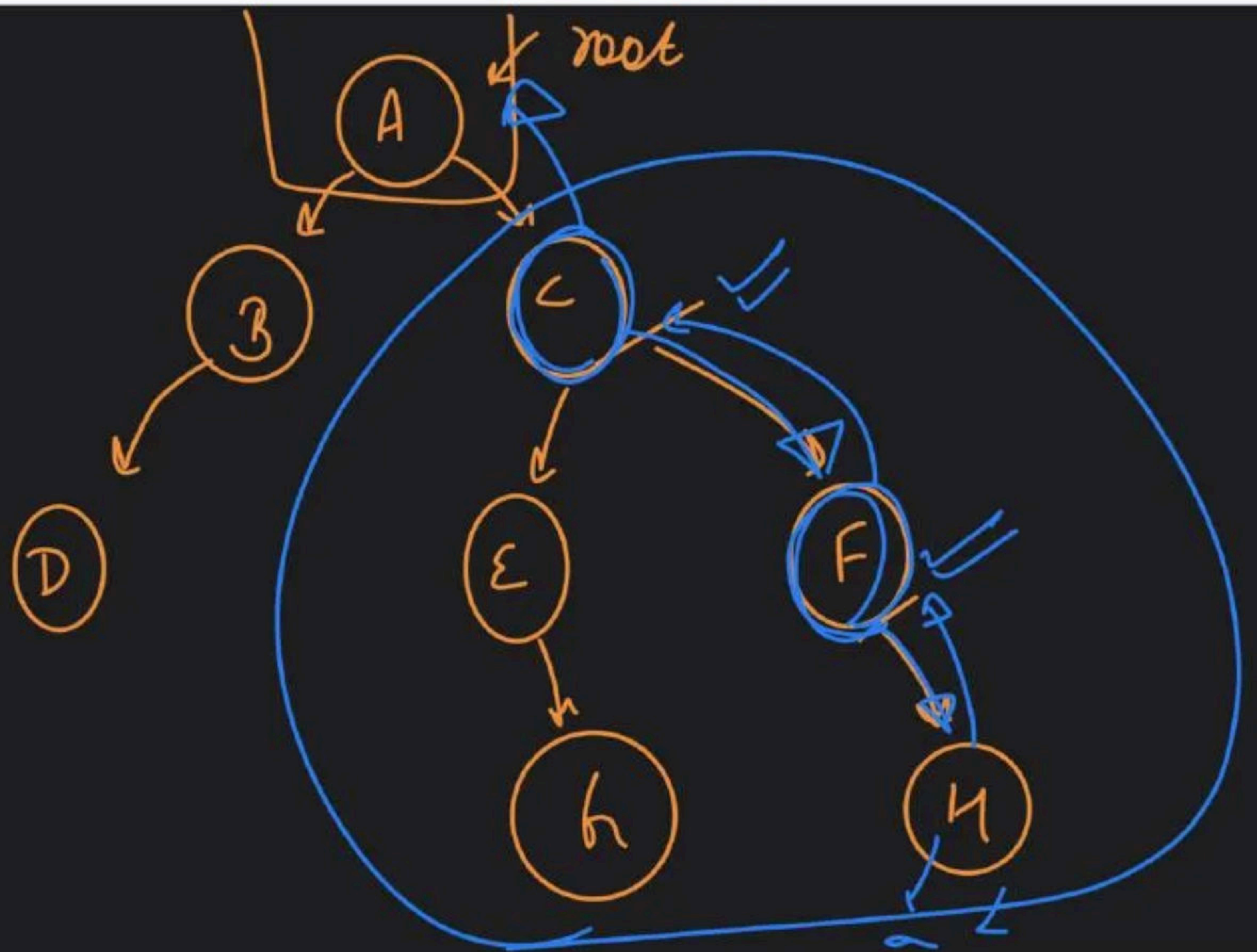


Right B
↓
do not print
leaves

30 50

Left
leaf node





< - >

[1]

Lvl 0

[10]

Lvl 1

[10, 40]

Lvl 2

60

[10, 40, 60]

Lvl 3

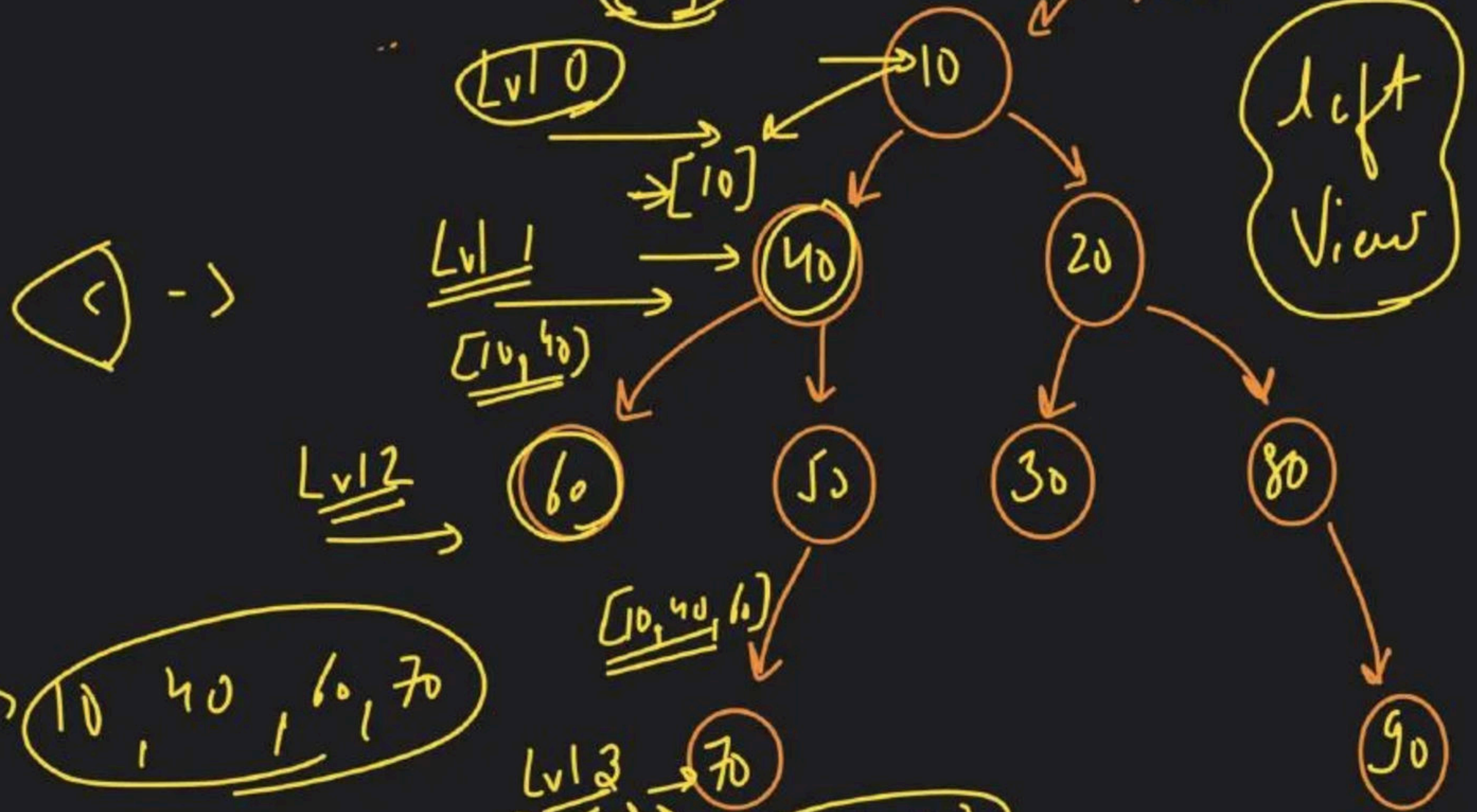
70

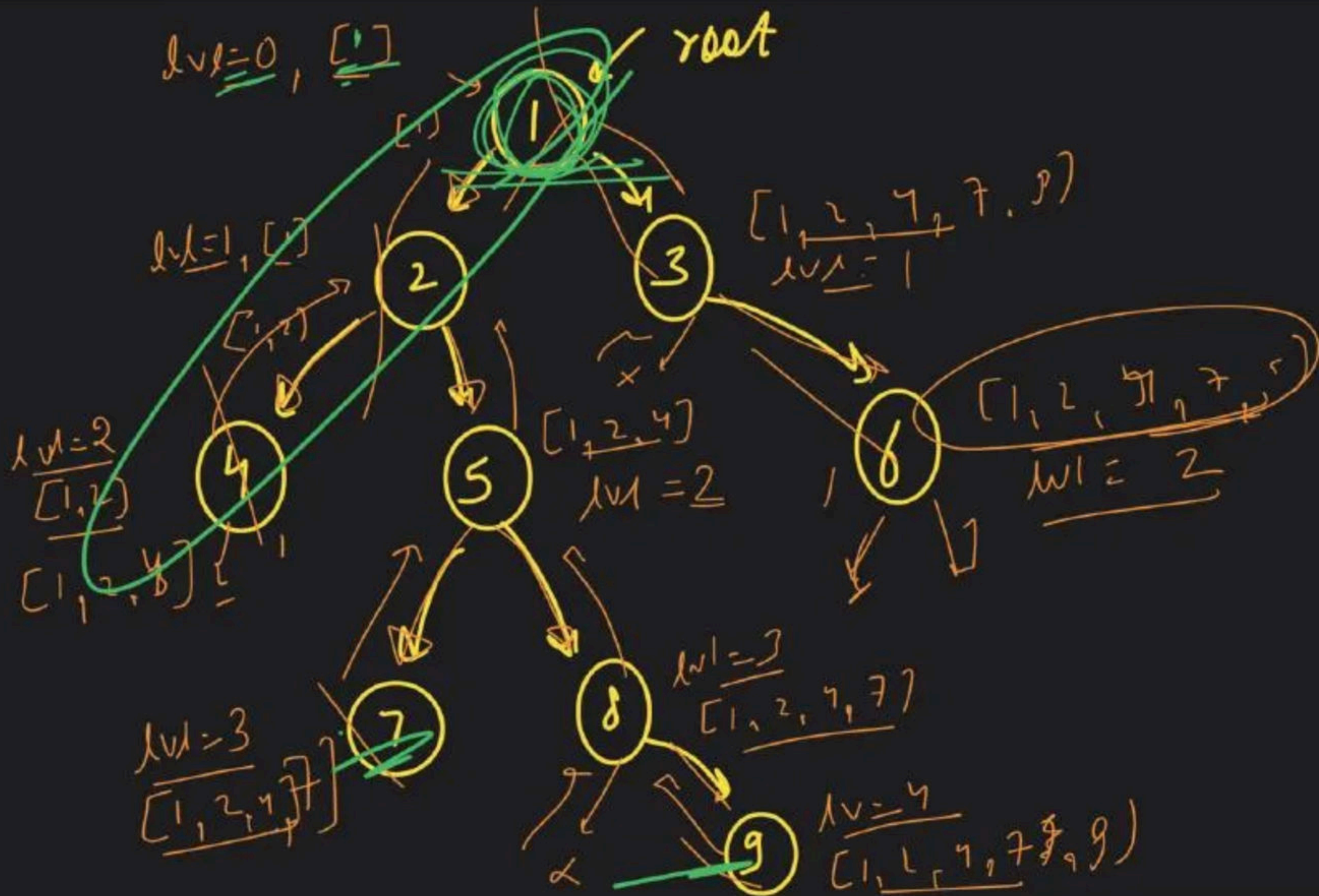
[10, 40, 60, 70]

root

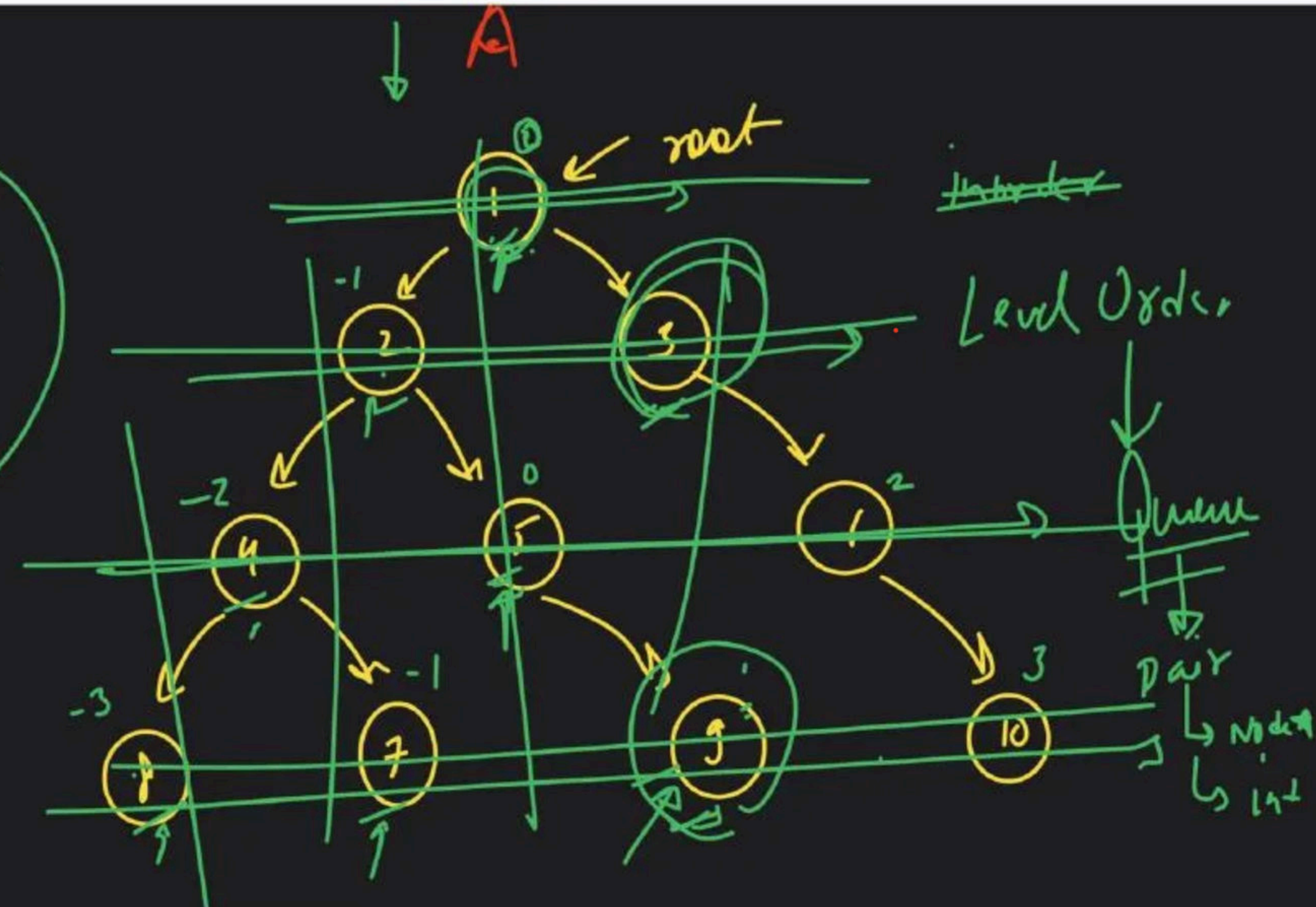
Left View

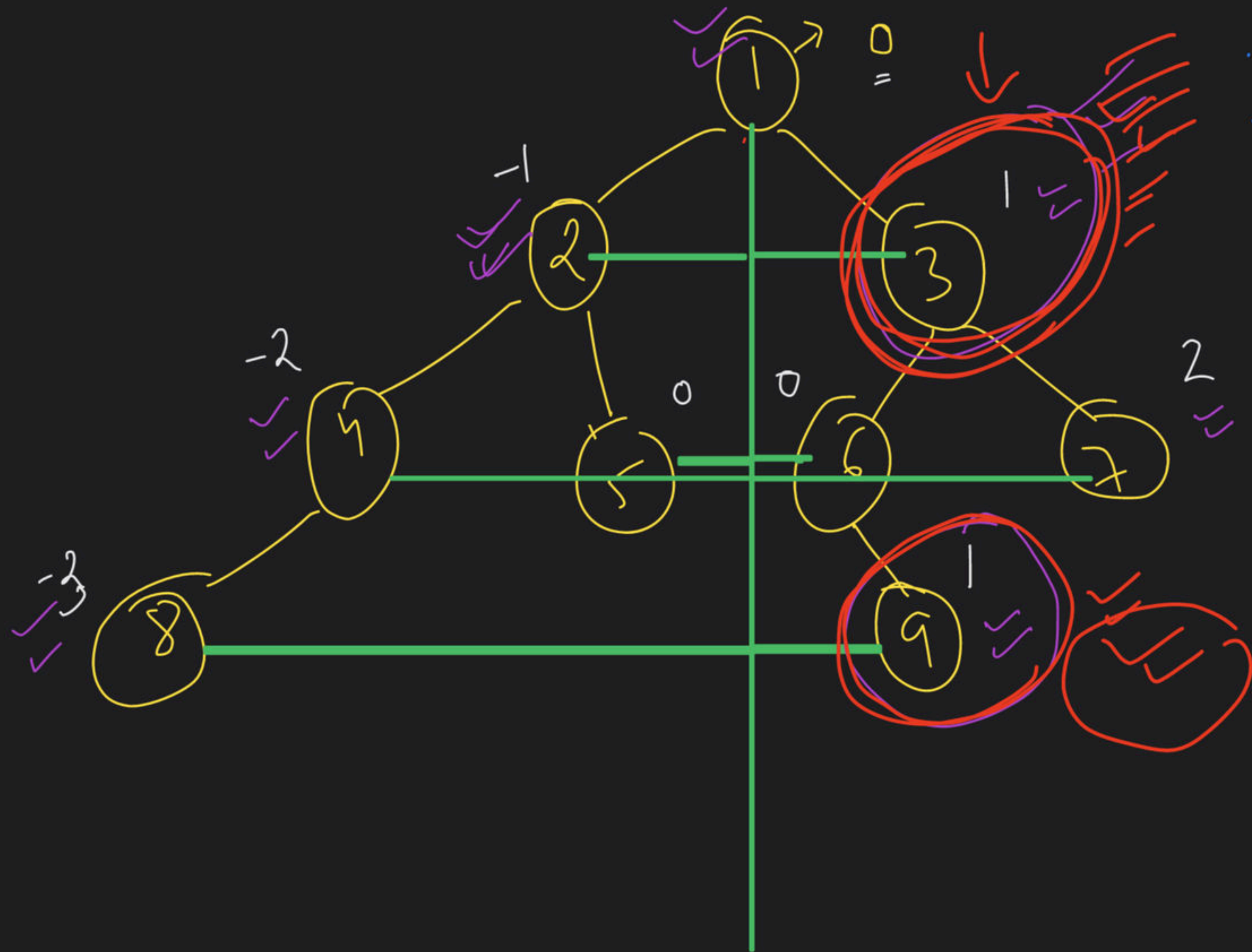
ans -> 10, 40, 60, 70





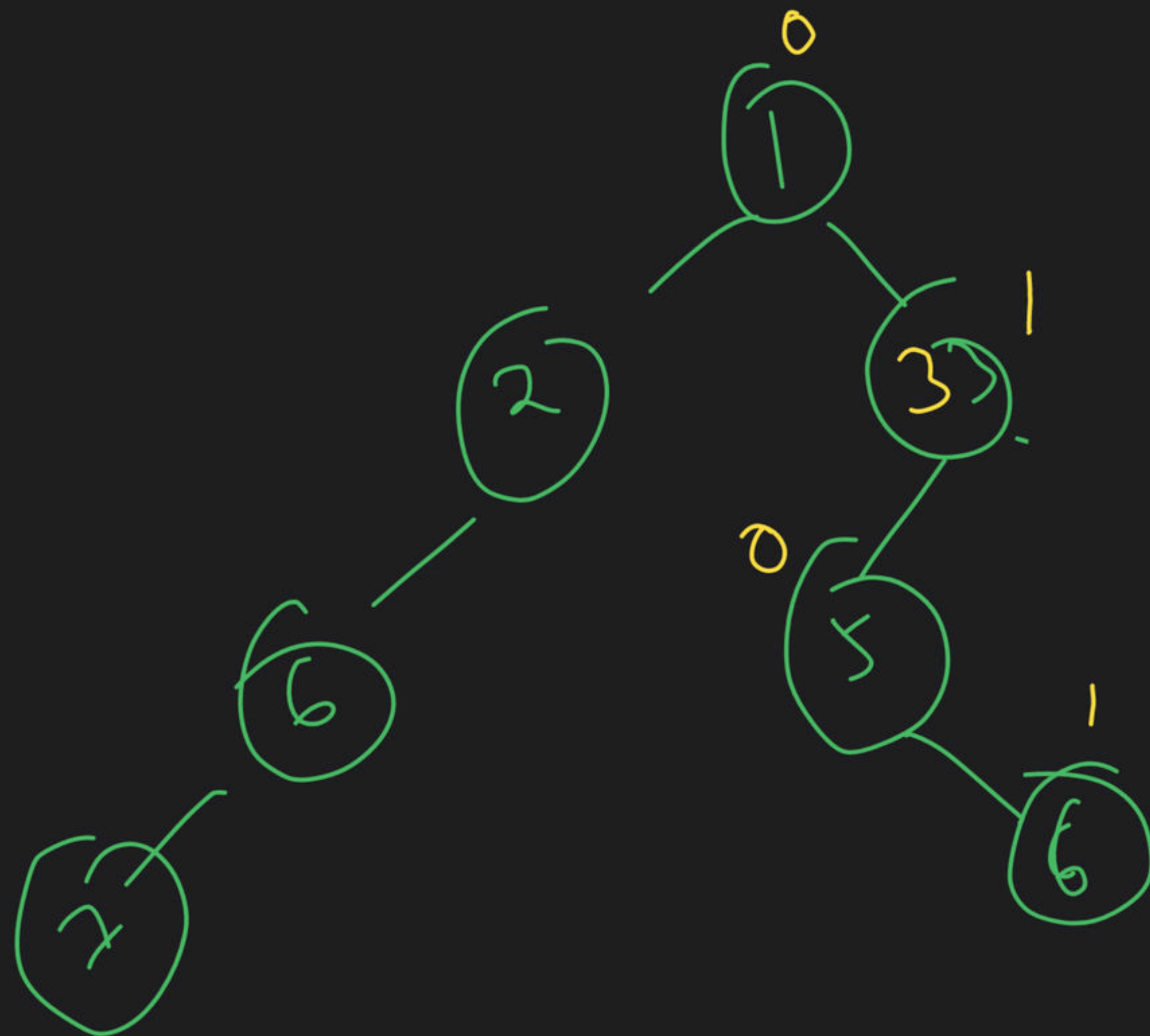
In ϕ
 pre ϕ
 post ϕ





jisBhi
 Kur. Dist
 Par sbse
 phle visit
 krunga wahi
 node dikhai
 krunga





1 3 5 6





